

Fraud, Evasion, and Enforcement in Decentralized Environmental Regulation

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Abstract

Delegating enforcement to private inspectors can create a market for fraud. We develop a method to estimate fraud rates from statistical indicators and apply it to 100 million Texas vehicle emissions tests. Approximately 11% of inspections are fraudulent, and fraudulently passed vehicles emit over triple the pollution of compliant vehicles, imposing over \$100 million in annual health damages. We estimate a structural model of drivers' compliance choice exploiting variation in repair costs across vehicle defects. The median driver switches to noncompliance at \$501 in repair costs. We evaluate counterfactual enforcement that shuts down corrupt inspectors. We find: (1) enforcement generates leakage: many drivers switch to unregistered driving rather than repair, offsetting pollution gains; (2) the pollution benefits exceed drivers' consumer surplus losses; (3) corrupt inspectors capture large rents in the status quo by allowing drivers to avoid costly repairs while remaining registered, and these rents exceed the environmental gains from enforcement.

JEL: K42, L11, L90, Q52, Q53

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1 Introduction

Environmental regulation creates incentives for evasion. When compliance requires costly action—repairing a vehicle, installing pollution controls, remediating a site—regulated individuals may pay to avoid compliance. When regulators delegate enforcement to private third parties (Dufflo et al., 2013; Laffont and Tirole, 1993; Mookherjee and Png, 1995), those inspectors face a choice: enforce the regulation honestly, or profit by selling compliance certificates to individuals who have not actually complied. The result is a market for fraud, with corrupt inspectors on the supply side and regulated individuals seeking to avoid costly compliance on the demand side.

We study this market in the context of vehicle emissions inspection programs, where the delegation problem is acute. Most U.S. states delegate emissions testing to privately operated inspection stations. Drivers whose vehicles fail may face repair bills of hundreds to thousands of dollars. Stations can sell fraudulent passes—“clean scanning”—for a fraction of that price (Bennett et al., 2013; Hubbard, 1998; Oliva, 2015). Fraud spares drivers—often lower-income—from costly repairs, but keeps high-emitting vehicles on the road, worsening local air pollution. The regulator can crack down on corrupt inspectors, but supply-side enforcement may be ineffective if drivers choose to drive unregistered rather than repair—leaving noncompliant vehicles on the road and the externality unresolved.

Whether tighter enforcement improves outcomes depends on how much fraud exists, how much pollution it causes, and how drivers respond when fraud access is reduced. We develop a framework to answer these questions, and apply it to 100 million Texas vehicle emissions inspection records from 2016–2024.

We make four contributions. First, we develop a new method for detecting fraud from administrative data that is applicable beyond our setting. A large literature identifies cheating, fraud, or manipulation by constructing statistical indicators and showing they are elevated in suspicious subsamples (DellaVigna and La Ferrara, 2010; Fisman and Wei, 2004; Ghanem and Zhang, 2014; Jacob and Levitt, 2003). This establishes that misconduct exists but not *how much*: any individual indicator is ambiguous, since it may reflect fraud or simply measurement noise. Our framework resolves this by exploiting the *covariance* across these indicators. The core intuition is that when indicators come from independent data streams, only a shared latent cause—fraud—can generate correlation across them, so the magnitude of cross-indicator covariance directly identifies the fraud rate. We formalize this in a mixture model that recovers station-specific fraud rates from observable moments, and find that 11.2 % of inspections over our sample period—roughly 10.7 million tests—are fraudulent, peaking at 13.4 % in 2022 before a major enforcement crackdown, with fraud concentrated

at a small tail of stations and among older vehicles.

Second, we show that this fraud creates major externalities. Using roadside remote sensing data from the Texas On-Road Vehicle Emissions Testing (TORVET) program, which measures tailpipe emissions under normal driving conditions, we show that vehicles whose most recent inspection is classified as fraudulent emit over triple the nitric oxide (NO) and hydrocarbons (HC) of similar vehicles that passed legitimately—despite all having “passing” inspection results. Applying county-level marginal damages from the AP4 integrated assessment model (Muller and Mendelsohn, 2009; Muller et al., 2011), we estimate that the average fraud-flagged vehicle generates \$121 /year in excess health damages, yielding cumulative statewide damages of \$1.1 billion over our sample period.

Third, we estimate a structural model of driver compliance choice—demand for fraud—that allows us to quantify driver welfare and evaluate counterfactual policy environments. Estimating demand is difficult in this setting: inspection fees are regulated and constant across stations, and fraud prices are not directly observed. We exploit the fact that while the compliance decision is binary, drivers face widely varying repair costs determined by the specific diagnostic issues flagged during inspection (e.g., a failing oxygen sensor versus a bad catalytic converter). Our identifying assumption is that among vehicles of the same make, model-year, and station, which diagnostic issue triggers the failure—and therefore the repair cost—is orthogonal to unobserved compliance preferences. We find that fraud demand is elastic but highly heterogeneous: the fraud rate rises 4.0 -fold across the repair cost distribution. For inexpensive repairs like a PCV valve (\$93), only 2% choose noncompliance; at the cost of a spark plug replacement (\$276), 24% do; and by the cost of a used catalytic converter (\$750), 68% choose noncompliance over repair.¹

Our fourth contribution evaluates a common policy response: supply-side enforcement that fully cracks down on fraudulent inspection stations. Our counterfactuals reveal an unintended consequence we call *enforcement leakage*. Many drivers displaced from fraud do not repair; instead, they substitute toward unregistered driving—fraud falls from over a 13% share of inspections to zero but the share driving unregistered rises by at least 5 percentage points, with only 4 percentage points converting to repair—partially offsetting the pollution gains from enforcement. The welfare implications are mixed. On one hand, the air pollution benefits of eliminating fraudulent inspections exceed the consumer surplus losses borne by drivers who lose fraud access. On the other hand, the status quo generates substantial rents for criminal fraud suppliers—rents that reflect drivers’ willingness to pay to avoid costly repairs and continue to drive registered. The environmental gains from enforcement are

¹Noncompliance here includes both fraud and unregistered driving; the structural model distinguishes the two margins.

smaller than the rents captured by corrupt inspectors—surplus that fraud suppliers can invest in sustaining the market, consistent with the persistence of fraud over time.

Our paper contributes to three literatures. First, we contribute to work on fraud and corruption in delegated enforcement. A theoretical literature emphasizes that illegal markets have both supply and demand sides, so enforcement against suppliers reshapes but need not eliminate the underlying activity (Becker et al., 2006; Becker and Stigler, 1974; Shleifer and Vishny, 1993). Empirical work has largely studied one side at a time: Oliva (2015) exploits fee variation to identify an aggregate fraud rate in Mexico City’s I/M program; Duflo et al. (2013) and Duflo et al. (2018) randomize auditor assignment to study incentive design; Greenstone et al. (2022) show that automated monitoring can overcome the delegation problem; and Hubbard (1998), Pierce and Snyder (2008), Bennett et al. (2013), and Habte and Holm (2022) document how competition among inspectors encourages leniency. We build on this literature in two ways. First, we develop a method that recovers station-specific fraud rates from cross-indicator covariance, without requiring price or policy variation. Second, we use these estimates to study how a major regulatory crackdown reshapes the market—tracing the effects through to pollution damages, driver behavior, and welfare.

Second, we contribute to the literature on environmental enforcement by documenting *enforcement leakage*. Existing work studies how enforcement intensity affects compliance (Blundell et al., 2020; Shimshack and Ward, 2005; Zou, 2021), but treats noncompliance as a single margin. We show that supply-side enforcement can reduce one form of noncompliance while increasing another, limiting the effectiveness of crackdowns that target inspectors without addressing the underlying demand for noncompliance.

Third, we contribute to work on noncompliance in vehicle emissions regulation. Exhaust standards have reduced new-vehicle emissions per mile by over 99% since the 1960s (Jacobsen et al., 2023), but these gains are vulnerable to noncompliance. Existing work documents the manufacturer side: Reynaert and Sallee (2021) show how automakers game emissions ratings, and Ale-Chilet et al. (2025) study collusion to weaken emissions control technology. We document a different channel: fraud by third-party inspectors that keeps noncompliant vehicles on the road, generating substantial health damages.

The paper proceeds as follows. Section 2 describes the institutional setting and data. Section 3 develops the fraud detection model. Section 4 presents descriptive evidence on fraud prevalence, pollution impacts, repair costs, and enforcement responses. Section 5 develops and estimates the structural demand model. Section 6 evaluates counterfactual enforcement policies. Section 7 concludes.

2 Background and Data

2.1 The Clean Air Act and Vehicle Inspections

The Clean Air Act requires regions that fail to meet federal air quality standards to implement vehicle inspection and maintenance (I/M) programs.² Most programs rely on decentralized enforcement: rather than operating government-run testing facilities, states delegate inspections to thousands of privately operated auto service stations. Texas operates one of the largest I/M programs in the country. Under the state’s Clean Air Act Implementation Plan, 17 counties in the Dallas-Fort Worth, Houston-Galveston-Brazoria, Austin, and El Paso metropolitan areas are designated as non-attainment areas for ozone and require annual vehicle emissions inspections.³ Owners of vehicles manufactured after 1996 must pass an emissions test annually to complete vehicle registration. Vehicles that fail must be repaired and retested before registration can be renewed.

This setting offers several advantages for studying regulatory enforcement and fraud. First, the market is large and policy-relevant: Texas conducts over 10 million emissions inspections per year across over 5,500 licensed stations. Second, fraud is well-documented. Investigative reporting and regulatory audits have revealed widespread “clean scanning”—the practice of passing vehicles without conducting legitimate inspections ([Eastern Research Group, 2024](#)). Third, the administrative data infrastructure is unusually rich: every inspection is recorded in real time with detailed electronic diagnostics, enabling us to construct novel fraud indicators and estimate a fraud detection model at the station-year level.

2.2 Drivers, Inspection Stations, and Regulators

The emissions inspection market involves three agents: regulators who oversee the program, inspection stations that conduct tests, and vehicle owners who must comply with testing requirements.

Two state agencies share responsibility for the program. The Texas Commission on Environmental Quality (TCEQ) sets program policy, maintains the central inspection database (known as TIMS), and contracts for periodic program evaluations. The Texas Department of Public Safety (DPS) licenses stations and inspectors, conducts audits, and enforces compliance through administrative sanctions.

Emissions inspections are conducted at privately licensed auto service stations. Inspection

²As of 2020, 31 U.S. states and the District of Columbia operate I/M programs covering over 100 million registered vehicles ([U.S. Environmental Protection Agency, 2024](#)).

³The 17 emissions counties are: Brazoria, Collin, Dallas, Denton, Ellis, Fort Bend, Galveston, Harris, Johnson, Kaufman, Montgomery, Parker, Rockwall, Tarrant, Travis, Williamson, and El Paso.

prices are regulated, with a maximum of \$18.50 for an emissions test as of 2024. Given the low implied margins, stations typically enter the inspection market not to profit from inspections directly, but to attract customers for more lucrative repair and service work.⁴ Approximately 5,500–6,100 stations perform emissions inspections each year, with modest entry and exit over our sample period (Appendix Figure A1). We obtain the Master Station List from DPS via a Public Information Request, which provides the universe of licensed stations along with location, license date, and enforcement history.

A driver brings their vehicle to a licensed inspection station for an annual emissions test. Approximately 94.6 % of vehicles pass on the initial attempt, allowing the driver to complete registration. If the vehicle fails, the driver learns the nature of the problem—and in many cases the inspection station itself offers repair services and may quote a price. The driver then faces a choice among three margins of compliance. First, they can pay for repairs, return for a retest, and complete registration. Second, if the repair cost is prohibitive, they can abandon the process—driving unregistered, scrapping the vehicle, or selling it outside the regulated area. Third, they can seek a fraudulent pass from a station willing to clean-scan the vehicle.⁵ Some drivers may know in advance that their vehicle will fail and go directly to a fraud station; these vehicles appear in the data as initial passes with fraud indicators rather than as failures followed by retests. The diagnostic data recorded during each inspection provide a detailed picture of the vehicle’s emissions condition, which we exploit in Section 4.1 to predict repair costs for all failed vehicles.

2.3 On-Board Diagnostic Inspections and Data

Modern emissions inspections in Texas and throughout the United States rely on On-Board Diagnostics (OBD-II), a standardized system mandated on all 1996 and later model year vehicles. OBD-II systems continuously monitor emissions-related components and store Diagnostic Trouble Codes (DTCs) when malfunctions are detected, illuminating the vehicle’s Malfunction Indicator Light (MIL)—commonly known as the “check engine light”—to alert the driver. Texas phased in OBD-based emissions testing between 2002 and 2004 across

⁴To become licensed, a station must apply through a local DPS Vehicle Inspection Office, meet facility and equipment standards, purchase an approved On-board Diagnostic (OBD) analyzer, and establish a data-sharing agreement with TCEQ. The station certification fee is \$100 per two-year period. The majority of stations (73%) are independent shops (including quick lube and inspection-focused outlets); the remainder are national chains (16%) and franchised or used dealers (11%). See Appendix Table A2 for station summary statistics.

⁵When a vehicle passes a retest, the inspector can manually enter information about the repair performed, including the type of issue addressed and the cost. This reporting is effectively voluntary: approximately 63% of entries are recorded as zero, reflecting data entry omissions, self-performed repairs, and warranty work (Appendix Table A4).

emissions counties, replacing earlier tailpipe-based methods.⁶

During an OBD inspection, the inspector connects an analyzer to the vehicle’s OBD-II diagnostic port, typically located under the dashboard. The analyzer communicates with the vehicle’s Engine Control Module (ECM) and downloads several pieces of information. First, it retrieves the vehicle’s *electronic profile*: the OBD_VIN or electronic Vehicle Identification Number (“eVIN”) stored in the ECM, calibration identifiers, the Powertrain Control Module identifier, and the count of supported parameters. Second, it reads the status of up to 13 *readiness monitors*—self-diagnostic tests that the vehicle runs to verify emissions system components are functioning properly. Third, it checks for stored DTCs and the current MIL status. A vehicle passes if the MIL is off, no critical DTCs are present, and a sufficient number of readiness monitors have completed.

We purchased the universe of emissions inspection records from TCEQ’s Texas Inspection and Maintenance System (TIMS). Our analysis sample covers 2016 through 2024, comprising nearly 100 million OBD tests. Each record includes the inspector-entered VIN, the electronically downloaded OBD_VIN, station and analyzer identifiers, date and time of the test, the full electronic profile and readiness monitor status, DTC counts, the overall result (pass or fail), and vehicle characteristics including make, model, model year, and odometer reading. For vehicles that fail and return for a retest, we observe repair information for a subset of records, including the category of issue repaired and the repair cost. We define an *inspection cycle* as the sequence of tests on the same VIN within a registration period.⁷

2.4 Fraudulent Inspections and Indicators

Emissions inspection fraud plagues I/M programs across the United States.⁸ When fraudulent inspections allow high-emitting vehicles to remain on the road, they undermine program effectiveness and shift the burden of pollution reduction onto compliant drivers.

Fraud in this market takes the form of “clean scanning”—passing a vehicle without conducting a legitimate test. Two clean scanning methods are particularly widespread. The

⁶Pre-1996 vehicles may lack OBD-II systems, but these vehicles are now over 30 years old and few remain on the road. Texas exempts them from emissions testing, so all current inspections use OBD.

⁷While we observe every test in a cycle, our primary interest is the cycle-level outcome: whether each vehicle ultimately passes, fails, or exits the inspection system each year. See Appendix Table A3 for a comparison of vehicle and station characteristics by sanction status.

⁸In California, a 2024 federal indictment charged 12 defendants with operating a multi-year scheme using “OBDNator” simulator devices to falsify smog check results (U.S. Department of Justice, Eastern District of California, 2024). In 2020, the California Bureau of Automotive Repair filed accusations against nine stations for “clean gassing”—introducing surrogate gas into exhaust samples (California Bureau of Automotive Repair, 2020). In 2016, inspectors in New Jersey were indicted for using OBD simulators to falsify over 130 emissions tests (New Jersey Office of the Attorney General, 2016). In Georgia, inspectors were convicted for a racketeering scheme issuing over 1,400 fraudulent certificates by testing surrogate vehicles (Georgia Office of the Attorney General, 2015).

first employs an *OBD simulator*, a device that plugs into the analyzer in place of an actual vehicle and mimics a passing OBD system. The simulator may not report a valid VIN or may report it in a format that is not properly captured, resulting in a missing eVIN. The second uses a *surrogate vehicle*: the inspector connects the analyzer to a different, clean vehicle but enters the VIN of the failing vehicle being “inspected.” The surrogate reports its own eVIN, which does not match the entered VIN. Neither method requires the target vehicle to be physically present, enabling high-volume “assembly line” operations where stations process dozens or even hundreds of fake inspections per day. Complete bypass of the OBD system is impractical because the vehicle’s readiness monitors would not report as “ready,” triggering an automatic failure.

Clean scanning leaves electronic fingerprints in the TIMS data. We construct eight fraud indicators, organized into five measurement groups that capture logically distinct dimensions of the testing process: (1) *throughput* indicators flag assembly-line operations based on rapid test succession and high daily analyzer volume⁹; (2) *eVIN identity* flags tests where the electronic VIN downloaded from the vehicle’s computer is missing or does not match the inspector-entered VIN; (3) *OBD recycling* flags consecutive tests on the same analyzer that report identical electronic profiles for nominally distinct vehicles; (4) *operational* indicators flag manual VIN entry (not barcoded) and after-hours testing; and (5) *communication protocol* flags tests where the observed OBD protocol does not match the expected protocol for the vehicle’s make, model, and year. Appendix Table D1 reports incidence rates and construction details for each indicator; Appendix D.2 discusses additional candidate indicators that we evaluated and excluded. Because the five groups draw on distinct data sources and forensic signatures, cross-group indicators should be approximately conditionally independent given fraud status—a structure we formalize in Section 3.

Figure 1 plots the prevalence of six of our fraud indicators over the sample period. Two patterns stand out: indicators within the same measurement group move together—the two throughput indicators track each other closely, as do the two OBD recycling indicators—while indicators across groups diverge, consistent with each group capturing a distinct dimension of fraud activity. Most indicators rose through 2022 and then declined following the launch of Operation Cinderblock, the multi-agency enforcement campaign described below.

Fraud indicators are highly concentrated across stations: most stations have flag rates near zero, while a long right tail of stations exhibit rates above 50% (Appendix Figure H1). However, individual indicators cannot cleanly separate fraud from legitimate activity. Each

⁹OBD-II inspections are time-intensive and require several carefully sequenced steps. In Appendix D, we find that performing multiple inspections using the same equipment within a 5-minute interval and performing more than 100 inspections in a single day using the same equipment are highly correlated with our other fraud indicators.

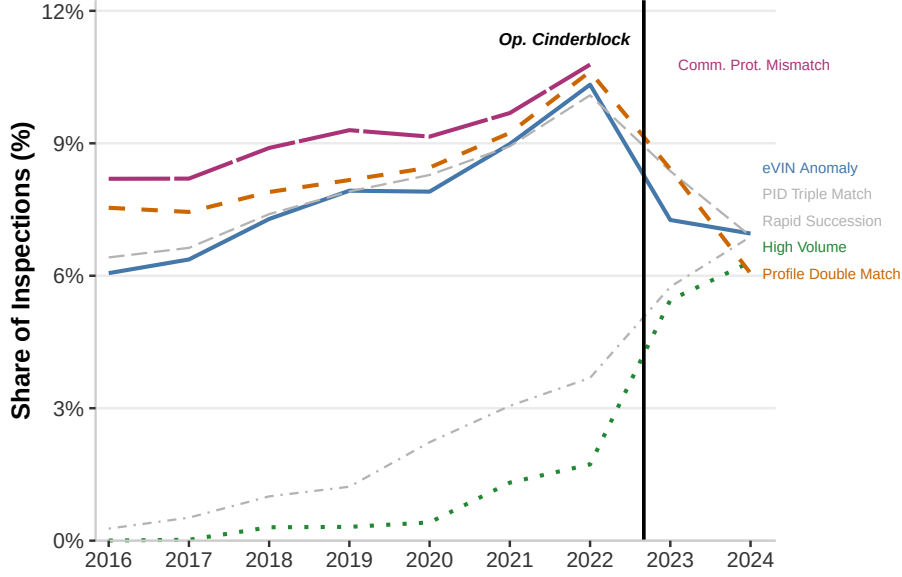


Figure 1. Fraud Indicators Over Time, 2016–2024

Note: Each line shows the share of emissions inspections flagged by the indicated fraud indicator. Four indicators representing distinct measurement groups are highlighted: High Volume (throughput, green), eVIN Anomaly (identity, blue), Profile Double Match (OBD recycling, orange), and Comm. Prot. Mismatch (communication protocol, purple). The remaining indicators are shown in grey. The vertical line marks Operation Cinderblock in August 2022. The Comm. Prot. Mismatch indicator only runs through 2022 due to changes in OBD-II communication protocols starting in 2023. The eVIN Anomaly indicator is a composite that flags tests where the electronic VIN is either missing or does not match the inspector-entered VIN. Sample includes all OBD-II emissions inspections in Texas emissions counties.

indicator has a non-trivial honest baseline—eVIN anomalies can arise from equipment issues and OBD communication errors, manual VIN entry reflects older scanning equipment, and so on—so auto-failing any flagged test would block many legitimate inspections. Rather than relying on any single indicator, we develop a mixture model that exploits *patterns* across multiple indicators to separate fraud from noise, classifying stations based on whether their indicator profiles are consistent with the “clean” or “fraud” cluster in the data.

2.5 Monitoring and Enforcement

DPS enforces fraud on the supply side by monitoring station compliance through multiple channels. The agency conducts approximately two announced (overt) audits per station per year, verifying equipment calibration, inspector credentials, and adherence to testing protocols. DPS also performs discretionary covert audits—undercover operations targeting stations with suspicious testing patterns. Violations of Texas Transportation Code Chapter 548 carry administrative and criminal penalties. DPS may suspend a station or inspector license temporarily or revoke it permanently. Civil penalties apply to procedural violations,

while falsifying government records—which includes issuing fraudulent inspection reports—constitutes a felony subject to criminal prosecution.

In August 2022, DPS launched “Operation Cinderblock,” a coordinated regulatory and criminal enforcement initiative targeting clean scanning. DPS initially identified hundreds of stations with high volumes of questionable inspections indicating likely clean scan violations; by March 2023, the operation had suspended or revoked hundreds of stations and removed hundreds of inspectors from the program.¹⁰

We obtained data on regulatory effort and enforcement outcomes from DPS via Public Information Requests. The audit data consist of a county-year panel recording the number of overt and covert audits conducted. Disciplinary action records—which DPS posts publicly—include the date of violation, action type (suspension, revocation, or civil penalty), and a narrative describing the case. We document the impact of this enforcement shock on fraud supply and driver compliance choices in Section 4.2.

3 Fraud Detection Model

We now develop a mixture model that formalizes the empirical structure of fraud in our setting. The model classifies stations as fraud-capable or clean and estimates station-specific fraud rates as the share of inspections that are fraudulent. The key insight underlying our approach is that cross-indicator covariance for a given station-vehicle combination is the empirical signature of fraud: under conditional independence, indicators co-vary only because some inspections are fraudulent and others are not. This covariance-based identification strategy recovers fraud rates cell by cell from observable moments, without requiring any single indicator to be a perfect fraud detector.

3.1 Model Setup

We observe K binary fraud indicators $Y = (Y_1, \dots, Y_K)$ for each passing inspection.¹¹ The data are organized into cells (s, v) defined by station $s \in \{1, \dots, S\}$ and vehicle type $v \in \{1, \dots, V\}$, where vehicle types are defined by make and model-year bins (Appendix E.1). Within each cell, we observe n_{sv} inspections, with $c_{sv}(y)$ denoting the count of inspections with outcome pattern $y \in \{0, 1\}^K$. The model has two levels of latent structure.

¹⁰Texas Department of Public Safety, “DPS’ Operation Cinderblock Cracks Down on Fraudulent Vehicle Inspections,” Press Release, March 30, 2023, <https://www.dps.texas.gov/news/dps-operation-cinderblock-cracks-down-fraudulent-vehicle-inspections>.

¹¹For clarity, the exposition abstracts from market (region) and time (inspection-cycle year) dimensions. In practice, all objects—parameters, data moments, and latent variables—vary along these dimensions as well, and the model is estimated separately for each market-year.

Assumption 1 (Two-component mixture). *Each station is either clean ($b_s = 0$) or fraud-capable ($b_s = 1$), with prior probability $P(b_s = 1) = w$. At clean stations, all inspections are honest. At fraud-capable stations, each inspection of vehicle type v is independently fraudulent with probability $\pi_{sv} \in [0, 1]$.*

The station-level type b_s governs the extensive margin — whether a station participates in fraud at all — while the cell-level fraud rate π_{sv} governs the intensive margin — how much fraud occurs, allowing rates to vary across vehicle types within a station and across stations. The sharp clean/fraud-capable boundary is a modeling simplification; in practice, some nominally clean stations may conduct low-level opportunistic fraud. Monte Carlo simulations show that contaminating up to 50% of clean stations with low fraud rates ($\pi \leq 0.08$) has negligible effect on fraud-rate estimates (Appendix C).

Conditional on an inspection i 's fraud status $F_i \in \{0, 1\}$, each indicator j fires independently as a Bernoulli draw:

$$P(Y_j = 1 \mid F_i = 1, v) = \beta_{j,v}, \quad P(Y_j = 1 \mid F_i = 0, s, v) = \Lambda_{j,sv}, \quad (1)$$

where $\beta_{j,v}$ is the **detection rate** — the probability that indicator j fires on a fraudulent inspection of vehicle type v — and $\Lambda_{j,sv}$ is the **honest baseline** — the probability that indicator j fires on an honest inspection due to equipment issues, vehicle characteristics, or other non-fraud causes. Detection rates depend on the vehicle type (different fraud methods may be used for different vehicles) but not on the station.¹² Honest baselines vary by station because stations differ in equipment quality and testing practices. The marginal flag rate in cell (s, v) is:

$$p_{j,sv} = \pi_{sv} \cdot \beta_{j,v} + (1 - \pi_{sv}) \cdot \Lambda_{j,sv}. \quad (2)$$

Assumption 2 (Conditional independence). *For all indicator pairs $j \neq k$ and all cells (s, v) :*

$$Y_j \perp Y_k \mid F, s, v.$$

Conditional independence requires that, once we condition on whether an inspection is fraudulent, knowing that one indicator fired provides no information about whether another fires. This is a restriction on the measurement technology, not on fraud behavior: it says nothing about which stations commit fraud, at what rates, or which vehicle types are targeted. The assumption is most defensible when indicators are constructed from distinct data sources that do not share common noise. As described in Section 2.4, our five indicator

¹²Monte Carlo simulations show that introducing station-level heterogeneity in β causes only mild degradation (Appendix C).

groups — throughput patterns, eVIN identity, OBD data recycling, operational procedures, and communication protocol — draw on logically distinct aspects of the testing process, each capturing a different forensic signature of clean scanning (Section 2).

Conditional independence is violated when indicators share noise sources (e.g., two flags triggered by the same scanner malfunction) or when unobserved fraud subtypes create residual correlation within the fraud component. Monte Carlo simulations (Appendix C) show that moderate violations reduce precision but leave fraud-rate estimates largely intact, while severe violations cause the estimator to overestimate fraud by attributing spurious covariance at clean stations to the fraud component. With $K \geq 4$ indicators, the assumption generates testable overidentifying restrictions, which we discuss in Section 3.2.

Under Assumptions 1 and 2, the joint distribution of indicators at a fraud-capable station takes the product-mixture form:

$$P(Y = y \mid b_s = 1, s, v) = \pi_{sv} \prod_{j=1}^K \beta_{j,v}^{y_j} (1 - \beta_{j,v})^{1-y_j} + (1 - \pi_{sv}) \prod_{j=1}^K \Lambda_{j,sv}^{y_j} (1 - \Lambda_{j,sv})^{1-y_j}. \quad (3)$$

At clean stations, the distribution simplifies to the honest component alone. The entire joint distribution is determined by $1 + 2K$ parameters per cell, and the observable joint distribution provides $2^K - 1$ moments. Stations are observed across V vehicle types; cross-type variation within a station provides the covariance moments that identify π_{sv} , while pooling across vehicle types enables station-level classification. Table B1 summarizes the model’s key objects.

3.2 Identification

The target objects are the cell-level fraud rates π_{sv} , the station types b_s , the detection rates $\beta_{j,v}$, and the honest baselines $\Lambda_{j,sv}$. Before stating the formal results, we summarize the identification logic. Cross-indicator covariance within (s, v) cells identifies the fraud rates π_{sv} : under conditional independence, indicators covary only through the latent mixture, and the magnitude of this covariance pins down the mixing proportion. The remaining parameters — $\beta_{j,v}$, $\Lambda_{j,sv}$, and b_s — are then identified from the joint distribution of K indicators, which provides $2^K - 1$ moments per cell to pin down $1 + 2K$ parameters (Appendix B.2).

We now make this precise. In the remainder of this subsection, we suppress the cell index (s, v) where it is unambiguous.

Define the **detection gap** for indicator j :

$$D_j \equiv \beta_j - \Lambda_j, \quad (4)$$

the difference in flagging probability between fraudulent and honest inspections. Define the **excess flag rate**:

$$\delta_j \equiv p_j - \Lambda_j = \pi \cdot D_j, \quad (5)$$

the portion of the observed flag rate attributable to fraud.

Assumption 3 (Detection gaps). *For at least two indicators j , $D_j > 0$ — fraud stations trigger these indicators at higher rates than honest stations.*

The following result, proved in Appendix B, is the basis for identification.

Proposition 1 (Covariance under conditional independence). *Under Assumptions 1–3, the covariance between any two indicators j and k within a cell is:*

$$\text{Cov}(Y_j, Y_k) = \pi(1 - \pi) \cdot D_j \cdot D_k. \quad (6)$$

Under conditional independence, all covariance between indicators therefore arises from the latent mixture itself. The magnitude of this covariance is governed by the variance of the latent variable, $\pi(1 - \pi)$, scaled by each indicator’s “loading” on fraud, D_j and D_k . At clean stations, $\pi = 0$ and the covariance vanishes: indicators are unconditionally independent. The covariance also vanishes when $D_j \rightarrow 0$: that is, when fraud inspections are statistically indistinguishable from honest ones. This is a fundamental identification limit: the estimator can only detect fraud that leaves a measurable trace in the indicators. Monte Carlo simulations confirm that recall degrades smoothly as the detection gap shrinks (Appendix C, Figure C1).

Figure H2 provides visual evidence for this identification logic. Focusing on the 2022 inspection cycle in the Dallas-Fort Worth region, we plot station-level excess flag rates for two groups of indicators drawn from logically distinct measurement systems: identity and communication indicators on the x -axis, equipment and OBD indicators on the y -axis. Most stations cluster near the origin, with excess flag rates close to zero on both dimensions. A second group of stations is elevated on *both* axes, with excess rates rising along the diagonal. The cross-group correlation of 0.95 is consistent with a single latent factor (fraud) loading on all indicators, exactly the structure formalized in Proposition 1.

Identification with known baselines ($K = 2$). When the honest baselines Λ_j are known, two indicators suffice. The excess flag rates $\delta_j = p_j - \Lambda_j$ are observable, and the ratio of their product to the covariance eliminates the detection gaps:

$$\frac{\delta_1 \cdot \delta_2}{\text{Cov}(Y_1, Y_2)} = \frac{\pi^2 \cdot D_1 \cdot D_2}{\pi(1 - \pi) \cdot D_1 \cdot D_2} = \frac{\pi}{1 - \pi}, \quad (7)$$

yielding the closed-form fraud rate:

$$\pi = \frac{\delta_1 \cdot \delta_2}{\delta_1 \cdot \delta_2 + \text{Cov}(Y_1, Y_2)}. \quad (8)$$

This identifies π cell by cell from observable quantities, given known baselines.

Identification with unknown baselines ($K \geq 5$). In practice, we use $K = 5$ indicators, and the model does not require known baselines. With K conditionally independent binary indicators and a binary latent variable, the observable joint distribution provides $2^K - 1$ moments to identify $1 + 2K$ parameters per cell. For $K \geq 3$, the system is at least just-identified without any baseline assumptions; for $K \geq 4$, it is overidentified (Appendix B.2). With our $K = 5$ indicators, we have $2^5 - 1 = 31$ observable moments and $1 + 2(5) = 11$ parameters per cell, yielding 20 overidentifying restrictions — testable implications of conditional independence. Finite-sample power is limited by cell sizes, so we treat overidentification primarily as a source of estimation efficiency rather than a formal specification test; Appendix C examines finite-sample power via Monte Carlo simulation.

3.3 Estimation

The identification results motivate a likelihood-based approach that jointly recovers all model parameters. Direct maximization of the observed-data likelihood is intractable because the latent station types b_s and inspection fraud statuses F_i are unobserved. We therefore use an Expectation-Maximization (EM) algorithm that iterates between computing posterior probabilities — $\gamma_s \equiv P(b_s = 1 \mid \text{data})$ for station types and $\tau_{sv}(y) \equiv P(F_i = 1 \mid Y = y, b_s = 1)$ for inspection-level fraud status — and updating model parameters via weighted sufficient statistics. Honest baselines follow an additive logit structure $\Lambda_{j,sv} = \text{logit}^{-1}(\alpha_{j,v} + \epsilon_{j,s})$, with the restriction that $\epsilon_{j,s} = 0$ at fraud-type stations so that the fraud and clean components remain non-nested. Full likelihood expressions and update equations are in Appendix B.4.

To avoid label switching, we initialize station classifications using a Cochran-Mantel-Haenszel-type test statistic that pools the sample cross-indicator covariance across vehicle types within each station, augmented with average excess flag rates to handle near-100% fraud stations (Appendix B.3). We run 50 warmup iterations with these initial classifications held fixed before releasing γ_s to the likelihood.

The estimator uses $K = 5$ indicators corresponding to the five measurement groups described in Section 2.4. Cells with fewer than 5 inspections are excluded, stations with fewer than 30 total inspections are dropped, and convergence is declared when the change in log-likelihood falls below 10^{-6} . Stations with $\gamma_s > 0.9$ are classified as fraud-capable.

Table 1. Classification Results by Region, 2022

Region	Stations			Inspections		
	Total	Fraud (%)	High-int.	Total	Fraudulent	IFR (%)
Dallas–Fort Worth	2,416	1,483 (61.4)	212	5,141,594	746,667	14.5
Houston	2,312	1,585 (68.6)	364	4,408,517	597,496	13.6
Austin	452	334 (73.9)	69	1,201,152	118,632	9.9
El Paso	239	159 (66.5)	15	564,900	54,625	9.7
Total	5,419	3,561 (65.7)	660	11,316,163	1,517,420	13.4

Notes: Regions are the four Texas metropolitan areas subject to the state’s vehicle emissions inspection program: Dallas–Fort Worth (9 counties), Houston (8 counties), Austin (2 counties), and El Paso (1 county). Fraud stations are those with posterior $\hat{\gamma}_s > 0.9$. High-intensity stations are fraud stations with station-average $\hat{\pi}_s > 25\%$. IFR (inspection fraud rate) is the estimated share of all passing inspections that are fraudulent: $\sum_{s,v} \hat{\pi}_{sv} n_{sv} \mathbf{1}[\hat{\gamma}_s > 0.9] / N_{\text{total}}$. The denominator includes all passing tests.

Monte Carlo simulations (Appendix C) confirm that the estimator recovers true fraud rates with RMSE 0.039 and near-zero bias, achieving 99% recall and 94% precision under baseline conditions, with robust performance across misspecifications including endogenous sorting, contaminated clean components, and station-level detection heterogeneity.

3.4 Classification Results

We apply the estimator to 36 region-year samples spanning 9 years (2016–2024) across 4 Texas metropolitan areas: Dallas-Fort Worth, Houston, Austin, and El Paso. All estimations converge. The headline result is the statewide inspection fraud rate (IFR), which is the estimated share of all passing inspections that are fraudulent. Pooling across all four regions, the statewide IFR rose from 11.8 % in 2016 to a peak of 13.4 % in 2022, before declining to 7.7 % in 2024 following the launch of Operation Cinderblock. Over the full sample period, the model identifies a cumulative 10.7 million fraudulent inspections statewide.

Table 1 summarizes the 2022 cross-section. Regional IFRs range from 9.7 % in El Paso to 14.5 % in Dallas-Fort Worth. While a substantial share of stations are classified as fraud-capable across all regions, the vast majority of these stations have low fraud rates; only 660 stations statewide have average fraud rates exceeding 25%.

The posterior $\hat{\gamma}_s$ is sharply bimodal: of 48,985 station-year observations, 15,032 have $\hat{\gamma}_s < 0.05$ and 28,532 have $\hat{\gamma}_s > 0.95$, with only 5,421 in between (Appendix Figure H3). Results are insensitive to threshold choices between 0.5 and 0.99. Among fraud-classified stations, fraud rates vary substantially: the median station-average $\hat{\pi}_s$ is just 1.2 %, with an inspection-weighted mean of 16.8 %, but the 90th percentile reaches 63.9 %. Most fraud-

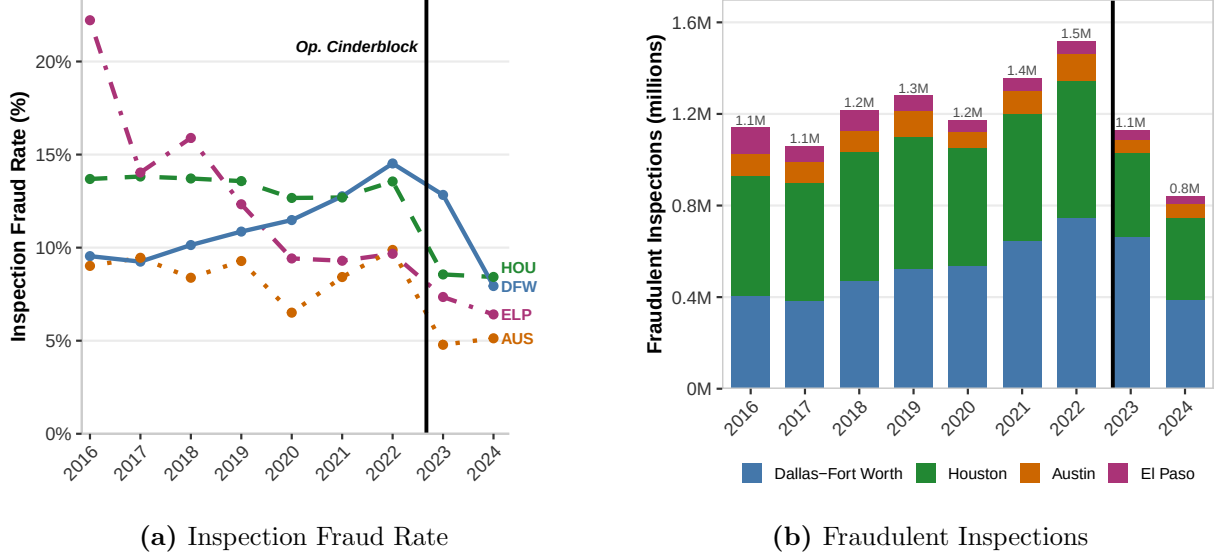


Figure 2. Estimated Inspection Fraud Over Time

Note: Panel (a) shows the estimated inspection fraud rate (IFR) by region and year, defined as $\sum_{s,v} \hat{\pi}_{sv} n_{sv} \mathbf{1}[\hat{\gamma}_s > 0.9] / N_{\text{total}}$, where the denominator includes all passing inspections (including those at stations excluded by sample size screens). Panel (b) shows the corresponding absolute number of fraudulent inspections. The vertical line marks the launch of Operation Cinderblock in August 2022. The four regions—Dallas-Fort Worth (9 counties), Houston (5), Austin (2), and El Paso (1)—cover all 17 Texas counties where emissions inspections are required.

capable stations conduct a mix of legitimate and fraudulent inspections; only 660 stations statewide have average fraud rates exceeding 25%, though these “high-intensity” stations account for a disproportionate share of total fraudulent inspections.

Figure 2 disaggregates the statewide trend by region. Dallas-Fort Worth and Houston drive the bulk of the pattern, with IFRs rising through 2022 and declining after Operation Cinderblock. Austin and El Paso have lower but non-trivial fraud rates. Panel (b) shows the corresponding volume of fraudulent inspections, which peaked at 1.5 million in 2022.

Estimated detection rates $\hat{\beta}_{j,v}$ range from a median of 0.26 for the high-volume indicator to near 1.00 for eVIN anomaly, consistent with the eVIN being a direct digital record of vehicle identity that is difficult to fabricate (Appendix Table H1).

External validation. We validate the model’s classifications against Texas DPS enforcement records, which include 2,305 suspension and revocation actions against 1,887 unique stations over 2016–2024. Figure H4 presents the results. 75.6 % of sanctioned stations appearing in our estimation samples are classified as fraud-capable by the model (Panel a, “Recall” line). Restricting to stations sanctioned specifically for emissions-related violations — clean piping, clean scanning, and failure to perform required emissions tests — recall rises

to 77.0 %.¹³ Moreover, sanctioned stations are disproportionately high-intensity: Panel (b) shows that the distribution of $\hat{\pi}_s$ among sanctioned fraud stations is shifted substantially rightward relative to all fraud-classified stations, confirming that enforcement targets the most egregious offenders.

Conversely, precision — the share of fraud-classified stations that have ever been sanctioned — is much lower (Panel a, “Precision” line). This asymmetry is expected and informative. Enforcement is severely resource-constrained: as documented in Section 2, DPS conducts only a few hundred covert audits per year across the entire state, while the model identifies thousands of fraud-capable stations across the four metropolitan areas. The gap between model-identified fraud and observed enforcement is precisely the phenomenon the estimator is designed to measure — the vast majority of fraudulent inspections occur at stations that have not been (and may never be) sanctioned.

4 Descriptive Evidence

4.1 Repair Costs and Compliance Choices

As described in Section 2, a driver whose vehicle fails its emissions inspection faces a choice: pay for repairs and retest, seek a fraudulent pass, or abandon the registration process.¹⁴ Because the inspection fee itself is small and fixed, repair cost is the primary source of variation in compliance costs across drivers, and we use it to identify the elasticity of compliance and demand for fraud.¹⁵ We first discuss what drives variation in repair costs and which sources of variation are useful for identification, then describe our approach to predicting repair costs from inspection data, and present the reduced-form relationship between predicted cost and fraud.

Identification strategy. Repair costs vary with vehicle make, age, geographic market, and station—but each of these dimensions is correlated with owner characteristics and fraud access, confounding the compliance cost channel. We instead exploit the *specific malfunction* that caused the vehicle to fail. Conditional on vehicle type, age, and location, the particular diagnostic issue provides plausibly exogenous variation in compliance costs: two otherwise identical vehicles visiting the same station may require very different repairs—one needing

¹³These correspond to violation Categories C and E under 37 TAC §23.62.

¹⁴Abandonment encompasses several distinct outcomes: scrapping the vehicle, relocating outside the regulated area, or continuing to drive unregistered. These have different welfare implications, which we discuss in the following sections.

¹⁵This contrasts with settings where inspection fees themselves vary across jurisdictions or time periods—as in [Oliva \(2015\)](#), who uses variation in inspection fees to identify demand. In Texas, the regulated fee is constant, so identification comes entirely from variation in the cost of the underlying repair.

a \$75 oxygen sensor replacement, the other a \$600 EGR valve or catalytic converter—depending on which component happened to fail. We operationalize this using the OBD diagnostic data recorded at the initial failure—check engine light status, diagnostic trouble code counts, and readiness monitor failures—which are mechanically linked to the type and severity of repair needed.

Measuring and predicting repair costs. We do not directly observe each driver’s repair cost. Inspectors can manually record repair costs at retest, but reporting is effectively voluntary: approximately 63 % of entries are recorded as zero (Appendix Table A4). Among retests with positive recorded costs, the average is \$275 in 2024 dollars.¹⁶ More importantly, we observe OBD diagnostic data for *all* vehicles that fail their initial test—including those that subsequently obtain a fraudulent pass or abandon registration.

We estimate a model of log repair costs on diagnostic variables using retests with positive recorded costs, then predict costs for all 4,922,414 inspection cycles beginning with a failure:¹⁷

$$\log(c_i) = \mathbf{x}'_i \boldsymbol{\beta} + \delta_{\text{make}} + \delta_{\text{MY}} + \delta_{\text{station}} + \delta_{\text{year}} + \varepsilon_i \quad (9)$$

where $\mathbf{x}_i$ includes check engine light status, diagnostic trouble code counts, readiness monitor failures, and gas cap test result (Table A5), and δ terms are fixed effects for make, model year, station, and year. Because $\hat{\boldsymbol{\beta}}$ is estimated conditional on these fixed effects, predicted repair cost varies only with the diagnostic profile—determined by the vehicle’s mechanical condition rather than the driver’s compliance decision.¹⁸ We convert the index to dollar terms using $\hat{r}_i = \bar{c} \cdot \exp(\mathbf{x}'_i \hat{\boldsymbol{\beta}}) / \mathbb{E}[\exp(\mathbf{x}' \hat{\boldsymbol{\beta}})]$, where $\bar{c} = \$275$ is the mean reported repair cost in 2024 dollars.

Fraud increases with predicted repair cost. Figure 3a plots outcome shares—repair, fraud, and abandonment—across 100 equal-frequency bins of predicted repair cost (in dollars). The fraud rate rises from 9.7% at the bottom of the distribution to 39.0% at the top—a 4.0 -fold increase. Abandonment follows a parallel trajectory at similar levels, while the repair rate declines correspondingly. As predicted repair costs rise, drivers substitute away from repair and toward both non-compliance alternatives. That the ratio of fraud to abandonment remains approximately constant across the cost distribution is consistent

¹⁶Repair costs are deflated to 2024 dollars using the BLS motor vehicle maintenance and repair price index (FRED series CUUR0000SETD). See Appendix A.3 for details.

¹⁷Predicted repair costs are defined only for vehicles with an observed initial failure. Some drivers may proceed directly to a fraud station without first attempting a legitimate test. We discuss this limitation and the additional assumptions required in Section 5.

¹⁸In this sense, $\mathbf{x}_i$ serves as an instrument for actual repair cost. See Appendix A.3 for details on the estimation and dollar conversion.

with diagnostic severity being orthogonal to the factors that determine the choice between fraud and abandonment—such as proximity to a willing station or the outside option of driving unregistered—an assumption we formalize in the structural model (Section 5). Were diagnostic profiles systematically correlated with access to fraud, we would expect the fraud share of non-compliance to shift with predicted cost rather than remain stable. We therefore take the parallel trends as supporting the hypothesis that vehicle-specific diagnostics provide exogenous variation in compliance costs. Section 5 uses this variation to estimate the structural price sensitivity.

4.2 Supply-Side Enforcement and Compliance Responses

The previous subsection established that drivers’ compliance choices respond to the cost of repair—the demand side of the fraud market. We now turn to the supply side: how does regulatory enforcement against fraudulent stations affect the availability of fraud and, in turn, driver compliance choices? The answer matters for policy design, since enforcement resources are finite and supply-side crackdowns may generate unintended substitution if drivers respond to reduced fraud access by abandoning registration rather than complying.

In late 2022, DPS launched Operation Cinderblock—a coordinated regulatory and criminal enforcement campaign targeting stations engaged in systematic clean scanning (Section 2). Figure 3 documents the enforcement shock and its consequences for fraud supply and noncompliance.

Panel (b) displays enforcement activity over our sample period. Prior to 2022, DPS conducted approximately 3,500 covert audits annually but imposed relatively few sanctions—only 18 enforcement actions in 2019 and 4 in 2021. Following Operation Cinderblock, enforcement actions increased dramatically: 173 in 2022, 821 in 2023, and 609 in 2024—roughly a five-fold increase in annualized enforcement intensity. Covert audit activity also increased, peaking at nearly 7,000 in 2023 before declining to 4,300 in 2024.

Panel (c) shows that this campaign was highly effective at eliminating the worst offenders. We define a station as “high-intensity” if more than 25% of its passing inspections are estimated to be fraudulent by our classification model. The share of high-intensity fraud stations fell from approximately 13% in 2016 to 2% by 2024, with most of the decline occurring after Operation Cinderblock. The enforcement campaign thus succeeded in shutting down the vast majority of the most prolific fraud operations.

Despite this success, the net impact of the crackdown on overall noncompliance was partially offset by behavioral responses. Panel (d) plots the estimated fraud inspection rate alongside the share of vehicles observed on the road with lapsed registration (from the TORVET remote sensing program, described in Section 4.3). The fraud rate fell from a

peak of over 14% in 2022 to approximately 8% by 2024—a decline of roughly 6 percentage points. However, the unregistered share rose by more than 2 percentage points over the same period, from approximately 3% to over 4%. The combined noncompliance rate (fraud plus unregistered) therefore fell by only about 4 percentage points, suggesting that some drivers who previously obtained fraudulent inspections substituted toward abandoning registration entirely when fraud stations became less accessible. We exploit this cross-market variation in fraud availability—driven by the enforcement campaign’s differential impact on local fraud station supply—to identify the parameters governing drivers’ tradeoffs between compliance, fraud, and abandonment in the structural model of Section 5, and evaluate counterfactual enforcement policies in Section 6.

4.3 Environmental Impacts

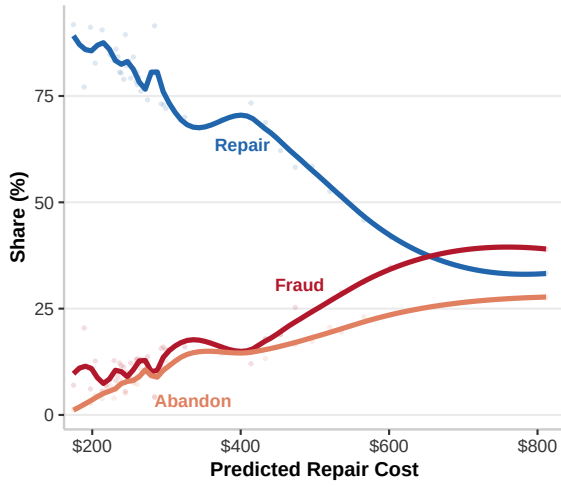
Fraudulent inspections allow high-emitting vehicles to pass and remain on the road without repair, generating environmental externalities through elevated local air pollution. To quantify this impact, we compare the on-road emissions of vehicles whose most recent inspection exhibits indicators of fraud to those of otherwise similar vehicles that passed through the standard inspection process.

Remote sensing data. We use data from the Texas On-Road Vehicle Emissions Testing (TORVET) program, which deploys roadside remote sensing equipment to measure tailpipe concentrations of CO, HC, NO, and CO₂ from passing vehicles under normal driving conditions, along with speed, acceleration, and license plate identification linked to VIN. We match each TORVET capture to the vehicle’s most recent OBD-II emissions inspection within 365 days and convert concentrations to emission rates (g/kg-fuel) using the carbon balance method of [Eastern Research Group \(2024\)](#). Appendix F provides details on the program, formula, and sample construction.

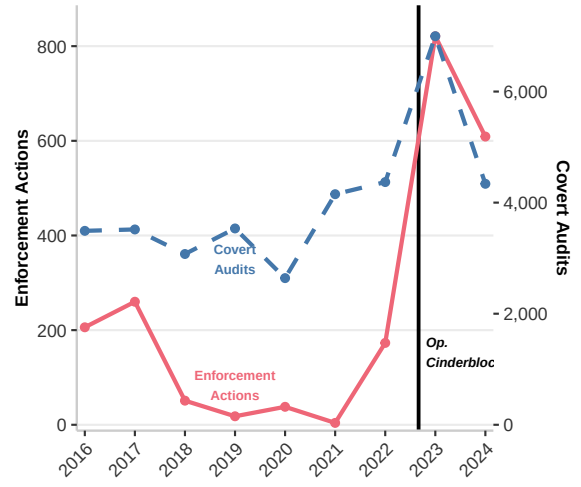
Regression specification. We estimate the emissions impact of fraud using the following specification:

$$y_{ivt} = \beta_1 \cdot \text{Fraud}_i + \beta_2 \cdot \text{Unregistered}_i + \gamma \mathbf{X}_{ivt} + \delta_{\text{make}} + \delta_{\text{MY}} + \delta_{\text{date}} + \varepsilon_{ivt} \quad (10)$$

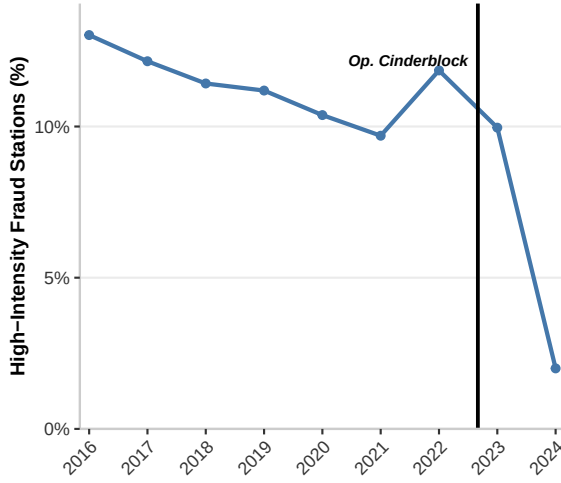
where y_{ivt} is the emission rate (g/kg-fuel) for vehicle i captured at TORVET site-time t , Fraud_i is the fraud indicator from the vehicle’s most recent OBD inspection, Unregistered_i indicates whether the vehicle’s registration expired 6 or more months before capture, $\mathbf{X}_{ivt}$ includes driving controls (speed, acceleration, vehicle specific power), and δ terms are fixed



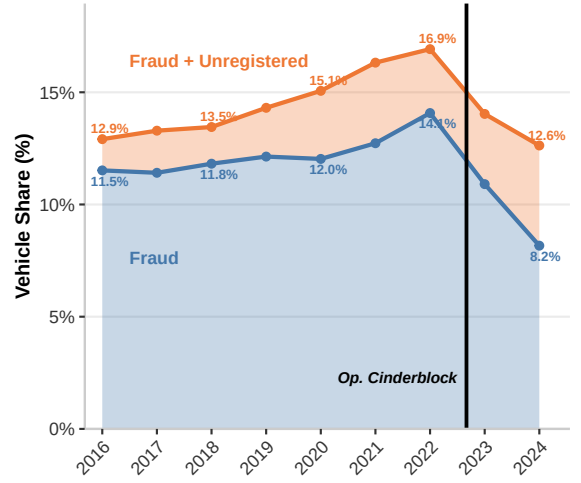
(a) Compliance Choices by Predicted Repair Cost



(b) Enforcement Actions and Covert Audits



(c) High-Intensity Fraud Station Share



(d) Fraud and Unregistered Vehicle Shares

Figure 3. Compliance Choices, Enforcement, and Noncompliance Responses

Note: Panel (a): each point is a binned scatter of outcome shares within equal-frequency bins of predicted repair cost (2016–2024); lines are LOESS smooths; percentages at the right margin show smoothed shares at the top of the observed cost distribution. Predicted cost equals \$275 (mean TIMS-recorded repair cost in 2024 dollars) multiplied by an individual scaling factor estimated from the initial failure profile (MIL status, DTC count, gas cap test, and readiness monitor failures), normalized by the overall sample mean so that variation reflects within-type diagnostic severity conditional on make, model year, and inspection station. Fraud is defined as an inspection cycle ending in a pass with estimated fraud posterior $\hat{\tau}_i > 0.5$ from the fraud detection model (Section 3). Panels (b)–(d): sample is 14 DFW and Houston-Galveston counties, 2016–2024. Panel (b): solid line shows station license suspensions and revocations (left axis); dashed line shows covert audits (right axis). Data from DPS disciplinary records and audit counts obtained via Public Information Requests. Panel (c): share of stations classified as high-intensity fraud (station-level $\hat{\pi}_s > 0.25$, weighted mean across counties by total inspections). Panel (d): blue line and shaded area show the estimated fraud inspection rate from the classification model; orange line and shaded area show the additional unregistered vehicle share from TORVET remote sensing (vehicles with registration ≥ 12 months overdue). Labels show values every two years. Vertical lines mark the start of Operation Cinderblock (August 2022).

effects for vehicle make, model year, and capture date. The baseline category is registered vehicles without fraud flags. Capture-date fixed effects absorb site-level and weather variation, since TORVET typically operates one monitoring site per day. Our preferred specification includes all three sets of fixed effects and driving controls, restricted to Texas-registered vehicles in the 17 TCEQ emissions-testing counties.¹⁹

We classify an inspection as fraudulent using the inspection-level posterior $\hat{\tau}_i$ from the fraud detection model (Section 3), which combines information from all five indicator groups to estimate the probability that a given inspection was fraudulent. Our primary specification defines $\text{Fraud}_i = \mathbb{1}[\hat{\tau}_i > 0.5]$. As an alternative, we also estimate the specification replacing the binary indicator with the continuous posterior $\hat{\tau}_i \in [0, 1]$, which uses the full information in the posterior distribution rather than discretizing at a threshold. Appendix F.5 shows that results are additionally robust to using individual fraud indicators directly.

Results. Table 2 reports our estimates. Vehicles whose most recent inspection is classified as fraudulent ($\hat{\tau}_i > 0.5$) emit 2.23 g/kg-fuel more NO and 0.45 g/kg-fuel more HC on the road than registered vehicles that passed legitimately, representing increases of 204.6 % and 239.2 % relative to the control mean. Using the continuous posterior yields nearly identical coefficients (columns 4 and 8), confirming that the binary threshold captures the relevant variation. Vehicles with lapsed registration also emit significantly more: 0.94 g/kg-fuel more NO and 0.42 g/kg-fuel more HC than the registered baseline. These differentials are striking given that 100% of fraud-flagged vehicles received a passing result on their most recent OBD inspection—consistent with the fraud hypothesis that the inspection result was obtained without a legitimate test, so the vehicle’s actual emissions condition was never assessed or remediated. We focus on NO and HC because these ozone and PM_{2.5} precursors are the primary drivers of air quality damages from mobile sources; CO is also elevated but omitted from damage calculations because CO health damages per ton are orders of magnitude smaller.²⁰ Results are robust to alternative fraud indicators (Appendix Table F2).

Monetizing health damages. We translate emission rate differentials into dollar-valued health damages by combining the regression coefficients with vehicle-specific annual VMT (from odometer differences across inspections), EPA fuel economy ratings matched by make, model, and year, and county-level marginal damages per metric ton of ground-level NOx and VOC from the AP4 integrated assessment model (Muller and Mendelsohn, 2009; Muller

¹⁹We weight each observation by the inverse of the number of captures per vehicle per year so that each vehicle contributes equally, and cluster standard errors two-way by vehicle and capture date.

²⁰AP4 does not include CO as a criteria pollutant for health damage estimation. Ambient CO concentrations in U.S. metropolitan areas have been well below NAAQS thresholds since the 1990s.

Table 2. On-Road Emissions Impact of Fraudulent Inspections and Lapsed Registration

	NO (g/kg fuel)				HC (g/kg fuel)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Fraud Measures</i>								
$\mathbb{1}(\hat{\tau}_i > 0.5)$	3.46 (0.04)	2.22 (0.03)	2.23 (0.03)		0.69 (0.01)	0.46 (0.01)	0.45 (0.01)	
$\hat{\tau}_i$ (continuous)				2.28 (0.03)				0.46 (0.01)
Unregistered	1.80 (0.04)	0.95 (0.03)	0.94 (0.03)	0.95 (0.03)	0.60 (0.02)	0.42 (0.02)	0.42 (0.02)	0.43 (0.02)
Control mean	1.09	1.09	1.09	0.97	0.19	0.19	0.19	0.17
% effect (fraud)		204.6%				239.2%		
% effect (unreg.)		86.5%				223.6%		
Observations				1,884,938				
N fraud-flagged				234,563				
N unregistered				102,126				
Make FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Model-year FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Capture-date FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Driving controls	No	No	Yes	Yes	No	No	Yes	Yes

Note: Each column reports coefficients from a regression of on-road emissions (g/kg-fuel) on a fraud measure and an unregistered vehicle indicator. Columns (1)–(3) and (5)–(7) use the binary indicator $\mathbb{1}(\hat{\tau}_i > 0.5)$ from the fraud detection model. Columns (4) and (8) use the continuous posterior $\hat{\tau}_i \in [0, 1]$. The baseline category is registered vehicles without fraud flags. Columns (1) and (5) include no fixed effects or controls. Columns (2) and (6) include make, model-year, and capture-date fixed effects. Columns (3)–(4) and (7)–(8) add driving controls (speed, acceleration, vehicle specific power). Capture-date fixed effects absorb weather and site-level variation. The sample includes Texas-registered vehicles in the 17 TCEQ emissions-testing counties (excluding diesel and new-car exempt vehicles). Unregistered vehicles have lapsed registration (expired 6+ months or missing) and no OBD inspection within 365 days of capture. Vehicles with recently expired registration (0–6 months) and currently registered vehicles that we were unable to match to a recent emissions test are excluded from the sample. Emission rates are computed using the ERG/CARB (2024) full carbon balance formula. Our emission test sample begins in 2016, so we drop that year’s remote sensing captures: fraud flags such as the repeat electronic profile compare consecutive tests at the same analyzer, but no pre-2016 tests are available for comparison, understating fraud at the start of the sample. Observations are weighted by the inverse of the number of remote sensing captures per vehicle per year. Standard errors two-way clustered by vehicle and capture date in parentheses. % effects are from the preferred specification (columns 3 and 7).

et al., 2011).²¹

The average fraud-flagged vehicle generates \$121 /year in excess health damages from

²¹We map HC to VOC, which is standard for mobile source emissions. For each vehicle, we convert excess g/kg-fuel to annual metric tons via the chain: excess g/kg-fuel $\times$ kg-fuel/gallon $\times$ gallons/mile $\times$ miles/year $\div 10^6$, then multiply by the AP4 marginal damage for the vehicle’s registration county.

NOx and VOC, with substantial variation across counties driven by population exposure (AP4 damages for Harris County are roughly twice those for El Paso). Scaling to the full fraud-flagged fleet—approximately 1147929 vehicles per year classified as fraudulent across the 17 TCEQ counties—yields statewide annual damages of \$138.9 million, or \$1.1 billion cumulatively over 2017–2024. Dallas–Fort Worth and Houston account for the vast majority of aggregate damages, reflecting both high fraud rates and dense population exposure. To put these magnitudes in context, fraud-flagged vehicles constitute approximately 11.3 % of the tested fleet but contribute an estimated 18.8 % of total on-road NOx emissions, reflecting their disproportionately high emission rates. Table 3 reports the metropolitan-area breakdown, and Table F4 in the appendix provides county-level detail.

Unregistered vehicles represent an additional source of excess emissions. Using county-by-year unregistered shares from the TORVET sample for the 14 counties with deployed sensors (DFW and Houston), we estimate annual unregistered damages of \$13.2 million to \$26.5 million, depending on whether unregistered vehicles drive 50% or 100% as much as registered vehicles.²² Combining both channels, the cumulative health damages from inspection noncompliance over 2017–2024 are \$1.2 billion to \$1.3 billion.²³

A robustness check allowing the fraud coefficient to vary by model-year bin yields \$149.5 million in annual fraud damages, confirming that the baseline is conservative (Table F5).

These estimates measure how much lower pollution would be if all fraud-flagged and unregistered vehicles emitted at the rate of compliant vehicles—they are accounting exercises, not policy evaluations, because they do not account for how drivers would respond to changes in the policy environment. In Section 5, we develop a structural model that endogenizes compliance choices and use it in Section 6 to evaluate supply-side enforcement policies that account for these behavioral responses.

5 Structural Model

The descriptive evidence in Section 4.1 established that compliance choices respond elastically to repair costs, with fraud and abandonment rising in parallel as the cost of repair increases. To quantify the welfare effects of enforcement policy and conduct counterfactual simulations, we embed this reduced-form relationship in a discrete choice demand model. The model recovers the distribution of repair cost sensitivity across drivers, measures how fraud

²²We omit Austin and El Paso because TORVET sensors are not deployed there; vehicles from those counties appear in sensor data only when traveling through monitored areas, inflating observed unregistered rates.

²³All damages are expressed in 2020 USD using AP4 marginal damage rates. Cumulative totals are undiscounted sums.

Table 3. Health Damages from Excess Emissions by Metropolitan Area, 2017–2024

Metro Area	Fraud Vehicles/yr	Per-Vehicle Damage	Avg. Annual Fraud Damage	Total Fraud Damage	Avg. Annual Unreg. Damage
Dallas–Fort Worth	521,500	\$132	\$68.6 M	\$549.2 M	\$13.6 M
Houston–Galveston	485,194	\$123	\$59.5 M	\$475.6 M	\$12.9 M
Austin	85,842	\$91	\$7.8 M	\$62.6 M	— [†]
El Paso	55,394	\$53	\$3.0 M	\$23.7 M	— [†]
Total	1,147,929	\$121	\$138.9 M	\$1111.1 M	\$26.5 M

Note: Health damages from excess NOx and VOC emissions monetized using AP4 (Muller and Mendelsohn, 2009) county-level ground-source marginal damages (\$/metric ton, 2020 USD). Marginal damages reflect national health impacts from Texas-source emissions, including downwind populations. Fraud vehicle counts from inspection posteriors ($\hat{\tau}_i > 0.5$ from the fraud detection model). Per-vehicle damages computed using EPA fuel economy ratings and annual VMT from odometer differences. “Avg. Annual” columns report the mean across 2017–2024; “Total” sums undiscounted across all eight years. Unregistered shares computed from TORVET remote sensing captures by county and year. Unregistered damage assumes 100% of registered-vehicle VMT; at 50% VMT, unregistered damages are approximately half. [†]TORVET sensors are deployed only in the DFW and Houston metro areas. Vehicles registered in Austin and El Paso are captured only when traveling through sensor counties, producing upward-biased unregistered shares; unregistered damages are therefore omitted for these metros. DFW includes Collin, Dallas, Denton, Ellis, Johnson, Kaufman, Parker, Rockwall, and Tarrant counties. Houston includes Brazoria, Fort Bend, Galveston, Harris, and Montgomery counties. Austin includes Travis and Williamson counties.

station availability affects noncompliance, and provides the substitution patterns needed to evaluate the net effect of supply-side enforcement on pollution damages.

5.1 Compliance Choice

We model compliance choices among vehicles that fail their initial emissions inspection. A market m is a county-year pair. We restrict to the 17 TCEQ emissions-testing counties over 2017–2024, yielding 4,381,031 initial-fail vehicle-cycles. Each driver observes a repair cost $r_i > 0$, a set of fraud-capable stations in the local market, and an outside option of leaving the inspection system. The driver chooses to *repair* (pay r_i , obtain a legitimate pass, and maintain registration), obtain *fraud* (a fraudulent passing inspection at one of the available stations, without repairing), or *abandon* the inspection system entirely—encompassing both unregistered driving and vehicle exit.²⁴ In the estimation sample, 79.6 % repair, 11.9 % obtain fraud, and 8.5 % abandon.²⁵

²⁴We decompose abandon into unregistered driving versus exit post-estimation using TORVET roadside sensor data (Section 5.5), which we use to bound the pollution consequences of enforcement-driven substitution from fraud toward abandonment.

²⁵The initial-fail sample excludes drivers who proceed directly to a fraud station without first attempting a legitimate inspection, for whom we lack predicted repair costs. The structural parameters are therefore identified from the tested population. Appendix G.3 discusses this restriction.

Fraud station types. Although the driver chooses a specific fraud station, we treat fraud as a relatively homogeneous product—a fraudulent inspection certificate is identical regardless of which station issues it—and aggregate stations into three types that differ in their business model and accessibility. We classify fraud-capable stations using the fraud detection model from Section 3. *High-intensity in-county* stations ($N_{hi,m}$) are dedicated fraud operations where fraud constitutes a large share of business ($\hat{\pi}_s > 25\%$). *Opportunistic in-county* stations ($N_{opp,m}$) engage in occasional fraud that is not their primary business; this channel matters because opportunistic stations are far more common than dedicated fraud shops, making them more likely to be encountered by drivers seeking inspections for failing vehicles. *High-intensity out-of-county* stations ($N_{ho,m}$) are dedicated fraud operations ($\hat{\pi}_s > 25\%$) located outside the driver’s home county, where remote clean scanning reduces the relevance of physical distance. Within each type, each additional station of type k contributes the same utility δ_k to the driver’s choice set, so the fraud supply side of market m is summarized by three station counts. This setup allows us to investigate counterfactual supply-side enforcement policies that change the number and composition of fraud stations available to drivers.

Driver i in market m obtains utility from option j :

$$u_{ij} = V_j(\mathbf{x}_i, \alpha_i) + \varepsilon_{ij} \quad (11)$$

where V_j is the systematic component and ε_{ij} follows the nested logit GEV distribution.

Repair. The systematic utility of repair is:

$$V_{\text{repair},i} = \mathbf{x}'_i \boldsymbol{\beta}_r + \alpha_i \cdot r_i \quad (12)$$

where r_i is the predicted repair cost from Section 4.1, $\mathbf{x}_i$ includes vehicle age bin indicators, county fixed effects, and a quadratic time trend, and α_i is the individual-specific repair cost coefficient. Inspection fees are regulated and small relative to repair costs; vehicles that fail can retest for free after completing repairs, so the primary cost of compliance is the repair itself.²⁶ α_i is specified as:

$$\alpha_i = -\exp(\mu + \sigma \nu_i), \quad \nu_i \sim N(0, 1) \quad (13)$$

²⁶The regulated inspection fee is \$18.50 for the OBD-II emissions test plus a \$7.00 state fee. Retests after repair are free within the same inspection cycle.

The log-normal specification ensures $\alpha_i < 0$ for all agents—every driver dislikes paying for repairs—while allowing substantial heterogeneity in price sensitivity. The parameters μ and σ govern the location and scale of this heterogeneity.

Non-comply nest. The choice has a two-level structure. At the top level, the driver chooses between repair and a *non-comply* nest that groups fraud and abandon—options that allow the driver to avoid paying for repairs, whether through fraud, unregistered driving, or exiting the market. Conditional on choosing noncompliance, the within-nest split between fraud and abandon is determined by station availability and the attractiveness of the outside option. This structure is motivated by the descriptive evidence in Figure 3a: as predicted repair costs rise, the fraud rate and abandonment rate increase in parallel, while the ratio of fraud to abandonment remains approximately constant across the cost distribution. The constant ratio implies that repair cost governs the comply/non-comply margin, while within-nest factors—not repair cost—determine the fraud/abandon split.²⁷

The systematic utility of the non-comply nest at the top level is:

$$V_{nc,i} = \mathbf{x}'_i \boldsymbol{\beta}_{nc} + IV_i \quad (14)$$

The coefficient vector $\boldsymbol{\beta}_{nc}$ enters on the same covariates as the repair utility, allowing the effects of age, location, and time to differ between the repair and noncompliance equations. The inclusive value IV_i summarizes the attractiveness of all options available within the nest:

$$IV_i = \lambda \cdot \log \left(\underbrace{N_{opp,m} e^{\delta_{opp}/\lambda}}_{\text{opportunistic fraud stations}} + \underbrace{N_{hi,m} e^{\delta_{hi}/\lambda}}_{\text{high-intensity fraud stations (in-county)}} + \underbrace{N_{ho,m} e^{\delta_{ho}/\lambda}}_{\text{high-intensity fraud stations (out-of-county)}} + \underbrace{e^{\mathbf{z}'_i \boldsymbol{\beta}_a / \lambda}}_{\text{abandon}} \right) \quad (15)$$

The first three terms correspond to the fraud station types: each station of type k contributes $e^{\delta_k/\lambda}$ to the sum, weighted by the count $N_{k,m}$ of available stations, where δ_k captures the utility of obtaining fraud at each station type, net of the unobserved price charged for a fraudulent inspection.²⁸ The final term is the utility of abandonment. While the station type utilities δ_k are constants—reflecting the homogeneity assumption within type—the abandon utility $\mathbf{z}'_i \boldsymbol{\beta}_a$ varies across individuals, where $\mathbf{z}_i$ includes age bins, county fixed effects, and time trends. These controls govern the within-nest margin between fraud and abandon,

²⁷The alternative—grouping fraud with repair in a “register” nest—would predict that higher repair costs shift drivers from fraud toward abandonment within the nest, changing the fraud/abandon ratio. The observed constant ratio rejects this structure and is consistent with IIA across nests.

²⁸Arrest warrants from the Travis County Constable’s office indicate prices for fraudulent inspections in the \$100–\$200 range.

distinct from β_{nc} which governs the top-level comply/non-comply decision.

More fraud stations raise the inclusive value, which in turn raises $V_{\text{nc},i}$ and increases the probability of choosing noncompliance over repair. The nesting parameter $\lambda \in (0, 1)$ governs within-nest substitution: as $\lambda \rightarrow 0$, noncompliance alternatives become close substitutes and adding a fraud station primarily cannibalizes other fraud stations' business; as $\lambda \rightarrow 1$, adding a station draws more substitution from repair toward noncompliance as station-level fraud becomes more common.

Choice probabilities follow the standard nested logit form: at the top level, the driver chooses between repair and non-comply based on $V_{\text{repair},i}$ and $V_{\text{nc},i}$; conditional on non-comply, the within-nest shares are proportional to the terms in the inclusive value. Appendix G provides the explicit expressions.

5.2 Identification

Each structural parameter is identified by a distinct source of variation in the data.

Repair cost sensitivity (μ, σ). The mean and heterogeneity of repair cost sensitivity are identified by within-vehicle-type variation in predicted repair costs. As described in Section 4.1, predicted repair cost is constructed from OBD diagnostic indicators—check engine light status, diagnostic trouble code counts, and readiness monitor failures—estimated conditional on make, model-year, station, and year fixed effects. These fixed effects are absorbed by the controls in β_r and β_{nc} , so the identifying variation for α_i comes from the diagnostic profile: two otherwise similar vehicles visiting the same station may require very different repairs depending on which component failed. The mean parameter μ is pinned down by the average fraud rate gradient across the repair cost distribution; the scale parameter σ is identified by its *curvature*—how the gradient of fraud with respect to repair cost varies across the distribution.

Assumption 4 (Diagnostic exogeneity). *Conditional on vehicle characteristics $\mathbf{x}_i$ (age, location, etc.), the vehicle's diagnostic profile is independent of unobserved preferences for compliance: $r_i \perp (\nu_i, \varepsilon_i) \mid \mathbf{x}_i$.*

This assumption requires that, within a vehicle type and market, the specific malfunction is as-good-as-random with respect to the driver's inclination toward fraud. It would be violated if, for example, drivers planning to commit fraud deliberately triggered certain diagnostic codes, or if unobserved driver characteristics (such as vehicle maintenance habits) simultaneously determined both the malfunction type and the propensity for fraud. We view this as plausible because the specific component failure (oxygen sensor vs. catalytic converter

vs. EGR valve) is primarily determined by the vehicle’s mechanical condition and driving history rather than by owner characteristics that would also predict fraud demand.

Nesting parameter (λ). The nesting parameter is identified from panel variation in fraud station counts. For example, the post-2022 enforcement crackdown shut down many fraud stations in Dallas County; the substitution patterns in Dallas relative to counties with smaller changes in fraud availability help identify λ . The key variation is the *substitution pattern*—how much of the increased fraud in markets with more stations comes from repair versus from abandon. With λ close to 1, adding a fraud station draws equally from repair and abandon. With λ close to 0 (strong nesting), the new station primarily cannibalizes abandonment within the non-comply nest, with limited effect on the repair rate.

Station type utilities (δ_{opp} , δ_{hi} , δ_{ho}). Conditional on the number of stations available to drivers in each category, the relative choice probabilities across station types identify the δ_k parameters. For example, if markets with more high-intensity in-county stations have disproportionately higher fraud rates than markets with the same total number of fraud stations but more opportunistic ones, then $\delta_{\text{hi}} > \delta_{\text{opp}}$. The δ_k capture the value of fraud at each station type net of the unobserved price charged for a fraudulent inspection.

5.3 Estimation

We estimate the model by simulated maximum likelihood integrating over the log-normal distribution of α_i . The parameter vector includes the coefficient vectors (β_r , β_{nc} , β_a), log-normal parameters (μ , σ), the nesting parameter (λ), and station type utilities (δ_{opp} , δ_{hi} , δ_{ho}). Appendix G provides the full likelihood function. The preferred specification includes county fixed effects and quadratic time trends on the full 17-county sample. Appendix Table G1 reports robustness across alternative control sets and sample definitions.

5.4 Results

Table 4 reports the estimates from the preferred specification. The log-normal distribution of α_i has location $\hat{\mu} = 0.329$ and scale $\hat{\sigma} = 0.874$, implying a mean sensitivity of $\mathbb{E}[\alpha] = 2.04$ per \$100 of repair cost and a median of 1.39. The substantial dispersion indicates that drivers differ considerably in their willingness to pay for repairs: some switch to noncompliance at relatively low repair costs, while others repair even expensive malfunctions.

The nesting parameter $\hat{\lambda} = 0.857$ indicates moderate within-nest substitution. This value implies that adding a fraud station draws roughly 60% of its clientele from other fraud stations’ business and 40% from repair. The pattern is economically intuitive: fraud and

Table 4. Demand Model Estimates: Preferred Specification

	Estimate	Std. Error
<i>Repair cost sensitivity (log-normal)</i>		
μ (location)	0.3289	0.0751
σ (scale)	0.8737	0.0083
$\mathbb{E}[\alpha]$ (mean, per \$100)	-2.035	0.167
Median α (per \$100)	-1.389	0.104
<i>Nesting</i>		
λ (nesting parameter)	0.8570	0.0168
<i>Station type utilities (within nest)</i>		
δ_{opp} (opportunistic)	-6.2529	0.1418
δ_{hi} (high-intensity, in-county)	-5.2236	0.1079
δ_{ho} (high-intensity, out-of-county)	-6.1843	0.0845
<i>Implied elasticities (at mean repair cost)</i>		
$\varepsilon(\text{repair, RC})$		1.19
$\varepsilon(\text{non-comply, RC})$		4.41
$\varepsilon(\text{repair, } N_{\text{hi}})$		-0.0424
<i>Model fit</i>		
Log-likelihood		-2,583,036
AIC		5,166,219
Parameters		74
Observations		4,381,031
Controls	Age bins + county FE + time trend	
Sample	17 TCEQ counties, 2017–2024	
Halton draws	50	

Note: Standard errors computed from the inverse of the numerical Hessian of the log-likelihood evaluated at the maximum. Delta method standard errors for $\mathbb{E}[\alpha]$, median α , and λ . $\alpha_i = -\exp(\mu + \sigma\nu_i)$ with $\nu_i \sim N(0, 1)$; $\mathbb{E}[\alpha] = -\exp(\mu + \sigma^2/2)$. Station type utilities are relative to the abandon alternative within the non-comply nest. Controls include five age bins (≤ 5 , 6–10, 11–15, 16–20, 20+), 16 county fixed effects (17 counties, one absorbed), and a quadratic time trend. The same controls enter the repair equation, the non-comply nest equation, and the abandon utility within the nest.

abandonment are closer substitutes for each other than either is for repair, since both allow the driver to avoid paying for repairs.

Among fraud station types, high-intensity in-county stations are the most attractive ($\hat{\delta}_{\text{hi}} = -5.224$), followed by high-intensity out-of-county ($\hat{\delta}_{\text{ho}} = -6.184$) and opportunistic in-county ($\hat{\delta}_{\text{opp}} = -6.253$). The premium for high-intensity in-county stations relative to opportunistic in-county stations ($\Delta\delta = 1.03$) is consistent with dedicated fraud operations being easier to find or better marketed through online postings or word of mouth. The

out-of-county penalty ($\delta_{ho} - \delta_{hi} = 0.07$) captures the additional travel and search costs of visiting a fraud station outside the driver’s home county.

The elasticity of the repair rate with respect to repair cost is 1.19 at the mean repair cost, ranging from 0.65 at \$150 (where most drivers repair regardless) to 2.57 at \$600 (where marginal drivers are price-sensitive). Fraud station availability has a modest but significant effect: the elasticity of the repair rate with respect to the number of fraud stations is approximately -0.04 to -0.05 , similar across station types, confirming that enforcement-driven reductions in fraud supply shift some drivers toward repair.

To provide intuition for drivers’ compliance preferences, Figure 4 plots the CDF of the indifference repair cost—the cost at which each driver switches from repair to noncompliance. For any repair cost on the x-axis, the curve reads off the share of drivers who would choose noncompliance if they faced that cost. Vertical lines mark the median cost of common emissions repairs from RepairPal data, with dotted step lines showing the implied compliance rate at each. For example, for a \$93 PCV valve replacement, 98% of drivers would repair; however, for more expensive repairs, compliance drops sharply: at the cost of a used catalytic converter (\$750), only 32% would repair.²⁹ The median indifference cost is approximately \$501, with substantial dispersion: 25% of drivers switch to noncompliance below \$278, while the most price-insensitive quartile repairs even above \$903. This heterogeneity is a key input to counterfactual analysis: a policy that reduces fraud access will push some marginal drivers toward repair while pushing others toward abandonment.

The model fits the data well. Across 100 equal-frequency bins of predicted repair cost, the mean absolute error between predicted and observed fraud rates is 1.3 percentage points, and the model matches the overall choice shares exactly at the mean (repair: 79.6 %, fraud: 11.9 %, abandon: 8.5 %). Appendix Table G1 reports results across alternative specifications, confirming that the key structural parameters—particularly σ and λ —are stable across specifications.

5.5 Abandon Decomposition

The demand model estimates a single “abandon” probability that combines two distinct outcomes: unregistered driving (the vehicle remains on the road, generating pollution) and exit (the vehicle is scrapped or sold, generating none).³⁰ We decompose abandon using data from the TORVET roadside remote sensing program (Section 4.3), which uses license plate

²⁹Catalytic converters are among the most expensive emissions-related repairs. Prices vary widely depending on vehicle make and model and whether the part is new or aftermarket: new OEM converters can cost thousands of dollars, but used or aftermarket alternatives are often available for substantially less.

³⁰Vehicles sold out of the market may generate pollution damages elsewhere but are outside the scope of our analysis.

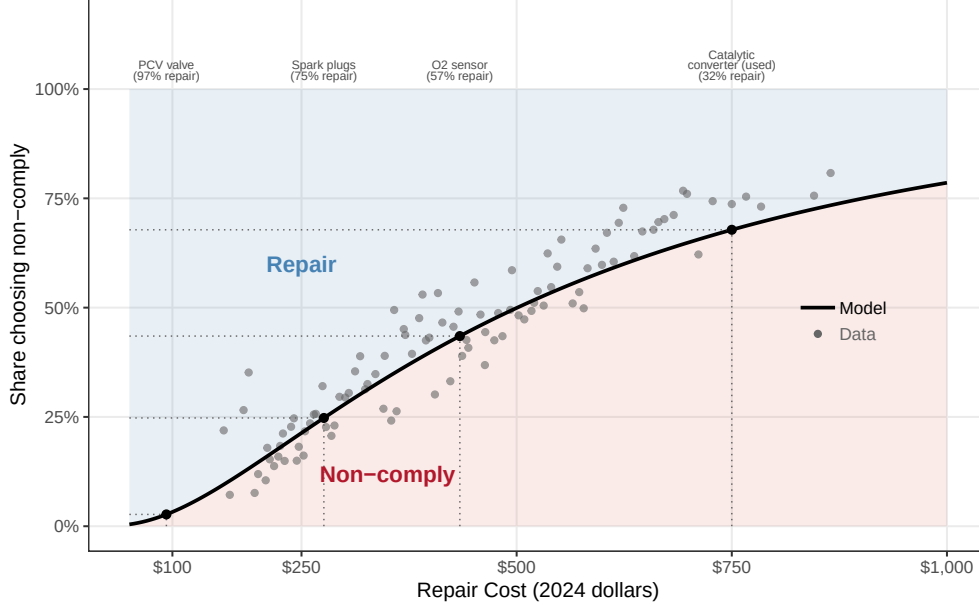


Figure 4. Distribution of Driver Indifference Costs: Repair vs. Non-Comply Choice

Note: The curve shows the CDF of the indifference repair cost r_i^* —the cost at which each driver is indifferent between repair and noncompliance—computed from the estimated log-normal repair cost sensitivity ($\hat{\mu} = 0.329$, $\hat{\sigma} = 0.874$) at population-mean controls. For a given repair cost, the height of the curve gives the share of drivers who would choose noncompliance; the area above the curve is the share who would repair. Vertical dashed lines mark median national repair costs for common emissions-related repairs from RepairPal data. Dotted step lines connect each repair cost to the implied noncompliance share on the y-axis.

imaging and infrared/ultraviolet spectroscopy to measure the tailpipe emissions of passing vehicles under normal driving conditions. Crucially, TORVET captures vehicles on the road regardless of their registration or inspection status, providing a window into the unregistered population that is invisible in the inspection data.

We estimate a logit regression of whether a vehicle’s VIN appears in TORVET data in the calendar year following its inspection cycle:

$$P(\text{in TORVET}_{t+1} \mid \mathbf{x}_i) = \Lambda(\boldsymbol{\phi}'\mathbf{x}_i + \text{abandon}_i \cdot (\boldsymbol{\phi}'_a\mathbf{x}_i)) \quad (16)$$

where $\mathbf{x}_i$ includes age bins and county $\times$ year fixed effects, and abandon_i indicates whether the vehicle abandoned the inspection system. Vehicles that are on the road—whether registered, fraud-flagged, or unregistered—appear in TORVET at a base capture rate of approximately 2%. Exited vehicles (scrapped or sold) do not appear. The probability of unregistered driving conditional on abandonment is therefore:

$$P(\text{unreg} \mid \text{abandon}, \mathbf{x}_i) = \frac{P(\text{in TORVET}_{t+1} \mid \text{abandon}, \mathbf{x}_i)}{P(\text{in TORVET}_{t+1} \mid \text{registered}, \mathbf{x}_i)} \quad (17)$$

The denominator captures the baseline TORVET detection rate for vehicles of the same type that remain on the road. This decomposition requires that TORVET captures are as good as random among on-road vehicles conditional on $\mathbf{x}_i$ —that is, unregistered vehicles are no more or less likely to pass a TORVET sensor than registered vehicles with the same characteristics. This is plausible because TORVET sensors are deployed on major roadways and capture vehicles passively through license plate imaging, without any selection based on registration status.

Overall, 55.9% of abandoning vehicles continue driving unregistered. This share has risen over the sample period, from 38 % in 2018 to 59 % in 2023—a pattern consistent with the enforcement crackdown pushing vehicles from fraud toward unregistered driving rather than exit. Younger vehicles are more likely to drive unregistered (they retain more value), and the rate varies substantially across counties. Appendix Table G2 reports the full decomposition by age and year.

Under counterfactual enforcement scenarios, the demand model predicts new abandon shares, which we decompose into unregistered driving and exit using the cell-specific $P(\text{unreg} | \text{abandon}, \mathbf{x}_i)$ estimated above. We do not treat this probability as structural: marginal drivers pushed from fraud into abandonment were actively seeking to avoid repair and may drive unregistered at higher rates than current abandoners. We therefore use the estimated decomposition as a *lower bound* and construct an *upper bound* by assuming all marginal abandoners drive unregistered.

6 Counterfactuals: Supply-Side Enforcement and Welfare

Supply-side enforcement can take several forms. Texas ramped up enforcement through Operation Cinderblock, using covert audits and criminal prosecution to shut down fraud stations (Section 4.2). Other states have adopted technological approaches that make clean scanning more difficult: California’s Bureau of Automotive Repair, for example, requires electronic VIN verification and stores the vehicle’s calibration identification and verification number at each inspection, automatically flagging mismatches that are indicative of surrogate-vehicle fraud.³¹ We use the estimated demand model to investigate the welfare effects of enforcement that reduces the supply of fraud-capable stations, which captures both direct enforcement (station closures) and technological deterrence (making clean scans infeasible).

³¹California regulations prohibit any “electronic device or software capable of simulating the OBD data stream from a vehicle or manipulating OBD VIN, calibration identification, calibration verification number, MIL-status, readiness, or diagnostic trouble codes” in the approved testing area (California Bureau of Automotive Repair, 2021). Several of the fraud indicators we identify in Section 3—missing eVIN, eVIN-VIN mismatch, calibration ID reuse—would trigger automatic flags under California’s system.

Table 5. Counterfactual Substitution Patterns (2022)

	Baseline	Remove High-Intensity	Remove All Fraud
Repair (%)	78.9	81.2	82.7
Fraud (%)	13.6	6.4	0.0
Abandon (%)	7.5	12.4	17.3
Unregistered (%)	3.9	6.4–8.7	9.0–13.7
Fraud + Unregistered (%)	17.4	12.8–15.1	9.0–13.7
N vehicles	522,946		

Note: Predicted choice shares for 2022 initial-fail vehicles in the 17 TCEQ emissions counties. “Remove High-Intensity” sets $N_{hi} = N_{ho} = 0$ (eliminates dedicated fraud operations, retains opportunistic stations). “Remove All Fraud” sets all station counts to zero. Unregistered decomposition uses cell-specific $P(\text{unreg} | \text{abandon}, \mathbf{x})$ from the TORVET logit (Section 5.5). Lower bound: marginal abandoners have the same unreg probability as average abandoners conditional on observables. Upper bound: all abandoners drive unregistered. Fraud + Unregistered is the pollution-relevant noncompliance share.

We consider two counterfactuals for the 522,946 initial-fail vehicles in 2022, the peak year before Operation Cinderblock. The first removes all high-intensity fraud stations ($N_{hi} = N_{ho} = 0$), retaining only opportunistic in-county stations—roughly what a targeted enforcement campaign like Operation Cinderblock achieves. The second removes all fraud stations ($N_{opp} = N_{hi} = N_{ho} = 0$), a benchmark corresponding to technological solutions that effectively eliminate clean scanning. The first counterfactual holds opportunistic stations’ behavior fixed; remaining stations may adjust pricing in response to reduced competition, a supply-side response we do not model here but address in ongoing work. This caveat does not apply to the second counterfactual, where no fraud stations remain.

6.1 Substitution Patterns

Table 5 reports the predicted choice shares under each scenario.

Removing high-intensity stations reduces the fraud rate by -7.2 percentage points but does not convert all displaced fraud into repair. The repair rate increases by +2.3 percentage points, while the abandon rate rises by +4.9 percentage points. This pattern reflects the nesting structure: within the non-comply nest, some drivers who would have chosen fraud at a high-intensity station substitute toward abandonment rather than repair, because fraud and abandonment are closer substitutes for each other than either is for repair ($\hat{\lambda} = 0.857$).

The substitution toward abandonment is the enforcement leakage channel. The “Fraud + Unregistered” rows in Table 5 report the pollution-relevant noncompliance share under both bounding assumptions from Section 5.5. Even under the lower bound, enforcement-induced unregistered driving partially offsets the pollution gains from reduced fraud; under

Table 6. Welfare Decomposition of Supply-Side Enforcement (2022)

	Remove High-Intensity Stations	Remove All Fraud Stations
<i>Panel A: Drivers and repair sector (per initial-fail driver)</i>		
Δ CS (drivers)	\$-5.25	\$-8.64
Δ PS repair sector	\$1.58	\$2.58
<i>Panel B: Pollution benefits (per initial-fail driver)</i>		
Lower bound	\$2.71	\$4.63
Upper bound	\$5.33	\$10.21
<i>Panel C: Fraud station surplus (per initial-fail driver)</i>		
Δ PS fraud stations	\$-10.76	\$-20.33
<i>Panel D: Net welfare excl. fraud PS ($A + B$, per driver)</i>		
Lower bound	\$-0.96	\$-1.43
Upper bound	\$1.66	\$4.15
<i>Panel E: Net welfare incl. fraud PS ($A + B + C$, per driver)</i>		
Lower bound	\$-11.73	\$-21.76
Upper bound	\$-9.10	\$-16.18

Note: Welfare decomposition for 2022. Panels A–C report per-driver values for the initial-fail estimation sample. Δ CS from the nested logit logsum scaled by $1/E[|\alpha|]$. Repair sector Δ PS at 25% profit margin (similar for 20–40%). Pollution benefits use age-heterogeneous fraud coefficients from TORVET remote sensing and county-specific AP4 marginal damages for NOx and VOC. Lower bound: all marginal abandon-switchers drive unregistered (maximum pollution leakage). Upper bound: marginal abandon-switchers have the same $P(\text{unreg} \mid \text{abandon}, \mathbf{x})$ as average abandoners conditional on observables (minimum leakage). Fraud station Δ PS at \$150/inspection with near-zero marginal cost (similar for \$100–\$200). All values per initial-fail driver. The initial-fail sample contains 522,946 vehicles in 2022.

the upper bound, the offset is substantially larger.

Removing all fraud stations amplifies both effects. The fraud rate falls to zero, repair increases by +3.8 percentage points, and abandon increases by +9.8 percentage points. Figure H5 illustrates the reallocation.

6.2 Welfare Decomposition

We decompose the welfare effect of each counterfactual into four components: driver consumer surplus, repair sector producer surplus, pollution externalities, and fraud station producer surplus. Table 6 reports the results.

Table 6 reports welfare components per initial-fail driver—the population for which we observe predicted repair costs and can compute counterfactual choice probabilities. Panel A reports surplus changes. Drivers are worse off when fraud stations are removed: those who

were choosing fraud are revealed-preferred to that option, and removing it forces them into less-preferred alternatives (repair at cost r_i or abandonment). Consumer surplus is computed from the nested logit logsum scaled by the mean marginal utility of income ($1/\mathbb{E}[|\alpha|]$). On the producer side, fraud stations earn approximately \$150 per fraudulent inspection with near-zero marginal cost, so their surplus loss is approximately equal to lost revenue.³² Auto repair shops gain business from drivers who switch to repair; assuming a 25% profit margin on repair revenue, the repair sector gain partially offsets the fraud station loss.³³ The contrast is stark: fraud is a high-margin service (near 100%) while legitimate repair is low-margin ($\sim 25\%$), so a dollar of fraud revenue destroyed is not offset by a dollar of repair revenue gained.

Panel B reports the pollution benefit—the primary welfare gain from enforcement. We compute excess emissions using vehicle age-specific fraud coefficients from the TORVET remote sensing regressions (Section 4.3) and convert to dollar-valued health damages using county-specific AP4 marginal damage rates for NOx and VOC. The bounds reflect the TORVET abandon decomposition from Section 5.5: the lower bound assumes all marginal drivers displaced into abandonment drive unregistered, while the upper bound applies the estimated unregistered rate conditional on observables.

Panel D reports net welfare excluding fraud station surplus ($A + B$), and Panel E includes it ($A + B + C$). Whether fraud station profits should count as a social cost of enforcement is a normative question we do not take a position on; we report both so the reader can apply either framework.

For the 522,946 vehicles in the initial-fail sample, removing high-intensity stations yields an aggregate net welfare change of \$-504,641 to \$867,512 excluding fraud station surplus. Including fraud station surplus, the net welfare change is \$-6.1 million to \$-4.8 million, reflecting the large criminal rents destroyed by shutting down fraud operations. These estimates apply only to vehicles observed failing a legitimate test. In 2022, an additional 1,366,567 vehicles obtained fraudulent inspections without first attempting a legitimate test—drivers who learned of their vehicle’s fail-state condition from a previous inspection cycle or a recent mechanic visit and proceeded directly to a fraud station. If these direct-to-fraud drivers respond similarly to enforcement, the aggregate welfare impacts could be an order of magnitude larger than those reported here. Extending the model to this population is an important direction for future work (Section 5.1).

³²The \$150 estimate is based on arrest warrants of stations accused of clean scanning and discussions with law enforcement agents involved in Operation Cinderblock. Results are similar for \$100–\$200.

³³We lack direct data on mechanics’ cost structure, so we assume a 25% profit margin reflecting the parts-intensive nature of emissions repairs (catalytic converters, oxygen sensors). Results are similar for 20–40%.

6.3 Discussion

Three findings emerge from the counterfactual analysis. First, supply-side enforcement generates substantial substitution toward abandonment. Not all displaced fraud converts to repair—the nesting structure implies that a significant fraction of drivers substitute toward the nearest non-comply alternative (abandonment) rather than crossing the nest boundary to repair. This substitution partially offsets the pollution benefits of enforcement.

Second, whether supply-side enforcement is welfare-improving depends on the normative treatment of fraud station profits. Excluding fraud station surplus (Panel D), full enforcement is likely welfare-improving—the pollution benefits to the broader population exceed the combined consumer surplus loss and repair sector gain. Including fraud station surplus (Panel E), enforcement generates a net welfare loss under all assumptions, because the rents destroyed by shutting down fraud operations are large. This finding sharpens the normative question at the heart of illegal market enforcement: if society does not value the profits of criminal enterprises, the case for enforcement is strong; if one applies the Beckerian framework that counts all surplus equally, the case is more nuanced. These rents represent surplus that fraud suppliers can invest in sustaining the market, consistent with the persistence of fraud over time. The welfare calculation also omits other benefits of enforcement such as reduced criminal activity and improved program integrity.

Third, the results highlight the complementarity between supply-side and demand-side interventions. Registration enforcement (targeting unregistered vehicles) would reduce the leakage channel, amplifying the net benefits of fraud station closures. Similarly, repair cost subsidies or waivers for low-income drivers would shift the repair/non-comply margin, converting more displaced fraud into repair rather than abandonment. The demand model provides the substitution elasticities needed to evaluate these combined interventions, which we are pursuing in ongoing work.

7 Conclusion

Delegated enforcement creates a market for fraud. We study this market empirically—measuring its size, tracing its environmental consequences, and evaluating what happens when the regulator intervenes.

A central finding is that fraud generates large rents for corrupt inspectors. Fraud stations charge a fraction of the cost of repair, yet drivers are willing to pay because fraud allows them to avoid costly compliance while maintaining a current vehicle registration. The rents captured by corrupt inspectors exceed the environmental gains from enforcement. Whether eliminating these rents counts as a social cost of enforcement is a normative question we leave

to policymakers and the public, but their magnitude is a positive fact that any enforcement design must confront, and helps explain why fraud has been persistent in this market despite periodic enforcement crackdowns.

If policymakers seek to improve air quality through enforcement, our findings suggest that isolated supply-side crackdowns may not be fully effective. Reducing fraud supply without addressing the underlying demand for noncompliance leaves the leakage channel open, and the most effective enforcement regimes likely combine fraud station closures with demand-side interventions that make compliance less costly. Increasing on-road registration enforcement would reduce the leakage channel, amplifying the pollution benefits of fraud station closures. Similarly, repair cost subsidies or waivers for low-income drivers could be complementary, shifting displaced drivers toward repair rather than unregistered driving.

A natural next step in our research is to model equilibrium in the market for inspections jointly, with stations choosing whether and how intensively to offer fraud and vehicle owners selecting where to inspect and whether they will comply with the inspection program. Estimating both sides of this market using our data would allow for counterfactual simulations that account for full equilibrium responses, including entry and exit of fraudulent stations and strategic reallocation of demand across stations, rather than the partial equilibrium patterns we document here. This follow up work could be used to design optimal monitoring and enforcement regimes: how should the regulator allocate audit resources, what penalty schedule maximizes deterrence per dollar of enforcement cost, and what impact does the leakage channel that we document have on the impact of these enforcement tools? Our findings demonstrate that answering these questions is of first-order importance.

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Online Appendix for “Fraud, Evasion, and Enforcement in Decentralized Environmental Regulation”

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The following appendices are **for online publication only**.

Appendix A Data Appendix

Appendix B Fraud Detection Model: Proofs and Derivations

Appendix C Fraud Detection Monte Carlo Analysis

Appendix D Fraud Indicator Selection, Construction, and Validation

Appendix E Fraud Identification: Robustness and Implementation

Appendix F Remote Sensing Data and Pollution Impacts

Appendix G Demand Model Details

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A Data Appendix

A.1 Data and Sample Construction

We purchased the universe of emissions inspection records from TCEQ’s Texas Inspection and Maintenance System (TIMS). The raw data comprise monthly bulk export files in parquet format, covering 2016 through 2024. Each monthly file contains all emissions tests conducted that month, with over 200 fields including station identifiers, test date and time, vehicle characteristics, OBD diagnostic data, and test results.

Column Selection. From the raw data, we retain 95 columns essential to our analysis: station information (station ID, name, analyzer ID, county), test metadata (date, time, sequence, type), pass/fail indicators for each component, OBD-II status and readiness monitor results, the full electronic profile (OBD_VIN, PCM identifier, calibration ID, CVN, PID count), gas cap test result (GAS_CAP_PF_FL, GAS_CAP_MISS), inspector MIL checks (OBD2_MIL_ON_RUN, OBD2_MIL_CHECK), repair information where reported, and vehicle characteristics (VIN, make, model, model year, odometer).

Sample Restrictions. Following [Eastern Research Group \(2024a\)](#), we apply the following filters to construct our analysis sample:

- Exclude safety-only and visual-only inspections (test types “H” and “P”); retain only OBD-II emissions tests
- Exclude aborted tests (abort codes “A” and “J”)
- Require valid 17-character VIN with no invalid VIN flag
- Require model year 1996 or later (when OBD-II became mandatory)
- Exclude implausible model years (more than 2 years in the future)
- Require valid test sequence number

Deduplication. After applying filters, we remove duplicate records. Duplicates arise from data transmission errors between analyzers and the central TIMS database. We identify duplicates using a six-field key: cleaned VIN, cleaned OBD_VIN, test end time, test sequence number, analyzer ID, and overall result. Records matching on all six fields are collapsed to a single observation.

Waivers. Texas allows vehicles that fail inspection to apply for a low-income waiver if repairs would impose undue financial burden. We identify waived vehicles using the `WAIVER_NO` field: any test with a non-blank waiver number is classified as “W” (waiver) rather than pass or fail. Waivers are rare in our data—only a few hundred are granted statewide each year—and we exclude them from most analyses following [Eastern Research Group \(2024a\)](#).

Fleet and Government Stations. We retain but flag tests conducted at Fleet (station ID containing “F”) and Government (station ID containing “G”) stations. These stations inspect only their own vehicles and are not open to the public. They exhibit higher rates of certain fraud indicators (particularly missing eVIN) for legitimate technical reasons: commercial and fleet vehicles often predate eVIN requirements or use internal tracking systems.

A.2 Inspection Cycle Identification

We define an *inspection cycle* as the sequence of tests on the same VIN within an annual registration period. Texas requires annual emissions inspections, so cycle boundaries occur when a vehicle passes, receives a waiver, or exhibits a gap of more than 365 days between

tests. This definition captures the relevant decision unit for drivers: each cycle represents an attempt to achieve compliance, possibly involving multiple failed tests and retests before the vehicle ultimately passes, receives a waiver, or is abandoned.

Cycle ID Algorithm. We assign cycle identifiers as follows. First, we sort all tests by VIN, then by test date and time. For each VIN, we initialize a cycle counter at 1. A new cycle begins after any test that results in a pass (“P”) or waiver (“W”), or after a gap of more than 365 days since the previous test. Within each cycle, we number tests sequentially (1, 2, 3, ...) to identify initial tests versus retests.

Station-Specific Cycles. For some analyses, we further partition cycles by station-analyzer combination. The *VIN-ETS cycle* groups tests on the same VIN, at the same analyzer, within the same registration cycle. This allows us to track whether a vehicle that fails at one station returns to the same station for a retest (enabling within-station comparisons of electronic profiles) or switches to a different station.

Summary Statistics. Applying this algorithm to our 2016–2024 sample yields approximately 95.9 million unique VIN-cycles. The median cycle contains 1 test(s), reflecting that most vehicles pass on the initial attempt. Cycles with multiple tests typically involve an initial failure followed by one or more retests until the vehicle passes, receives a waiver, or the driver abandons the registration attempt.

Table A1. Vehicle Test Cycle Summary Statistics

	Mean	Median	Std. Dev.
<i>Panel A: Full Sample (2016–2024)</i>			
Number of unique VINs	22,682,838		
Cycles observed per vehicle	4.23	4	2.73
Tests per cycle	1.05	1	0.25
<i>Panel B: Cycle Outcomes (2023)</i>			
Number of unique VINs	10,659,379		
Pass on initial test (%)	94.8		
Pass after retest (%)	4.5		
Abandon / never pass (%)	0.8		
Vehicle age (years)	9.3	8	5.5

Note: Sample includes all vehicle inspection cycles in Texas emissions counties. A cycle begins with each initial test attempt and ends when the vehicle passes, receives a waiver, or is abandoned. Panel B reports the share of cycles in each mutually exclusive outcome category. Panel A covers all years 2016–2024; Panel B restricts to 2023.

Table A2. Station Summary Statistics (2023)

	Mean	Median	Std. Dev.
<i>Panel A: Station Characteristics</i>			
Number of stations	5,834		
Vehicles inspected	1,932	1,142	2,687
Number of analyzers	1.2	1	0.4
Years in operation	6.3	8	2.9
<i>Business type (%)</i>			
Independent	72.5		
National chain	16.1		
Franchised dealer	10.8		
Used dealer	0.4		
<i>Regulatory history</i>			
Sanctioned in 2023 (%)	11.9		
Matched to Google Places (%)	93.7		
<i>Panel B: Google Places Characteristics (Matched Stations)</i>			
Number of matched stations	5,468		
<i>Google business category (%)</i>			
Car repair	61.6		
Dealer	14.5		
Tire shop	10.2		
Car wash / gas station	2.4		
Other / unclassified	11.2		
Rating (1–5)	4.44	4.5	0.43
Number of reviews	754	259	1,595
Has website (%)	94.1		
Has phone listing (%)	92.9		
Open Saturday (%)	74.1		
Open Sunday (%)	20.8		

Note: Sample includes all licensed private emissions inspection stations that performed at least one inspection in Texas emissions counties, 2023. Business type classified from station name using keyword matching; independent includes quick lube and inspection-focused outlets. Panel B reports characteristics from the Google Places API for the subset of stations successfully matched to a Google business listing. Sanction statistics restricted to TCEQ disciplinary actions during 2023.

A.3 Vehicle Repair Costs

This section provides estimation details and validation for the repair cost model described in Section 4.1.

Table A3. Station and Vehicle Characteristics by Sanction Status (2023)

	Not Sanct.	Sanctioned	Difference
<i>Panel A: Vehicle Characteristics (2023)</i>			
Number of stations	5,139	695	
Number of unique VINs	10,296,790	976,961	
Pass on initial test (%)	94.4	97.0	2.6*** (0.02)
Pass on retest (%)	66.7	76.6	9.8*** (0.17)
Abandon / never pass (%)	0.8	0.3	-0.5*** (0.01)
Vehicle age (years)	9.0	12.1	3.15*** (0.01)
Quick lube / inspection (%)	6.3	7.9	1.6 (1.08)
Matched to Google Places (%)	94.8	85.6	-9.2*** (1.37)
<i>Panel B: Google Places Characteristics (Matched Stations)</i>			
Number of stations	4,873	595	
<i>Google business category (%)</i>			
Car repair	62.6	53.4	-9.2*** (2.16)
Dealer	14.7	12.9	-1.7 (1.47)
Tire shop	10.7	6.4	-4.3*** (1.10)
Car wash / gas station	2.4	2.2	-0.3 (0.64)
Other / unclassified	9.6	25.0	15.5*** (1.83)
Rating (1–5)	4.45	4.37	-0.07*** (0.03)
Number of reviews	788	446	-341.92*** (66.24)
Has website (%)	95.3	84.2	-11.1*** (1.53)
Open Saturday (%)	75.2	64.9	-10.3*** (2.05)
Open Sunday (%)	21.5	15.3	-6.2*** (1.59)

Note: Sample includes all private licensed stations that performed at least one emissions inspection in Texas emissions counties, 2023. Sanctioned stations received at least one TCEQ disciplinary action during 2023. Panel A reports vehicle-level test characteristics; Panel B reports station-level Google Places characteristics for the subset matched to a Google business listing. Differences are sanctioned minus not sanctioned; standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

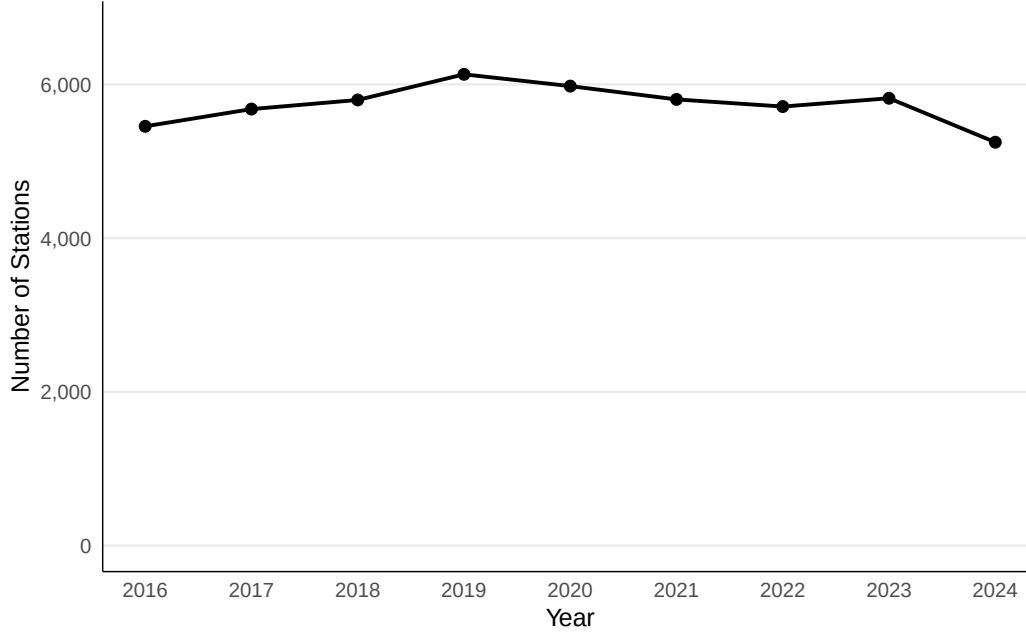


Figure A1. Number of Stations Performing Inspections by Year

Note: Figure plots the number of private licensed stations that performed at least one emissions inspection in each year, 2016–2024. The count is stable between approximately 5,200 and 6,100 stations throughout the sample period.

Fail cycle definition. We define a *fail cycle* as an inspection cycle (see Section A) whose initial test results in a failure. Over 2017–2024, we identify 4,922,414 fail cycles.¹ Each cycle terminates in one of three mutually exclusive outcomes. A cycle ends in *repair* (3,917,260 cycles, 84%) if the vehicle subsequently passes a retest that is not flagged as fraudulent by the detection model described in Section 3. A cycle ends in *fraud* (591,541 cycles, 7.5%) if the vehicle passes a retest that is flagged by our fraud identification model. A cycle ends in *abandonment* (413,613 cycles, 8.5%) if the vehicle never passes within the registration period.

Repair cost model. Table A4 reports summary statistics for all individual test records with a positive inspector-recorded repair cost; the unit of observation in that table is a single test, and the sample spans 2016–2024. For the regression model, we collapse multiple tests within the same fail cycle into a single observation, restrict to cycles ending in repair (excluding fraud and abandonment outcomes), and drop 2016 as described above. Among these fail cycles, approximately 7% have a positive inspector-recorded cost. The regression sample (Table A5) is therefore smaller than the count of tests with positive costs in Table A4,

¹We exclude 2016 because classifying abandoned cycles requires observing the full registration year. Late-2016 fail cycles extend into November 2017, which is missing from our bulk data export; we therefore cannot reliably determine whether those cycles resulted in a pass or were abandoned.

Table A4. Repair Cost Summary Statistics

	Mean	Std. Dev.
Total tests	100,960,972	
Tests with repair reported	664,104	
Tests with positive repair cost	248,108	
Repair cost (2024 \$)	\$268	\$1,474
<i>Repair Type (share)</i>		
Fuel system	29.9%	
Ignition/electrical	12.2%	
Emissions system	25.4%	
Engine mechanical	2.8%	
Misc.	29.6%	

Note: Sample includes all emissions tests (2016–2024). A repair is identified when the repair group code (REP_GRP) is 1–5. Repair cost statistics are computed among tests with positive reported repair costs, expressed in 2024 dollars using the BLS motor vehicle maintenance and repair price index (CUUR0000SETD). Repair type shares are computed among tests with positive repair costs.

which reports individual test records across all years and outcomes rather than deduplicated fail cycles restricted to the repair outcome. We estimate the model on this subset:

$$\begin{aligned} \log(c_i) = & \alpha + \beta_1 \cdot \text{MIL}_i + \beta_2 \cdot \text{MIL_Run}_i + \sum_{k=1}^6 \beta_{3k} \cdot \mathbb{1}\{\text{DTC}_i = k\} \\ & + \beta_4 \cdot \text{GasCap}_i + \sum_j \beta_{5j} \cdot \text{Monitor}_{ij} + \delta_{\text{make}} + \delta_{\text{MY}} + \delta_{\text{station}} + \delta_{\text{year}} + \varepsilon_i \quad (\text{A1}) \end{aligned}$$

where the variables are as defined in Table A5.² Odometer reading is excluded from the preferred specification but included as a robustness check (column 2 of Table A5). The odometer coefficient is positive and statistically significant, consistent with higher-mileage vehicles facing more expensive repairs, but its inclusion leaves all other coefficients essentially unchanged and does not meaningfully affect the R^2 , suggesting that make and model-year fixed effects already absorb most mileage-correlated cost variation. We therefore exclude it from the preferred specification to guard against attenuation from measurement error in self-reported odometer readings.

Table A5 reports the estimated coefficients. The key results confirm that diagnostic severity predicts repair costs: an illuminated MIL is associated with approximately 69% higher costs ($e^{0.53} - 1$), and DTC counts exhibit a monotone gradient from 17% (1 DTC) to 70% (6+ DTCs) relative to vehicles with zero stored codes. The gas cap failure indicator is significant

²A/C refrigerant and PCV readiness monitors are excluded: A/C does not count toward the Texas I/M not-ready threshold, and PCV readiness data were not reported in our data.

and negative (-19%), confirming that this cheap repair is associated with below-average costs. Among readiness monitors, catalyst, evaporative system, O_2 sensor heater, and secondary air failures are all significant and positive, while several rarely-triggered monitors (misfire, heated catalyst, comprehensive component) have large standard errors reflecting low prevalence. Adding log odometer (column 2) has negligible impact on the diagnostic coefficients, confirming that make and model-year fixed effects absorb mileage-correlated cost variation.

External validation. Because repair cost records are self-reported and voluntary—only a subset of retests include a recorded cost—we cross-check the scalar $\bar{c} = \$275$ against two independent benchmarks.

The first benchmark is RepairPal, an independent online repair cost estimator. We scraped RepairPal quotes for seven common emissions-related repair types (oxygen sensor, catalytic converter, EGR valve, evaporative canister, MAF sensor, spark plugs, and PCV valve), across a representative set of makes and model years. The simple average across all seven types is approximately \$660. Weighting by the observed distribution of repair types in our data (Table A4) reduces this to approximately \$540. The remaining gap reflects two factors: the repair type codes in our data map imprecisely to specific RepairPal repairs, and RepairPal covers only professionally diagnosed jobs at retail labor rates, while our records include many lower-cost self-performed or code-clearing repairs.

The second benchmark is a survey of the same Texas inspection stations conducted by Eastern Research Group (2024b). Station owners reported average repair costs following a failed emissions inspection of \$293–\$379 across metros (nominal 2023 dollars), with medians of \$150–\$250, consistent with our mean of \$275 in 2024 dollars.

Taken together, these benchmarks confirm that \$275 is a reasonable calibration of the scalar for the typical Texas emissions repair. We note, however, that the demand elasticity estimated in Section 5 is invariant to the scalar: because α scales inversely with $\bar{c}$, any rescaling of the scalar leaves relative rankings and elasticities unchanged. Counterfactual results that depend on dollar-valued cost levels—such as welfare calculations—do depend on the scalar being correctly calibrated, and readers should interpret those results with the uncertainty in $\bar{c}$ in mind.

B Fraud Detection Model: Proofs and Derivations

This appendix collects proofs, identification arguments, and estimation details for the fraud detection model developed in Section 3. We use the notation established there: π_{sv} is the

Table A5. Repair Cost Prediction Model

	(1)	(2)
Check engine light (MIL) on	0.5479*** (0.0294)	0.5475*** (0.0294)
DTC count = 1	0.1597*** (0.0284)	0.1590*** (0.0284)
DTC count = 2	0.2937*** (0.0322)	0.2924*** (0.0322)
DTC count = 3	0.3880*** (0.0383)	0.3875*** (0.0383)
DTC count = 4	0.4042*** (0.0460)	0.4027*** (0.0461)
DTC count = 5	0.5079*** (0.0557)	0.5063*** (0.0557)
DTC count = 6+	0.5305*** (0.0493)	0.5281*** (0.0493)
Catalyst not ready	0.1236*** (0.0103)	0.1230*** (0.0103)
Evaporative system not ready	0.0917*** (0.0141)	0.0916*** (0.0141)
Oxygen sensor not ready	0.0087 (0.0105)	0.0100 (0.0105)
EGR system not ready	0.0224** (0.0100)	0.0230** (0.0100)
Fuel system not ready	-0.0547*** (0.0180)	-0.0551*** (0.0179)
O ₂ sensor heater not ready	0.1206*** (0.0108)	0.1185*** (0.0108)
Misfire not ready	-0.0539 (0.0888)	-0.0551 (0.0883)
Secondary air system not ready	0.1329*** (0.0254)	0.1330*** (0.0254)
Heated catalyst not ready	-0.1089 (0.2100)	-0.1093 (0.2097)
Comprehensive component not ready	-0.1622 (0.1091)	-0.1613 (0.1092)
MIL on (inspector run check)	0.0433 (0.0290)	0.0440 (0.0289)
Gas cap failure	-0.2234*** (0.0334)	-0.2230*** (0.0334)
Log odometer		0.0236*** (0.0038)
Observations	152,225	152,225
R^2	0.565	0.565
Within R^2	0.041	0.041
Make FE	✓	✓
Model year FE	✓	✓
Station FE	✓	✓
Year FE	✓	✓

Note: Dependent variable: log repair cost (2024 dollars). Sample restricted to fail cycles ending in repair with positive inspector-recorded cost. Standard errors clustered by station in parentheses. Column (1) is the preferred specification used to construct the diagnostic multiplier; column (2) adds log odometer as a robustness check. DTC = diagnostic trouble code. MIL = malfunction indicator light. Omitted DTC category is zero stored codes.

Table B1. Fraud Detection Model Notation

Symbol	Description	Varies by
<i>Observed data</i>		
Y_j	Binary fraud indicator j	Inspection
$p_{j,sv}$	Observed flag rate for indicator j	Indicator $\times$ station $\times$ vehicle type
<i>Parameters to estimate</i>		
w	Prior fraud-capable share	Scalar
π_{sv}	Fraud rate (share of fraudulent inspections)	Station $\times$ vehicle type
$\beta_{j,v}$	Detection rate (flag prob. if fraudulent)	Indicator $\times$ vehicle type
$\Lambda_{j,sv}$	Honest baseline (flag prob. if honest)	Indicator $\times$ station $\times$ vehicle type
<i>Unobserved (latent) objects</i>		
b_s	Station type (0 = clean, 1 = fraud-capable)	Station
F_i	Inspection fraud status (0 = honest, 1 = fraud)	Inspection
γ_s	Posterior $P(b_s = 1 \mid \text{data})$	Station
$\tau_{sv}(y)$	Posterior $P(F_i = 1 \mid Y=y, b_s=1)$	Station $\times$ vehicle type $\times$ pattern

Notes: Indices are s (station), v (vehicle type), j (indicator), i (inspection). Detection gaps $D_{j,sv} \equiv \beta_{j,v} - \Lambda_{j,sv}$ and excess flag rates $\delta_{j,sv} \equiv p_{j,sv} - \Lambda_{j,sv}$ are derived quantities defined in Section 3.2.

fraud rate in cell (s, v) , $\beta_{j,v}$ is the detection rate for indicator j on vehicle type v , $\Lambda_{j,sv}$ is the honest baseline, $D_{j,sv} \equiv \beta_{j,v} - \Lambda_{j,sv}$ is the detection gap, and $\delta_{j,sv} \equiv p_{j,sv} - \Lambda_{j,sv} = \pi_{sv} \cdot D_{j,sv}$ is the excess flag rate.

B.1 Proof of Proposition 1

Proof. By the law of total expectation, the joint probability of indicators j and k both firing in cell (s, v) is:

$$\begin{aligned} \mathbb{E}[Y_j Y_k \mid s, v] &= \pi_{sv} \cdot P(Y_j = 1, Y_k = 1 \mid F = 1, s, v) \\ &\quad + (1 - \pi_{sv}) \cdot P(Y_j = 1, Y_k = 1 \mid F = 0, s, v). \end{aligned} \quad (\text{B1})$$

Under conditional independence (Assumption 2), the joint probabilities factor:

$$\mathbb{E}[Y_j Y_k \mid s, v] = \pi_{sv} \cdot \beta_{j,v} \beta_{k,v} + (1 - \pi_{sv}) \cdot \Lambda_{j,sv} \Lambda_{k,sv}. \quad (\text{B2})$$

The product of marginal means is:

$$\begin{aligned} \mathbb{E}[Y_j \mid s, v] \cdot \mathbb{E}[Y_k \mid s, v] &= [\pi_{sv} \beta_{j,v} + (1 - \pi_{sv}) \Lambda_{j,sv}] \\ &\quad \cdot [\pi_{sv} \beta_{k,v} + (1 - \pi_{sv}) \Lambda_{k,sv}]. \end{aligned} \quad (\text{B3})$$

Taking the difference and expanding:

$$\begin{aligned}
\text{Cov}(Y_j, Y_k \mid s, v) &= \mathbb{E}[Y_j Y_k] - \mathbb{E}[Y_j] \mathbb{E}[Y_k] \\
&= \pi_{sv}(1 - \pi_{sv}) [\beta_{j,v} \beta_{k,v} - \beta_{j,v} \Lambda_{k,sv} \\
&\quad - \Lambda_{j,sv} \beta_{k,v} + \Lambda_{j,sv} \Lambda_{k,sv}] \\
&= \pi_{sv}(1 - \pi_{sv}) (\beta_{j,v} - \Lambda_{j,sv}) (\beta_{k,v} - \Lambda_{k,sv}) \\
&= \pi_{sv}(1 - \pi_{sv}) \cdot D_{j,sv} \cdot D_{k,sv}.
\end{aligned}
\tag*{$\square$}$$

B.2 Identification with $K = 3$: Nonparametric Recovery

With $K = 3$ conditionally independent binary indicators and a binary latent variable, all model parameters are identified without any assumptions about the honest baselines $\Lambda_{j,sv}$.

Parameter count. The model has $1 + 2K = 7$ unknowns per cell: π_{sv} plus $(\beta_{j,v}, \Lambda_{j,sv})$ for $j = 1, 2, 3$.

Observable count. The joint distribution of three binary variables has $2^3 - 1 = 7$ free parameters.

With 7 equations and 7 unknowns, the system is just identified. Intuitively, the three pairwise covariances $\text{Cov}(Y_j, Y_k)$ for $j < k$ identify products of detection gaps via Proposition 1. The ratios of these covariances identify the ratios of detection gaps:

$$\frac{\text{Cov}(Y_1, Y_2)}{\text{Cov}(Y_1, Y_3)} = \frac{D_{2,sv}}{D_{3,sv}}, \quad \frac{\text{Cov}(Y_1, Y_2)}{\text{Cov}(Y_2, Y_3)} = \frac{D_{1,sv}}{D_{3,sv}}.$$

Combined with the three marginal flag rates $p_{j,sv}$ and the product structure $\delta_{j,sv} = \pi_{sv} \cdot D_{j,sv}$, this system pins down all seven unknowns cell by cell. The honest baselines $\Lambda_{j,sv}$ are recovered as residuals: $\Lambda_{j,sv} = p_{j,sv} - \pi_{sv} \cdot D_{j,sv}$.

More generally, with K conditionally independent binary indicators:

K	Moments ($2^K - 1$)	Parameters ($1 + 2K$)	Status
2	3	5 (3 with known Λ)	Just-identified with known baselines
3	7	7	Just-identified, no baselines needed
4	15	9	Overidentified (6 restrictions)
5	31	11	Overidentified (20 restrictions)

Each additional indicator either eliminates an assumption or provides testable restrictions. The overidentifying restrictions for $K \geq 4$ are testable implications of conditional

independence: after estimating π_{sv} and $D_{j,sv}$ from any subset of $\binom{K}{2}$ pairwise covariances, the remaining covariances should be consistent with the estimated values.

B.3 Augmented Station Classification

The CMH-type test statistic $\tilde{Z}_s$ pools cross-indicator covariance across vehicle types within each station. By Proposition 1, $\text{Cov}(Y_j, Y_k \mid s, v) = \pi_{sv}(1 - \pi_{sv})D_{j,sv}D_{k,sv}$, which is maximized at $\pi_{sv} = 0.5$ and vanishes as $\pi_{sv} \rightarrow 0$ or $\pi_{sv} \rightarrow 1$. This creates an identification challenge: stations that fraudulently pass nearly all inspections ($\pi_{sv} \approx 1$) generate little covariance, making them difficult to distinguish from clean stations using $\tilde{Z}_s$ alone.

To address this, we augment the classification with a second signal: the station’s average excess flag rate,

$$\bar{\delta}_s = \frac{1}{K \cdot |\mathcal{V}_s|} \sum_{v \in \mathcal{V}_s} \sum_{j=1}^K (p_{j,sv} - \bar{\Lambda}_{j,v}),$$

where $\bar{\Lambda}_{j,v}$ is the cross-station average flag rate for indicator j and vehicle type v , serving as a proxy for the honest baseline. Stations with $\pi_{sv} \approx 1$ have low $\tilde{Z}_s$ but high $\bar{\delta}_s$; stations with moderate π_{sv} have high $\tilde{Z}_s$ and moderate $\bar{\delta}_s$; clean stations have both $\tilde{Z}_s \approx 0$ and $\bar{\delta}_s \approx 0$.

The bivariate signal $(\tilde{Z}_s, \bar{\delta}_s)$ is modeled as a constrained Gaussian mixture:

$$(\tilde{Z}_s, \bar{\delta}_s) \sim (1 - w) \cdot N_2(\mu_0, \Sigma_0) + w \cdot N_2(\mu_1, \Sigma_1), \quad (\text{B4})$$

where the clean component $N_2(\mu_0, \Sigma_0)$ has its mean and variance fixed from the null distribution (analogous to the local false discovery rate framework), while the fraud component and mixing weight are estimated via EM on the $(\tilde{Z}_s, \bar{\delta}_s)$ values alone. A station is classified as fraud-capable if its bivariate posterior exceeds the threshold $\bar{\gamma}$.

B.4 EM Update Equations

This section provides the complete E-step and M-step update equations for the EM algorithm described in Section 3.3. All expectations are taken with respect to the current parameter values $\Theta^{(t)}$.

E-step: station posteriors. For each station s , compute the posterior probability of being fraud-capable:

$$\gamma_s^{(t)} = \frac{w^{(t)} \cdot \mathcal{L}_s^F(\Theta^{(t)})}{(1 - w^{(t)}) \cdot \mathcal{L}_s^H(\Theta^{(t)}) + w^{(t)} \cdot \mathcal{L}_s^F(\Theta^{(t)})}, \quad (\text{B5})$$

where $\mathcal{L}_s^F$ and $\mathcal{L}_s^H$ are the station-level likelihoods under the fraud-capable and clean components, computed as products of cell-pattern probabilities across vehicle types:

$$\mathcal{L}_s^C = \prod_v \prod_{y \in \{0,1\}^K} [P(Y = y \mid b_s = C, s, v)]^{c_{sv}(y)}, \quad C \in \{H, F\}, \quad (\text{B6})$$

with $P(Y = y \mid b_s = F, s, v)$ given by equation (3) and $P(Y = y \mid b_s = H, s, v)$ equal to the honest component alone.

E-step: inspection posteriors. For each outcome pattern $y \in \{0, 1\}^K$ in cell (s, v) at a fraud-capable station, compute the posterior probability that the inspection is fraudulent:

$$\tau_{sv}^{(t)}(y) = \frac{\pi_{sv}^{(t)} \prod_{j=1}^K \beta_{j,v}^{(t)y_j} (1 - \beta_{j,v}^{(t)})^{1-y_j}}{\pi_{sv}^{(t)} \prod_j \beta_{j,v}^{(t)y_j} (1 - \beta_{j,v}^{(t)})^{1-y_j} + (1 - \pi_{sv}^{(t)}) \prod_j \Lambda_{j,sv}^{(t)y_j} (1 - \Lambda_{j,sv}^{(t)})^{1-y_j}}. \quad (\text{B7})$$

Since inspections sharing the same outcome pattern y share the same posterior, only 2^K posteriors need be computed per cell (32 with $K = 5$), rather than one per inspection.

M-step: mixing weight.

$$\hat{w}^{(t+1)} = \frac{1}{S} \sum_{s=1}^S \gamma_s^{(t)}. \quad (\text{B8})$$

Fraud rates. The expected fraction of fraudulent inspections in cell (s, v) :

$$\hat{\pi}_{sv}^{(t+1)} = \frac{\sum_y c_{sv}(y) \cdot \tau_{sv}^{(t)}(y)}{n_{sv}}. \quad (\text{B9})$$

Detection rates. Among inferred-fraudulent inspections of type v , the fraction triggering indicator j :

$$\hat{\beta}_{j,v}^{(t+1)} = \frac{\sum_s \gamma_s^{(t)} \sum_{y: y_j=1} c_{sv}(y) \cdot \tau_{sv}^{(t)}(y)}{\sum_s \gamma_s^{(t)} \sum_y c_{sv}(y) \cdot \tau_{sv}^{(t)}(y)}. \quad (\text{B10})$$

Clean-component baselines. For each indicator j , update the vehicle-type effects $\alpha_{j,v}$ and station effects $\epsilon_{j,s}$ by fitting a weighted logistic regression:

$$\Lambda_{j,sv}^H = \text{logit}^{-1}(\alpha_{j,v} + \epsilon_{j,s}), \quad (\text{B11})$$

where station s receives weight $(1 - \gamma_s^{(t)}) \cdot n_{sv}$ and the outcome is the marginal flag rate $\hat{p}_{j,sv} = n_{sv}^{-1} \sum_{y: y_j=1} c_{sv}(y)$.

Fraud-component baselines. Under the middle-ground restriction ($\epsilon_{j,s} = 0$ at fraud stations):

$$\Lambda_{j,v}^F = \text{logit}^{-1}(\alpha_{j,v}). \quad (\text{B12})$$

The $\alpha_{j,v}$ parameters are shared across components. In practice, both the clean-component logistic regression and the fraud-component moment condition

$$\hat{\Lambda}_{j,v}^F = \frac{\sum_s \gamma_s^{(t)} \sum_{y:y_j=1} c_{sv}(y) \cdot (1 - \tau_{sv}^{(t)}(y))}{\sum_s \gamma_s^{(t)} \sum_y c_{sv}(y) \cdot (1 - \tau_{sv}^{(t)}(y))} \quad (\text{B13})$$

can be combined in a single weighted regression with appropriate offsets.

Convergence. The log-likelihood $\ell(\Theta^{(t)}) = \sum_s \log[(1-w)\mathcal{L}_s^H + w\mathcal{L}_s^F]$ is monitored at each iteration. EM guarantees monotonic increase. The algorithm stops when $|\ell(\Theta^{(t+1)}) - \ell(\Theta^{(t)})| < 10^{-6}$ or when parameter changes fall below a threshold.

B.5 CMH Test Statistic Construction

The test statistic $\tilde{Z}_s$ is constructed as follows. For each cell (s, v) and indicator pair (j, k) with $j < k$, the sample covariance is:

$$\widehat{\text{Cov}}(Y_j, Y_k \mid s, v) = \frac{1}{n_{sv}} \sum_{i=1}^{n_{sv}} Y_{j,i} Y_{k,i} - \hat{p}_{j,sv} \hat{p}_{k,sv},$$

where $\hat{p}_{j,sv} = n_{sv}^{-1} \sum_i Y_{j,i}$ is the marginal flag rate. The numerator of $\tilde{Z}_s$ is a weighted sum of these covariances across all cells and indicator pairs:

$$\text{Num}_s = \sum_v \sum_{j < k} n_{sv} \widehat{\text{Cov}}(Y_j, Y_k \mid s, v).$$

The denominator normalizes by the expected variance under the null of independence:

$$\text{Den}_s = \sqrt{\sum_v \sum_{j < k} n_{sv} \hat{\sigma}_{j,sv}^2 \hat{\sigma}_{k,sv}^2},$$

where $\hat{\sigma}_{j,sv}^2 = \hat{p}_{j,sv}(1 - \hat{p}_{j,sv})$ is the Bernoulli variance. The test statistic is:

$$\tilde{Z}_s = \frac{\text{Num}_s}{\text{Den}_s} = \frac{\sum_v \sum_{j < k} n_{sv} \widehat{\text{Cov}}(Y_j, Y_k \mid s, v)}{\sqrt{\sum_v \sum_{j < k} n_{sv} \hat{\sigma}_{j,sv}^2 \hat{\sigma}_{k,sv}^2}}. \quad (\text{B14})$$

The weighting by n_{sv} naturally downweights sparse cells where the sample covariance is noisy. Cells where $\hat{p}_{j,sv} \in \{0, 1\}$ contribute zero to both numerator and denominator, so no explicit exclusion of degenerate cells is needed. By the central limit theorem aggregating across cells, $\tilde{Z}_s \rightarrow N(0, 1)$ under the null without requiring a per-cell Gaussian approximation.

C Fraud Detection Monte Carlo Analysis

We evaluate the finite-sample performance of the hybrid CEM–EM estimator via Monte Carlo simulation. Table C1 reports fraud-rate recovery (π RMSE, bias) and station classification (recall, precision, F1) averaged over 50 Monte Carlo draws. A station is classified as fraudulent when its posterior exceeds $\bar{\gamma} = 0.9$, matching the threshold used in the empirical application (Section E.2).

Data-generating process. The baseline DGP draws $S = 300$ stations, of which a fraction $w = 0.25$ are fraudulent ($b_s = 1$), with $V = 10$ vehicle types and $K = 5$ indicators. Cell sizes are drawn from $\text{Poisson}(n)$ with $n = 50$, approximating the median cell size in the DFW estimation sample. Honest baseline rates are generated on the logistic scale as $\Lambda_{j,sv} = \text{logit}^{-1}(\alpha_{jv} + \varepsilon_{js})$, where α_{jv} are indicator-by-vehicle-type intercepts and $\varepsilon_{js} \sim N(0, 0.10)$ are station-level shifts (set to zero at fraud stations under the middle-ground restriction). Detection rates β_{jv} are drawn around base values that range from 0.30 to 0.60 across indicators, with small vehicle-type shifts (± 0.05). At fraud-capable stations, each inspection is fraudulent with probability π_{sv} and honest otherwise; fraudulent inspections draw indicators independently with $\Pr(Y_j=1) = \beta_{jv}$ and honest inspections with $\Pr(Y_j=1) = \Lambda_{j,sv}$. Station-level fraud propensity $\psi_s \sim \text{Beta}(2, 8)$ generates cell-level fraud rates π_{sv} with mean approximately 0.20.

Baseline performance. Under correct specification, the estimator recovers cell-level fraud rates with RMSE 0.039 and near-zero bias. Station classification achieves 99% recall and 94% precision (F1 = 0.96).

Finite sample. Cell size is the binding constraint on estimator performance: RMSE halves from 0.064 at $n = 20$ to 0.019 at $n = 200$. Precision improves from 79% to 99% as cell size grows, reflecting fewer noisy classification decisions at small sample sizes.

Number of indicators. We fix $K_{\text{dgp}} = 5$ indicators in the data-generating process and estimate with $K_{\text{est}} \in \{2, 3, 4, 5\}$, subsetting the available indicators. This tests whether the

estimator degrades gracefully when the researcher uses fewer indicators than are available—the relevant empirical scenario. The move from $K = 2$ (just-identified) to $K = 3$ (overidentified) produces the largest gain: RMSE drops from 0.067 to 0.053 and recall rises from 71% to 91%. With $K = 2$, the estimator exhibits a persistent downward bias (-1.7pp) because noise in $\hat{\Lambda}$ propagates directly through the single identifying covariance moment.

Conditional independence. We introduce a vehicle-level random effect $u_i \sim N(0, 1)$ that shifts all indicator probabilities on the logistic scale:

$$\Pr(Y_j = 1 \mid u_i) = \text{logit}^{-1}(\text{logit}(p_j) + \sigma_u \cdot u_i),$$

where $p_j = \beta_{jv}$ for fraudulent inspections and $p_j = \Lambda_{j,sv}$ for honest inspections, and σ_u governs the severity of the violation. The random effect u_i is drawn once per inspection and shared across all K indicators, generating positive cross-indicator covariance at both clean and fraud stations. This captures unobserved within-type vehicle heterogeneity—a vehicle that is borderline on one emissions indicator is likely borderline on others because combustion quality is common across pollutants. At $\sigma_u = 0.50$, precision drops modestly to 89%. At $\sigma_u \geq 1.0$, precision collapses to approximately 25%—the estimator classifies most clean stations as fraudulent—while recall remains at 100%. The bias direction is upward: spurious covariance at clean stations mimics the fraud signal, causing the estimator to overestimate fraud.

Identification frontier. The estimator’s ability to detect fraud depends on the detection gap $\beta_{jv} - \Lambda_{jv}$. We parameterize this gap with a scaling factor $\kappa \in [0, 1]$ that shrinks the detection rates toward the honest baseline:

$$\beta_{jv}^* = \Lambda_{jv} + \kappa \cdot (\beta_{jv} - \Lambda_{jv}),$$

where $\Lambda_{jv} = \text{logit}^{-1}(\alpha_{jv})$ is the baseline absent station-specific shifts, $\kappa = 1$ yields the baseline detection rates, and $\kappa = 0$ makes fraud inspections identical to clean (Figure C1). Panel (a) varies κ for a single fraud technology. Recall degrades smoothly: 92% at $\kappa = 0.5$, 50% at $\kappa = 0.2$, and below 5% as $\kappa \rightarrow 0$. Panel (b) mixes obvious ($\kappa = 1$) and stealth ($\kappa \approx 0$) technologies across stations, varying the fraction of stations that use the obvious technology. Even when 75% of fraud stations use stealth methods, the remaining obvious-fraud stations anchor $\hat{\beta}$ and the estimator achieves 47% recall overall. When fraud inspections are statistically indistinguishable from clean inspections, the identifying covariance moment $\text{Cov}(Y_j, Y_k \mid s, v) \propto \pi(\beta_j - \Lambda_j)(\beta_k - \Lambda_k)$ vanishes regardless of estimation method.

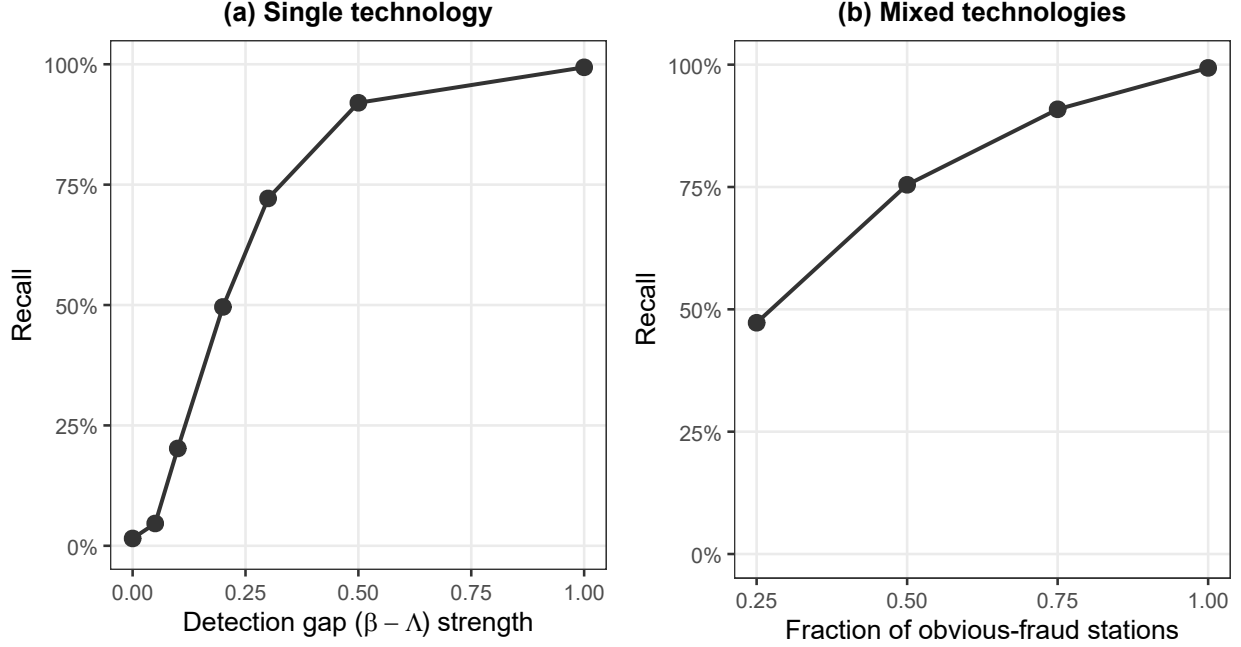


Figure C1. Identification Frontier

Note: Panel (a) varies a single fraud technology, scaling the detection gap $\beta - \Lambda$ from 0 (fraud inspections identical to clean) to 1 (baseline detection rates). Panel (b) mixes obvious ($\beta \gg \Lambda$) and stealth ($\beta \approx \Lambda$) fraud technologies across stations, varying the fraction of stations that use the obvious technology. Both panels report recall (fraction of true fraud stations correctly classified) averaged over 50 Monte Carlo draws.

Additional robustness. The estimator is robust to several other forms of misspecification. *Endogenous sorting* ($\zeta = 2.0$): vehicle-type shares at fraud stations are reweighted by $\exp(\zeta \cdot (\phi_v - \bar{\phi}))$, where ϕ_v is the vehicle type's fraud propensity, so that high-propensity types disproportionately sort to fraud stations. This has negligible effect on $\hat{\pi}$ (RMSE 0.048). *Station-specific detection rates* ($\sigma_\beta = 0.20$): each fraud station draws $\beta_{jv}^s = \beta_{jv} + \eta_s + \xi_{js}$, where $\eta_s, \xi_{js} \sim N(0, \sigma_\beta/2)$, introducing cross-station heterogeneity in detection technology. RMSE increases modestly to 0.042. *Contaminated clean component*: a fraction (50%) of nominally clean stations conduct low-level fraud ($\pi_{sv} \leq 0.08$), blurring the clean/fraud boundary. Fraud-rate estimates are nearly unaffected (RMSE 0.044, zero bias), though recall drops to 94% as contaminated stations are classified as clean. *Within-fraud indicator correlation* ($\rho = 0.50$): among fraudulent inspections, a shared uniform draw determines all indicator outcomes through correlated thresholds, inducing positive within-inspection dependence. This introduces modest downward bias (-2.1pp).

Taken together, these results indicate that the estimator performs well under conditions calibrated to the empirical application, with the detection gap $\beta - \Lambda$ as the binding constraint on identification power.

Table C1. Monte Carlo Results for the Fraud Identification Estimator

		Fraud rates		Classification		
	Variation	π RMSE	π Bias	Recall	Precision	F1
<i>Baseline</i>	$K=5, S=300, n=50, \pi \approx 0.20$	0.039	-0.000	99.3	93.6	96.3
<i>Sample size</i>	$n = 20$	0.064	+0.001	96.9	78.5	86.1
	$n = 50$	0.039	-0.000	99.3	93.6	96.3
	$n = 100$	0.027	-0.000	99.8	96.7	98.2
	$n = 200$	0.019	-0.000	99.8	99.0	99.4
<i>Indicators</i>	$K = 2$	0.067	-0.017	70.6	86.2	76.8
	$K = 3$	0.053	-0.004	90.6	95.9	93.1
	$K = 4$	0.042	-0.000	98.2	91.7	94.8
	$K = 5$	0.039	+0.000	99.2	93.6	96.3
<i>CI violation</i>	$\sigma_u = 0$ (baseline)	0.038	+0.000	99.3	93.1	96.1
	$\sigma_u = 0.25$	0.039	-0.001	99.4	93.3	96.2
	$\sigma_u = 0.50$	0.039	-0.003	99.5	89.0	93.9
	$\sigma_u = 1.00$	0.046	+0.001	100.0	27.5	43.0
	$\sigma_u = 2.00$	0.107	+0.070	100.0	24.7	39.6
<i>Robustness</i>	Sorting ($\zeta = 2.0$)	0.048	+0.001	99.7	94.2	96.9
	Station β ($\sigma_\beta = 0.20$)	0.042	-0.003	98.1	94.0	96.0
	Contaminated clean (50%, $\pi \leq 0.08$)	0.044	+0.000	93.9	92.5	92.9
	Within-fraud correlation ($\rho = 0.50$)	0.059	-0.021	99.7	99.3	99.5

Note: Baseline configuration: $K=5$ indicators, $S=300$ stations, $n=50$ inspections per cell, 25% fraud prevalence ($\pi \approx 0.20$). Recall and precision are percentages. A station is classified as fraudulent when $\hat{\gamma}_s > 0.9$. All results are averaged over 50 Monte Carlo draws.

D Fraud Indicator Selection, Construction, and Validation

We construct fraud indicators based on data abnormalities documented in the 2024 ERG emissions inspection audit report (Eastern Research Group, 2024a) and through consultation with the Texas Department of Public Safety and relevant law enforcement officials. This appendix provides construction details for all indicators included in the analysis, documents the complete candidate set, presents validation evidence, and discusses threshold selection for volume- and time-based measures. Indicator construction rules are defined based on the full Texas emissions testing dataset; all validation analyses—cross-correlation, concentration, threshold calibration, and robustness across indicator combinations—use the nine-county Dallas–Fort Worth (DFW) sample of 2022 passing tests (Collin, Dallas, Denton, Ellis, Johnson, Kaufman, Parker, Rockwall, and Tarrant counties). We focus on DFW because conversations with Texas DPS officials and contemporaneous media reporting identify the metropolitan area as a high-density fraud market during this period, making it

the most informative setting for validating indicator performance and calibrating estimation parameters.

D.1 Indicator Construction Details

This section provides construction details for each indicator included in Table D1, organized by measurement group.

Group 1: Throughput and Timing. *Rapid Succession.* We compute the time gap between consecutive tests on the same analyzer by ordering tests by `TEST_END_TIME` within each analyzer-day. A test is flagged when: (1) the gap from the previous test is less than 5 minutes, (2) the current test results in a pass, and (3) the current and previous tests have different VINs. The different-VIN requirement distinguishes rapid succession from quick retests of the same vehicle. We evaluate thresholds of 3, 5, and 7 minutes; Section D.6 documents the threshold selection analysis.

High Volume. We compute daily test counts at the analyzer level (identified by `ANALYZER_ID`), not the station level, because a single station may operate multiple analyzers. A test is flagged when the analyzer processed more than 100 tests on that calendar day and the focal test resulted in a pass. We chose the analyzer rather than the station as the counting unit because fraudulent operations may concentrate activity on a single machine while maintaining lower volumes on others. We evaluate thresholds of 100, 150, and 200 tests per day; Section D.6 documents the threshold selection analysis.

Group 2: eVIN Identity. *eVIN Absent.* We flag a test as missing eVIN when the vehicle passes, has model year 2005 or later, and the `OBD_VIN` field is either empty or contains only whitespace. Federal regulations (40 CFR 86.1806-05) require model year 2005+ vehicles to report the VIN through the OBD-II diagnostic link connector. We clean the raw `OBD_VIN` field by trimming whitespace and converting empty strings to missing values before comparison. The model year restriction avoids flagging pre-2005 vehicles, which are not required to report eVIN and may legitimately return empty values.

eVIN-VIN Mismatch. We flag a mismatch when the cleaned `OBD_VIN` differs from the cleaned inspector-entered `VIN`, conditional on both fields being non-empty and the test resulting in a pass. String cleaning includes uppercase conversion, removal of leading/trailing whitespace, and standardization of common transcription errors (e.g., “O” vs “0”). A mismatch indicates the analyzer communicated with a vehicle other than the one identified by the inspector. In the analysis, eVIN Absent and eVIN-VIN Mismatch are combined via logical OR into a single composite indicator (`evin_anomaly`).

Group 3: OBD Data Recycling. *Profile Double Match.* We construct an electronic OBD profile for each test as a composite of key diagnostic fields that together identify a vehicle’s powertrain control module (PCM) configuration: `PCM_ID`, `CAL_ID`, `CVN`, and `PID_COUNT`. A test is flagged when this profile exactly matches the immediately preceding test on the same analyzer that involved a different VIN. Two consecutive tests sharing an identical PCM configuration for nominally distinct vehicles indicates that the same physical device—either a surrogate vehicle or an OBD simulator—was connected to the analyzer for both tests.

PID Triple Match. We flag a test when the `PID_COUNT` field matches the previous two consecutive tests on the same analyzer, but all three tests report different VINs. The `PID_COUNT` field reports the number of supported OBD parameter identifiers, which varies across vehicle makes, models, and ECM configurations and takes a wide range of integer values in the inspection population. The triple-match condition is considerably more restrictive than a pairwise match: even if a particular `PID_COUNT` value is common in the population, three consecutive matches at the same analyzer across nominally different VINs is unlikely under random vehicle arrival. The indicator is weakest when the matching value is very common (e.g., corresponding to a high-volume make such as Chevrolet Silverado), but robust to this concern because the mixture model conditions on vehicle type, so common values are evaluated against a vehicle-type-specific baseline rather than the pooled distribution.

Group 4: Operational Practices. *Not Barcoded.* We flag a test when the VIN is manually entered by the inspector rather than populated automatically via barcode scanner. The TIMS data record how the VIN was entered for each inspection. Barcode scanning reads the VIN directly from the vehicle’s physical label, making substitution impossible; manual entry allows the inspector to type any 17-character string. Approximately 39% of passing tests have manually entered VINs, reflecting the prevalence of older manual-entry equipment at established stations rather than widespread fraud.

Outside Hours. We extract the hour from `TEST_END_TIME` and flag passing tests conducted before 6:00 AM or at or after 9:00 PM. These cutoffs reflect typical station operating hours; tests outside this window occur at elevated rates among otherwise-flagged tests. The indicator is not individually definitive—some stations legitimately operate extended hours—but provides a corroborating operational signal.

D.2 Other Candidate Indicators

We constructed additional candidate indicators beyond those assigned to the four measurement groups. An indicator was assigned to a group only if it satisfies two criteria: (1) *sufficient incidence*—the indicator flags at least 0.5% of all passing tests statewide, provid-

ing enough flagged observations for station-level classification; and (2) *measurement independence*—the indicator’s measurement procedure does not share inputs or contexts with an existing group that would produce mechanical correlation conditional on fraud status. An indicator can detect the same type of fraud as an existing group and still qualify for a separate group, provided its measurement channel is sufficiently independent. The 0.5% threshold is motivated by the minimum cell-size requirement ($n \geq 5$ flagged observations per station–vehicle-type cell) imposed during mixture model estimation. At typical DFW station sizes (approximately 2,000–3,000 passing tests per year) and with fraud-station flag rates that are substantially elevated above the statewide average—a pattern confirmed by the concentration analysis in Section D.5—a 0.5% statewide prevalence translates into enough flagged observations at fraud stations to satisfy this threshold across most vehicle-type cells. Table D1 summarizes all group-assigned indicators; the candidates that did not qualify for group assignment are described below.

MIL Mismatch flags tests where the malfunction indicator lamp (MIL) is commanded ON with stored diagnostic trouble codes but the inspector records the MIL as OFF during the key-on engine-running visual check. The indicator is an unambiguous signal of inspector misreporting but flags fewer than 0.1% of passing tests statewide, limiting its utility for station-level classification.

Excessive DTC flags passing tests with more than one stored diagnostic trouble code (model year 2001+), and *Excessive Readiness* flags passing tests with two or more unready OBD monitors (model year 2001+). Both indicators arise from legitimate causes—transient sensor faults, interrupted drive cycles, recent battery disconnection—as well as from fraud, making their interpretation ambiguous.

Short Time Retest, *LD-HD Switch*, *4th+ Test Pass*, and *Masked Repair* are defined only for vehicles with multiple test attempts within an inspection cycle. This scope restriction limits the applicable sample to a minority of all passing tests and introduces selection effects that complicate mixture model estimation.

Round Odometer flags passing tests where the inspector-recorded odometer reading is divisible by 1,000. Fabricated data tend to cluster on round numbers; however, this indicator’s statewide incidence falls below our threshold and its distribution across stations is diffuse, reflecting measurement rounding at established stations rather than systematic fraud at specific locations.

Two ERG-defined composite flags—*Electronic Profile Change* and *Readiness Profile Change*—capture changes in the OBD electronic profile or readiness monitor configuration between a prior failure and the subsequent pass. These retest-only indicators apply to a minority of inspections and are conceptually subsumed by the Group 3 indicators when evaluated across

Table D1. Complete Catalog of Fraud Indicators

Indicator	Description	Incidence (%)
Group 1 — Throughput / timing		
Rapid Succession	Consecutive tests on same analyzer fewer than 5 minutes apart, on distinct VINs	2.86
<i>3-min variant</i>	Stricter variant: same as Rapid Succession with a 3-minute threshold	1.05
<i>7-min variant</i>	Looser variant: same as Rapid Succession with a 7-minute threshold	6.34
High Volume	Analyzer performs more than 100 tests in a single calendar day	1.86
<i>>150 variant</i>	Stricter variant: same as High Volume with a 150-test-per-day threshold	1.37
<i>>200 variant</i>	Stricter variant: same as High Volume with a 200-test-per-day threshold	1.12
Group 2 — eVIN identity		
eVIN Absent	OBD system reports no vehicle identifier on a model-year 2005+ vehicle	6.75
eVIN–VIN Mismatch	Electronically reported VIN differs from inspector-entered VIN on a 2005+ vehicle	0.72
Group 3 — OBD data recycling		
PID Triple Match	OBD parameter identifier (PID) count repeats across 3 consecutive tests on distinct VINs; alternative to Profile Double Match	7.99
Profile Match	Double Full electronic OBD profile repeats across 2 consecutive tests on distinct VINs	8.29
Group 4 — Operational practices		
Not Barcoded	VIN manually entered by inspector rather than scanned from vehicle barcode; high baseline (39%) reflects manual-entry equipment at older stations	39.34
Outside Hours	Test completed before 6 am or after 9 pm	0.71
Group 5 — OBD communication protocol		
Comm. Prot. Mismatch	Observed OBD communication protocol does not match expected protocol for vehicle make, model, and year; incidence computed for 2016–2022 only (field unreliable in 2023–2024)	9.26*

Notes: Incidence is the share of all passing tests (statewide, 2016–2024) flagged by each indicator. Groups are defined by measurement dependence conditional on fraud status: indicators within the same group share a measurement input or context that produces mechanical correlation even after conditioning on fraud, and are therefore never combined in the same specification. **Bold** indicators are used in the preferred specification. Non-indented indicators without bold are alternative candidates evaluated in the robustness analysis (Table D3). Indented *italic* rows are threshold variants or duplicates of the reference indicator in the same group and are not separately evaluated. Group 2 exception: both eVIN Absent and eVIN–VIN Mismatch are combined into a single composite indicator (`evin_anomaly`) used in all specifications. * Group 5 incidence computed for 2016–2022 only; the communication protocol field is unreliable in 2023–2024. Indicators not assigned to a group are discussed in Section D.2.

all passing tests.

D.3 Five-Group Specification Logic

The preferred specification combines exactly one indicator from each of the four measurement groups. This design is motivated by the conditional independence assumption that underlies mixture model identification: conditional on a station’s fraud status, the K indicators must be mutually independent. The relevant grouping criterion is *measurement independence conditional on fraud status*—not whether two indicators detect the same type of fraud. Detecting the same underlying fraud act through different measurement channels is not only acceptable but desirable, because cross-group co-movement at fraud stations is precisely the source of identification power. What disqualifies two indicators from belonging to separate groups is a shared measurement input or shared measurement context that produces mechanical correlation even after conditioning on the binary fraud choice.

Profile Double Match and PID Triple Match belong to the same group not because both detect OBD data recycling—that is a feature, not a problem—but because both incorporate PID_COUNT as a direct measurement input. A device with a fixed PID_COUNT will simultaneously trigger both indicators whenever it is substituted, creating a correlation that persists conditional on fraud status. Using both in the same specification would therefore violate conditional independence through a shared measurement field, not through a shared fraud mechanism.

The four groups are organized by measurement channel rather than by fraud mechanism. Group 1 (throughput and timing) measures the operational intensity of fraud through counts and timing of tests on the same analyzer. Group 2 (eVIN identity) measures device identity spoofing through the OBD-II electronic VIN fields. Group 3 (OBD data recycling) measures profile reuse through the diagnostic fingerprint fields of consecutive tests. Group 4 (operational practices) measures enabling conditions through administrative records of how tests are recorded. Groups 2 and 3 both capture consequences of device substitution fraud, but through independent measurement channels: Group 2 detects that the connected device reports the wrong VIN, while Group 3 detects that it reports the same diagnostic profile as a prior test.

Because each group measures fraud through a distinct channel, a station engaged in systematic device substitution will generate signals across multiple groups simultaneously. This cross-group co-movement is the source of identification power. Within each group, the choice among candidate indicators is validated by the robustness analysis in Section D.7, which shows that the key economic estimates are insensitive to which indicator is selected within a group.

Before proceeding to estimation, we validate the measurement-channel grouping using two complementary analyses. First, the cross-correlation analysis in Section D.4 examines the station-level correlation structure across all indicator pairs, confirming that within-group pairs share measurement inputs (Groups 1 and 3) or are combined into a composite (Group 2), while cross-group correlations reflect common fraud causation rather than shared measurement. Second, the concentration analysis in Section D.5 examines whether each retained indicator is concentrated among a small subset of stations—consistent with systematic fraud—rather than diffusely distributed across the population.

D.4 Cross-Correlation Analysis

Figure D1 displays the pairwise Pearson correlation matrix for the eight Table D1 indicators, computed at the station level using 2024 passing tests. Indicators are ordered by measurement group (G1–G4) so that within-group pairs appear on the 2×2 diagonal blocks.

The heatmap reveals the expected group structure for Groups 1, 3, and 4. The Group 3 block—Profile Double Match and PID Triple Match—exhibits the strongest within-group correlation in the matrix ($r = 0.98$), consistent with both indicators responding to the same fraud act: a physical device (surrogate vehicle or OBD simulator) remaining connected to the analyzer across consecutive tests. The Group 1 block—Rapid Succession and High Volume—shows similarly strong within-group correlation ($r = 0.97$), reflecting that both throughput indicators co-occur at the same high-volume fraud operations. The Group 2 block is an exception: the station-level correlation between eVIN Absent and eVIN Mismatch is near zero ($r \approx 0$), because the two indicators capture complementary failure modes—the connected device either fails to report an eVIN at all, or reports one that disagrees with the inspector-entered VIN—and a single device will typically produce one or the other, not both simultaneously. This near-orthogonality within Group 2 motivates combining the two components into a single composite indicator (eVIN Anomaly = Absent OR Mismatch) rather than treating them as a within-group pair.

The most prominent cross-group correlations link Group 1 (throughput) to Group 3 (OBD data recycling): High Volume correlates with PID Triple Match at $r = 0.85$ and with Profile Double Match at $r = 0.83$. Group 2–Group 3 cross-block correlations are also elevated: eVIN Absent correlates with Profile Double Match at $r = 0.71$ and with PID Triple Match at $r = 0.70$. Both patterns are expected—a station cycling physical devices through the analyzer generates simultaneous throughput anomalies, identity spoofing, and data recycling signals. The cross-group co-movement reflects different measurement channels observing the same underlying fraud event—not a shared data-generating process—and is precisely the source of identification power in the mixture model.

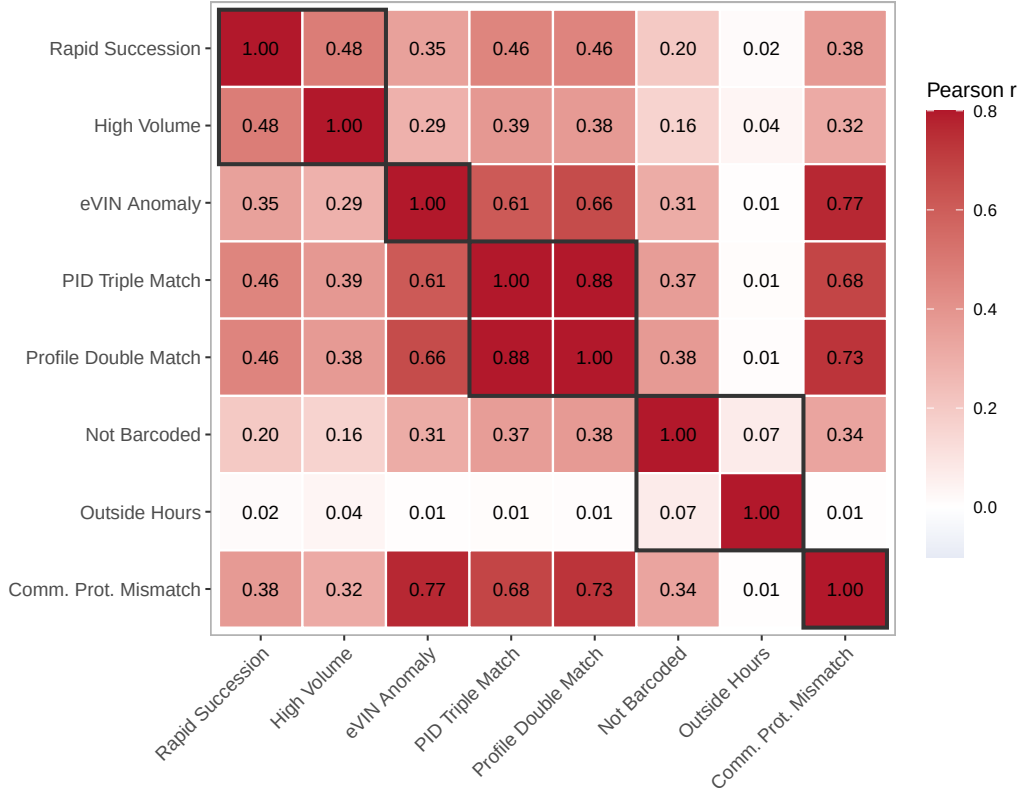


Figure D1. Cross-Correlation Matrix of Fraud Indicators

Note: Pairwise Pearson correlation coefficients between fraud indicator rates at the station level (2024, passing tests, stations with ≥ 100 tests). Indicators are ordered by measurement group (G1–G4) so that within-group pairs appear on the 2×2 diagonal blocks. The Group 3 within-block (Profile Double Match, PID Triple Match) shows the strongest correlation ($r = 0.98$), consistent with both indicators detecting the same OBD data-recycling mechanism. The Group 1 within-block (Rapid Succession, High Volume) shows strong correlation ($r = 0.97$). The Group 2 within-block (eVIN Absent, eVIN Mismatch) is near zero ($r \approx 0$) because the two components are complementary failure modes that rarely co-occur; in the analysis they are combined via logical OR into a single composite indicator. Elevated cross-block correlations between Group 1 and Group 3 ($r \approx 0.83$ – 0.85) reflect co-occurring fraud mechanisms, not a shared data-generating process.

Group 4 indicators (Not Barcoded and Outside Hours) show comparatively weak cross-group correlations, consistent with their role as enabling conditions and operational signals rather than direct mechanical signatures of data fabrication.

D.5 Concentration Analysis

Figure D2 presents Lorenz curves for all eight Table D1 indicators, computed at the station level using 2024 passing tests. Panels are ordered by measurement group (G1–G4, left to right), and curves are colored by specification role: blue for preferred-specification indicators and orange for alternative candidates.

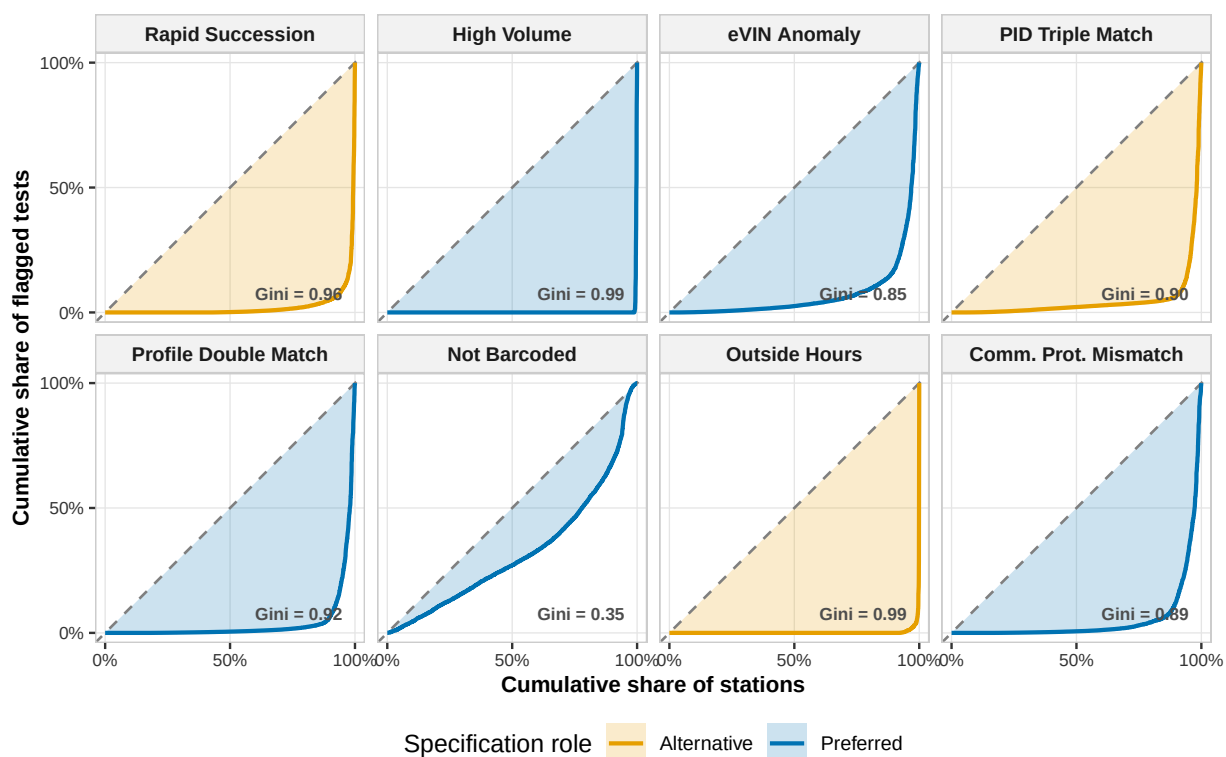


Figure D2. Lorenz Curves for Fraud Indicators

Note: Faceted Lorenz curves showing the cumulative distribution of flagged tests across stations, with stations ranked by flag rate (lowest to highest). The dashed diagonal represents perfect equality; curves bowing toward the upper-left indicate concentration among a small number of high-rate stations. Gini coefficients are reported in each panel. Blue curves are indicators in the preferred specification; orange curves are alternative candidates evaluated in the robustness analysis. Panels are ordered by measurement group (G1–G4). Sample: 2024 passing tests, stations with ≥ 100 inspections.

Six of the eight indicators are highly concentrated ($\text{Gini} \geq 0.93$): Outside Hours (0.99), High Volume (0.99), Rapid Succession (0.98), Profile Double Match (0.96), PID Triple Match (0.94), and eVIN Absent (0.93). For these indicators, a small fraction of stations accounts for the vast majority of flagged tests, consistent with systematic fraud operations at specific locations rather than diffuse opportunistic behavior.

The two exceptions are eVIN–VIN Mismatch ($\text{Gini} = 0.81$) and Not Barcoded ($\text{Gini} = 0.31$). The moderate concentration of eVIN–VIN Mismatch reflects that identifier discrepancies can arise from transcription errors as well as fraud, producing a longer tail of occasional flags at otherwise-clean stations. The low concentration of Not Barcoded reflects the equipment-age explanation noted in Table D1: stations with older manual-entry equipment flag every test regardless of fraud, so the indicator is widespread by construction.

rather than by fraud incidence. The low Gini for Not Barcoded is therefore not a validity concern—it is a feature of the indicator’s mechanism.

D.6 Threshold Selection: Rapid Succession and High-Volume Analyzer

Two core indicators—Rapid Succession and High-Volume Analyzer—require threshold selection. We considered multiple candidate thresholds for each and selected values that balance coverage (flagging sufficient tests for statistical power) against signal strength (maintaining high rates of corroborating fraud signals among flagged tests).

Figure D3 summarizes our threshold analysis using two corroborating signals—eVIN Absent and VIN Not Barcoded—which are independently elevated among fraudulent tests and serve as natural benchmarks for threshold calibration. We note that eVIN Absent also appears in the preferred specification; however, the benchmark role here is based solely on raw flag rates within each threshold band, computed prior to any mixture model estimation, so the calibration does not depend on the specification’s classification outputs. Panel (a) shows signal rates by time gap between consecutive tests on the same analyzer. Panel (b) shows the analogous analysis by daily analyzer volume.

Rapid Succession. We considered thresholds of 3, 5, and 7 minutes. Tests in the 0–3 minute band show the strongest fraud signal (74% eVIN Absent), flagging 535,000 tests. Signal drops sharply in the 3–5 minute band (16% eVIN Absent) and is near baseline in the 5–7 minute band (4% eVIN Absent, versus a baseline of 1.3%). We select 5 minutes as the threshold, retaining the high-signal 0–3 minute core together with the marginal 3–5 minute tests that still exceed baseline, for a total of 759,000 flagged tests.

High-Volume Analyzer. We considered thresholds of 75, 100, 150, and 200 tests per day. Tests at analyzers exceeding 100 per day show substantially elevated eVIN Absent rates (63% cumulative across all >100/day tests), consistent with systematic fraud. The 75–100 tests per day band shows a substantially weaker signal (49% eVIN Absent), indicating a mix of fraudulent and high-throughput legitimate operations. We therefore select 100 tests per day as our threshold. Tightening to 150 or 200 per day yields only modest gains in signal purity (64–67%) at the cost of reduced coverage.

Table D2 summarizes the threshold analysis.

D.7 Robustness Across Indicator Combinations

Our identification strategy requires selecting exactly one indicator from each of five measurement categories defined in Section D.1: throughput (G1), eVIN identity (G2), OBD reading

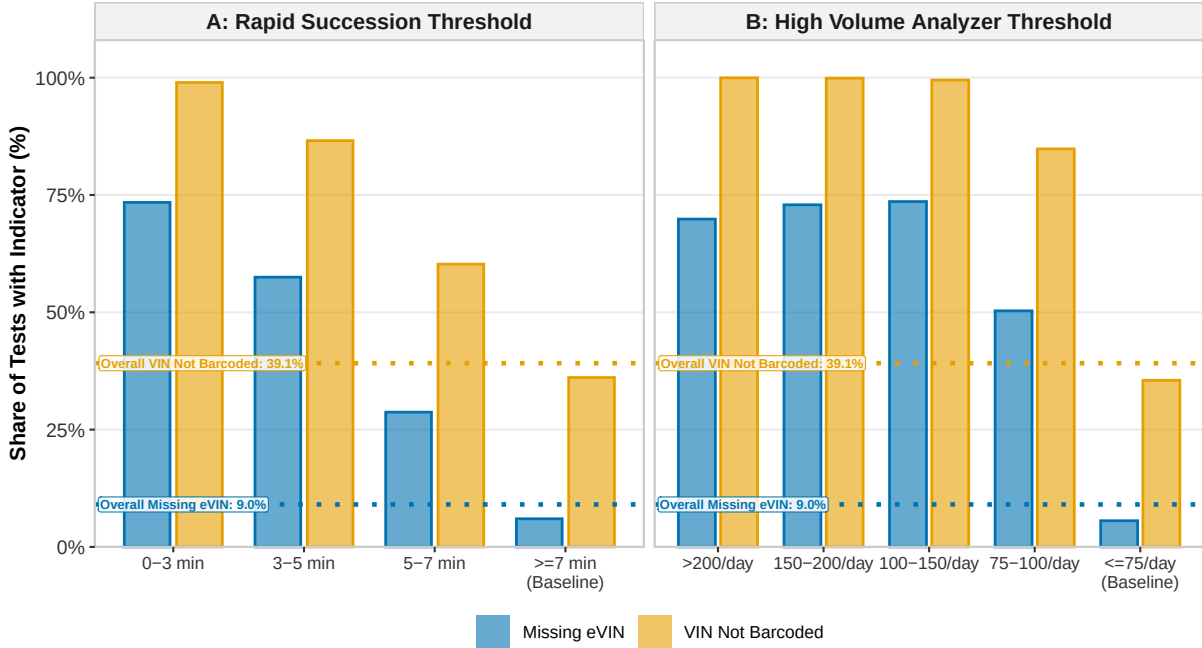


Figure D3. Threshold Selection for Rapid Succession and High-Volume Analyzer

Note: Bars show eVIN Absent rate (blue) and VIN Not Barcoded rate (orange) within each threshold band. Dotted reference lines show overall sample baseline rates. Panel (a): Signal is highest in the 0–3 minute band (74% eVIN Absent) and falls sharply in the 3–5 minute band (16%) and 5–7 minute band (4%, near the baseline of 1.3%). We select 5 minutes as the threshold to retain the high-signal 0–3 minute core together with the marginal 3–5 minute tests that still exceed baseline. Panel (b): Tests at analyzers exceeding the 100/day threshold show substantially stronger eVIN Absent rates (63% cumulative) than the 75–100 tests per day band (49%), while the baseline below 75/day is 1.2%. We select 100 tests per day as the threshold. Sample: 2024 passing tests.

Table D2. Summary of Threshold Selection

Indicator	Threshold	N Flagged	Missing eVIN	Corr.	Sel.
Rapid Succession	3 min	98,000	73%	0.33	
Rapid Succession	5 min	421,000	61%	0.41	✓
Rapid Succession	7 min	923,000	44%	0.52	
High-Volume Analyzer	75/day	779,000	56%	0.41	
High-Volume Analyzer	100/day	197,000	73%	0.31	✓
High-Volume Analyzer	150/day	56,000	72%	0.29	
High-Volume Analyzer	200/day	10,000	70%	0.21	

Note: N Flagged is the number of passing tests flagged at each threshold (2022). Missing eVIN is the share of flagged tests with missing electronic VIN. Correlation is the mean Pearson correlation with other core fraud indicators at the station level (stations with ≥ 100 tests). Selected thresholds balance coverage, signal strength, and correlation with other indicators.

pattern (G3), documentation (G4), and OBD communication protocol (G5). Two candidate indicators are available for each of G1, G3, and G4, while G2 is always *eVIN Anomaly* and G5 is always *Comm. Protocol Mismatch*, yielding eight cross-category combinations in total. We evaluate all eight and report two key outcomes: the number of high-intensity fraud stations and the overall inspection fraud rate.

Table D3 presents results for all eight combinations, ordered by G1 choice (Rapid Succession first, High Volume second), then G3 choice (PID Triple first, Profile Double second), then G4 choice (Not Barcoded first, Outside Hours second). The preferred combination used in the main analysis is shown in bold.

Table D3. Fraud Estimates Across All $K = 5$ Indicator Combinations

Throughput (G1)	OBD pattern (G3)	Documentation (G4)	High-intensity stations	Inspection fraud (%)
Rapid succ.	PID triple	Not barcoded	201	14.7
Rapid succ.	PID triple	Outside hrs.	203	14.7
Rapid succ.	Profile dbl.	Not barcoded	214	14.9
Rapid succ.	Profile dbl.	Outside hrs.	213	14.9
High vol.	PID triple	Not barcoded	205	14.7
High vol.	PID triple	Outside hrs.	204	14.7
High vol.	Profile dbl.	Not barcoded	215	14.9
High vol.	Profile dbl.	Outside hrs.	213	14.9

Note: All combinations include *eVIN Anomaly* (G2) and *Comm. Protocol Mismatch* (G5). Preferred combination (bold) is used in the main analysis. High-intensity stations: number of stations with estimated station-level fraud rate $\hat{\pi}_s > 25\%$. Inspection fraud rate: $\hat{\pi}$ -weighted share of inspections at stations classified as fraudulent ($\hat{\gamma} > 0.9$).

Both metrics are stable within each G1 (throughput) group. Combinations using *High Volume* identify 204–215 high-intensity fraud stations and yield inspection fraud rates of 14.7–14.9 %, while those using *Rapid Succession* identify 201–214 high-intensity stations with fraud rates of 14.7–14.9 %. This difference reflects the throughput indicator’s role in defining the fraud/clean boundary—*Rapid Succession* is more prevalent and pulls more marginal stations into the fraud component. Within each G1 group, both the high-intensity station count and the fraud rate are insensitive to the G3 and G4 choices.

The preferred specification uses *High Volume* as G1 and *Profile Double Match* as G3. *Profile Double Match*—requiring identical PCM configuration, calibration ID, CVN, and PID count across consecutive distinct-VIN tests—is more directly tied to the OBD simulator patterns documented in the ERG report (Eastern Research Group, 2024a) and provides greater specificity than the PID count alone. The stability of both metrics across indicator combinations within each G1 group supports the credibility of our identification approach.

E Fraud Identification: Robustness and Implementation

This appendix examines robustness across key design choices in the fraud identification model. All analyses in this section use the nine-county DFW 2022 sample described in Section D. The baseline specification uses $K = 5$ indicators from five measurement groups (Section D.3), the hybrid CEM–EM estimator described in Section 3.3, and the preferred indicator combination identified in Section D.7. Here we examine sensitivity to three implementation choices that interact with identification: the discretization of vehicle types, the classification threshold γ for fraud stations, and the minimum cell and station size thresholds that govern sample trimming.

E.1 Vehicle Type Discretization

The mixture model conditions on vehicle type v when decomposing station-level indicator rates into fraud and clean components. The definition of vehicle types therefore affects both the precision of the clean-component baseline Λ_{jv} and the validity of the conditional independence assumption. We evaluate 19 specifications that vary along two dimensions: the model-year (MY) bin width (make only, 5-year, 3-year, 2-year, or annual) and the make dimension (none, top-5, top-10, or top-15 makes by volume, with remaining makes pooled as “Other”).

Table E1 reports two key outcomes for each specification: the share of stations classified as high-intensity fraud (estimated station-level fraud rate $\hat{\pi}_s > 25\%$) and the overall inspection fraud rate. The high-intensity share is computed as a fraction of all stations in each specification’s estimation sample, which varies modestly across specifications as finer discretizations drop small cells.

The key finding is that the fraud estimates are remarkably stable across all 19 specifications. The inspection fraud rate ranges from 14.7 % to 15.6 %, and the number of high-intensity stations ($\hat{\pi}_s > 25\%$) ranges from 188 to 225. Neither the inclusion of the make dimension nor the granularity of model-year bins produces economically meaningful variation in the estimated fraud rate.

Two subsidiary patterns are worth noting. First, the make dimension has little effect on aggregate fraud estimates: MY-only specifications recover inspection fraud rates of 14.9 % and 215–225 high-intensity stations, while make-inclusive specifications span 14.7–15.6 % and 188–225 high-intensity stations. This near-equivalence suggests that the nine-county DFW sample is large enough that make-level heterogeneity in the clean component averages out across stations, so conditioning on make provides minimal additional identifying variation at the aggregate level.

Table E1. Fraud Estimates Across Vehicle Type Specifications

MY bins	Make dim.	V	Stations	Inspections	High-intensity stations (%)	Inspection fraud (%)
Make only	Top 5	6	2,487	5,139,983	9.0	14.8
	Top 10	11	2,479	5,137,167	9.0	14.7
	Top 15	16	2,476	5,132,617	9.0	14.8
5-year	—	6	2,491	5,138,243	9.0	14.9
	Top 5	36	2,459	5,103,191	9.0	14.9
	Top 10	65	2,446	5,057,517	8.8	15.0
	Top 15	95	2,435	5,010,995	8.7	15.1
3-year	—	10	2,486	5,136,721	9.0	14.9
	Top 5	55	2,444	5,072,900	8.9	14.8
	Top 10	100	2,416	4,990,399	8.8	15.0
	Top 15	144	2,391	4,911,486	8.7	15.1
2-year	—	14	2,478	5,133,987	9.0	14.9
	Top 5	84	2,421	5,035,562	8.7	14.9
	Top 10	150	2,386	4,913,585	8.7	15.1
	Top 15	214	2,347	4,802,661	8.8	15.3
Annual	—	28	2,457	5,121,672	8.8	14.9
	Top 5	159	2,365	4,894,477	8.6	15.1
	Top 10	284	2,304	4,668,082	8.4	15.4
	Top 15	407	2,255	4,483,639	8.3	15.6

Note: Each row is a separate estimation of the $K = 5$ preferred indicator specification using the hybrid CEM–EM estimator. MY bins: width of model-year bins used to define vehicle types. “Make only” uses make with no MY dimension. Make dim.: number of top makes included (by inspection volume); remaining makes pooled as “Other.” Dash indicates no make dimension. V : number of vehicle types in the resulting discretization. Stations and Inspections: estimation sample size after dropping station–vehicle-type cells with fewer than 5 observations and stations with fewer than 30 total inspections. High-intensity stations: share of stations with estimated station-level fraud rate $\hat{\pi}_s > 25\%$. Inspection fraud rate: $\hat{\pi}$ -weighted share of inspections at stations classified as fraudulent ($\hat{\gamma}_s > 0.9$).

Second, finer discretizations come at the cost of smaller estimation samples, as station–vehicle-type cells that fall below the minimum cell-size threshold are dropped. The coarsest specifications retain nearly the full sample (2,491 stations), while the finest (annual $\times$ top-15, $V = 407$) drops roughly 9 % of stations. Despite this sample reduction, the fraud estimates remain stable, indicating that the dropped cells are predominantly small, low-fraud vehicle types that contribute little to the identification of fraud stations. Finer make-inclusive specifications do push the IFR slightly upward (from 14.7 % to 15.6 %), likely because the additional conditioning reveals fraud signal in cells that would otherwise be diluted by pooling across makes.

Given this evidence, the choice of vehicle type specification has minimal impact on the key economic estimates. We adopt 3-year model-year bins $\times$ top-10 makes as our preferred specification, which balances cell-level precision against sample-size requirements: it retains 2,416 stations with $V = 100$ vehicle types and produces an inspection fraud rate of 15.0 %, squarely in the middle of the range across all specifications.

E.2 Sensitivity to Classification and Sample Trimming Parameters

The estimation procedure requires three tuning parameters beyond the choice of indicators and vehicle types: the classification threshold γ above which a station is labeled fraudulent, the minimum number of inspections per station–vehicle-type cell (min cell), and the minimum number of inspections per station (min station). We conduct independent one-at-a-time sweeps over each parameter, holding the other two at their baseline values ($\gamma = 0.9$, min cell = 5, min station = 30). Table E2 reports the results.

The classification threshold γ (Panel A) governs how much posterior evidence is required to label a station as fraudulent. The number of classified fraud stations falls monotonically from 1,750 at $\gamma = 0.50$ to 1,460 at $\gamma = 0.95$ —a 17 % reduction—but the economically meaningful quantities are nearly invariant. The number of high-intensity stations ($\hat{\pi}_s > 25\%$) ranges from 212 to 212, and the inspection fraud rate is stable at 15.0 %. This pattern arises because the marginal stations excluded by tighter thresholds have low estimated fraud rates: they contribute to the fraud station count but negligibly to the inspection-weighted fraud share. In other words, the classifier’s extensive margin (how many stations are labeled) is sensitive to γ , but its intensive margin (how much fraud is attributed to those stations) is not.

The minimum cell size (Panel B) determines the finest station–vehicle-type cells retained in estimation. Increasing from 3 to 20 progressively trims small cells, reducing the station count from 2,457 to 2,131, but the inspection fraud rate remains in a narrow 14.9–15.9 % band. At min cell = 50, however, the sample drops sharply to 1,738 stations and the fraud rate rises to 16.8 %, as substantial trimming removes informative cells from moderate-fraud stations. The high- π station count similarly declines from 212 to 158 at this extreme, indicating that some genuinely fraudulent stations lose enough cells to be unidentifiable. Our baseline of 5 sits comfortably in the stable region.

The minimum station size (Panel C) has the least impact. Across a tenfold range from 10 to 100 total inspections per station, the inspection fraud rate is essentially constant at 15.0 % and the high- π count varies by only 6 stations (207–213). This insensitivity reflects the fact that very small stations contribute little to the inspection-weighted fraud rate regardless of their classification.

Table E2. Sensitivity to Classification and Sample Trimming Parameters

γ	Min cell	Min station	S	Fraud stations	High- π (>25%)	Inspection fraud (%)
<i>Panel A: Classification threshold (γ)</i>						
0.50	—	—	2,416	1,750	212	15.0
0.60	—	—	2,416	1,720	212	15.0
0.70	—	—	2,416	1,543	212	15.0
0.80	—	—	2,416	1,499	212	15.0
0.90	—	—	2,416	1,483	212	15.0
0.95	—	—	2,416	1,460	212	15.0
<i>Panel B: Minimum cell size</i>						
—	3	—	2,457	1,275	211	14.9
—	5	—	2,416	1,483	212	15.0
—	10	—	2,308	1,350	206	15.3
—	20	—	2,131	1,011	192	15.9
—	50	—	1,738	715	158	16.8
<i>Panel C: Minimum station size</i>						
—	—	10	2,461	1,484	213	15.0
—	—	20	2,439	1,477	213	15.0
—	—	30	2,416	1,483	212	15.0
—	—	50	2,370	1,479	210	15.0
—	—	100	2,295	1,460	207	15.0

Note: Each panel varies one parameter while holding the others at baseline (bold row: $\gamma = 0.9$, min cell = 5, min station = 30). Dashes indicate held-fixed parameters. S : number of stations in the estimation sample. Fraud stations: stations with $\hat{\gamma}_s$ above the classification threshold. High- π (>25%): stations with estimated inspection-weighted fraud rate $\hat{\pi}_s > 25\%$. Inspection fraud (%): $\hat{\pi}$ -weighted share of inspections at classified fraud stations.

F Remote Sensing Data and Pollution Impacts

F.1 The TORVET Remote Sensing Program

Texas operates the Texas On-Road Vehicle Emissions Testing (TORVET) program, which uses roadside remote sensing devices (RSD) to measure the exhaust emissions of passing vehicles under normal driving conditions. The program deploys OPUS RSD equipment at locations across the state’s emissions-testing counties, capturing instantaneous tailpipe concentrations of carbon monoxide (CO, in percent), hydrocarbons (HC, in ppm as propane equivalents), nitric oxide (NO, in ppm), and carbon dioxide (CO₂, in percent). Each measurement is paired with the vehicle’s speed, acceleration, and vehicle-specific power (VSP), as well as ambient conditions including temperature, relative humidity, and barometric pressure.

Vehicle identification is obtained through license plate imaging linked to DMV registration records, providing the Vehicle Identification Number (VIN). The program captures vehicles at 12 roadside sites across emissions-testing counties, with equipment rotating among sites on a non-continuous schedule. Our data span 2016–2024 and were obtained from the Texas Commission on Environmental Quality (TCEQ) via a Public Information Request.

Unlike dynamometer-based inspection tests, remote sensing measures actual on-road emissions under real-world driving conditions. This distinction is central to our analysis: a vehicle may pass an OBD-II inspection through fraud while still emitting elevated pollutants on the road. Remote sensing captures this discrepancy.

F.2 Carbon Balance Formula

We convert raw pollutant concentrations to mass-based emission rates (grams per kilogram of fuel burned) using the full carbon balance method described in [Eastern Research Group \(2024a\)](#). The formula for pollutant p is:

$$\text{ER}_p = \frac{M_p \cdot Q_p / Q_{\text{CO}_2}}{\frac{Q_{\text{CO}}}{Q_{\text{CO}_2}} + 1 + \frac{6 \cdot Q_{\text{HC}}}{Q_{\text{CO}_2}}} \cdot \frac{W_c}{M_c} \quad (\text{F1})$$

where ER_p is the emission rate in g/kg-fuel, M_p is the molecular weight of pollutant p (CO: 28, HC: 44 as propane, NO: 30), Q_p and Q_{CO_2} are the measured concentrations (in percent), $W_c = 860$ gC/kg-fuel is the carbon content of gasoline assuming CH_2 stoichiometry, and $M_c = 12$ is the molecular weight of carbon.

The denominator in Equation F1 accounts for exhaust carbon that exits as CO and HC rather than CO_2 . Without this correction, dividing by CO_2 alone understates total fuel consumption and overstates emission rates—an effect that is negligible for clean vehicles (where nearly all carbon exits as CO_2) but substantial for high emitters. For a vehicle with 3% CO, the denominator reaches approximately 1.25, implying that the simplified formula overestimates emissions by 20%.

The factor of 6 in the denominator encodes two corrections: a factor of 2 for undetected hydrocarbons ([Singer and Harley, 1998](#)), since remote sensing instruments calibrated with propane detect approximately half of total exhaust HC, and a factor of 3 for the carbon atoms per propane molecule (C_3H_8). The HC molecular weight of 44 (propane) in the numerator similarly reflects the propane-equivalent calibration standard.

F.3 HC Zero-Point Correction

Remote sensing HC measurements exhibit systematic site-by-day instrument drift that must be corrected before analysis. Following standard practice in the remote sensing literature (Bishop and Stedman, 2008; Eastern Research Group, 2024a), we compute a zero-point offset for each site-day as the median HC reading among reference vehicles—defined as vehicles with a model year within two years of the observation year (i.e., $\text{age} \leq 2$). Modern vehicles with functional catalytic converters emit effectively zero HC, so any nonzero median among these vehicles reflects instrument bias rather than real emissions.

We subtract this site-day offset from all HC readings at the same site-day. The correction is applied only to HC; we do not correct CO or NO, as even modern vehicles emit measurable quantities of these pollutants under load, making a zero-point reference infeasible.

After correction, approximately 44% of HC readings remain negative. We preserve these negative values rather than censoring them, because the measurement noise is approximately symmetric: truncating the left tail would bias fleet-average estimates upward.

F.4 Sample Construction

We begin with all TORVET remote sensing captures with valid VINs and apply several filters before analysis. First, we restrict to vehicles registered in Texas in one of the 17 TCEQ emissions-testing counties, dropping out-of-state vehicles and vehicles in counties not subject to mandatory testing. Second, we exclude diesel vehicles, which undergo a different testing regime and have fundamentally different emissions chemistry. Third, we drop vehicles within two model years of the capture year, as Texas exempts new vehicles from emissions testing. Fourth, we classify each remaining vehicle’s registration status using the DMV expiration date linked to the license plate: vehicles whose registration expired six or more months before capture (or with missing expiration data) are classified as having lapsed registration; those with recently expired registration (0–6 months) are excluded as ambiguous.

We then match each filtered TORVET capture to the vehicle’s most recent OBD-II emissions inspection using a rolling join on VIN and date. For each calendar year of TORVET data, we load emission test records from the current year and the prior year to capture tests that occurred near the boundary. A match is valid if the TORVET capture occurs 0–365 days after the most recent test date. Vehicles with current registration but no matching test are excluded, as the absence of a match may reflect data limitations rather than test avoidance.

The resulting analysis sample contains three mutually exclusive groups. The baseline consists of vehicles matched to a non-flagged OBD inspection (compliant). Fraud-flagged

vehicles are those matched to an inspection exhibiting the repeat electronic profile indicator. Unregistered vehicles have both lapsed registration and no OBD inspection within 365 days—these vehicles are driving with expired registration and no evidence of a recent emissions test.

Table F1 summarizes the sample construction process by year.

Table F1. TORVET Remote Sensing: Sample Construction

Year	TORVET Obs.	Analysis N	% Compliant	% Fraud Flagged	% Unreg- istered	Dropped
2017	59,549	36,166	83.7%	13.1%	3.2%	23,383
2018	176,902	103,707	83.5%	13.5%	3.0%	73,195
2019	544,694	310,859	84.1%	12.6%	3.3%	233,835
2020	352,523	208,266	83.0%	12.2%	4.8%	144,257
2021	346,742	222,525	80.5%	13.8%	5.7%	124,217
2022	486,710	302,818	80.4%	14.7%	4.9%	183,892
2023	509,789	318,713	81.7%	12.5%	5.7%	191,076
2024	603,275	381,889	83.6%	8.1%	8.3%	221,386
Total	3,080,184	1,884,943	82.4%	12.2%	5.4%	1,195,241

Note: Each row summarizes one year of TORVET remote sensing data (2017–2024). TORVET Obs. is the total number of roadside emission captures with valid VINs. The Dropped column reports observations excluded before analysis: out-of-state vehicles, vehicles in non-emissions counties, diesel vehicles, new-car exempt vehicles (age ≤ 2), vehicles with recently expired registration (0–6 months), and currently registered vehicles that we were unable to match to a recent emissions test. Analysis N is the remaining sample, decomposed into three mutually exclusive groups. Compliant vehicles matched to a non-flagged OBD inspection within 365 days. Fraud Flagged vehicles matched to an inspection with estimated fraud posterior $\hat{\tau}_i > 0.5$ from the detection model (Section 3). Unregistered vehicles have lapsed registration (expired 6+ months or missing) and no OBD inspection within 365 days. The three composition columns sum to approximately 100%.

We restrict the g/kg-fuel sample to observations with $\text{CO}_2 \geq 5\%$ to avoid extreme values from division by near-zero denominators. This filter retains virtually all valid readings, as the median CO_2 concentration in our sample is approximately 15%.

F.5 Robustness Results

Table F2 reports results using alternative fraud indicators under the preferred specification. The emissions impact is not specific to the repeat electronic profile indicator: vehicles flagged by other indicators—including eVIN–VIN mismatch, missing eVIN, rapid succession, and the combined eVIN indicator—also exhibit significantly elevated on-road emissions.

Table F3 examines sensitivity to the fixed effect structure. Column (2) replaces separate make and model-year fixed effects with a single combined make $\times$ model $\times$ model-year fixed effect (e.g., Honda Civic 1999), which absorbs vehicle-type heterogeneity more granularly. Column (3) adds registered-county fixed effects to the preferred specification. In both cases,

Table F2. On-Road Emissions Impact: Alternative Fraud Indicators

	NO (g/kg fuel)			HC (g/kg fuel)		
	Coef.	(SE)	N flagged	Coef.	(SE)	N flagged
Fraud Posterior ($\hat{\tau}_i > 0.5$)	2.23***	(0.03)	234,563	0.45***	(0.01)	234,563
Repeat Electronic Profile	2.21***	(0.03)	179,540	0.43***	(0.01)	179,540
Rapid Succession	2.01***	(0.05)	66,392	0.38***	(0.02)	66,392
eVIN–VIN Mismatch	1.22***	(0.06)	20,861	0.26***	(0.03)	20,861
Missing eVIN	1.89***	(0.03)	159,547	0.30***	(0.01)	159,547
eVIN Mismatch or Missing	1.86***	(0.02)	180,408	0.31***	(0.01)	180,408
Observations	1,884,938					

Note: Each row reports the coefficient on the indicated fraud flag from a separate regression using the preferred specification (make, model-year, and capture-date fixed effects with driving controls). All regressions include the unregistered vehicle indicator. We drop 2016 captures due to incomplete fraud flag lookback at the start of the sample. Observations are weighted by the inverse of the number of remote sensing captures per vehicle per year. Standard errors two-way clustered by vehicle and capture date in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the fraud coefficient remains large, positive, and statistically significant, and is similar in magnitude to the preferred specification.

F.6 Emissions by Vehicle Age

Figure F1 plots mean on-road emissions by vehicle age for each of the three treatment categories: compliant vehicles, vehicles flagged for fraud, and unregistered vehicles. Emissions increase with vehicle age across all groups, consistent with deteriorating emissions control systems. Fraud-flagged vehicles emit substantially more than compliant vehicles at every age, and the gap widens for older vehicles. Unregistered vehicles exhibit intermediate emission levels.

F.7 Fraud Effects by Model Year

Figure F2 plots the estimated fraud effect by model-year bin from an interaction regression that replaces the single fraud indicator with fraud $\times$ model-year bin interactions. The treatment effect is large and significant for older vehicles (≤ 1999 through 2005–09) and diminishes for newer vintages, consistent with the raw age profiles in Figure F1.

F.8 Health Damages Over Time

Figure F3 plots statewide health damages by year, distinguishing fraud-flagged and unregistered vehicle contributions. Figure F4 disaggregates by metropolitan area.

Table F3. On-Road Emissions Impact: Alternative Fixed Effect Specifications

	NO (g/kg fuel)			HC (g/kg fuel)		
	(1)	(2)	(3)	(4)	(5)	(6)
Fraud flag	2.23*** (0.03)	2.16*** (0.03)	2.23*** (0.03)	0.45*** (0.01)	0.46*** (0.01)	0.45*** (0.01)
Unregistered	0.94*** (0.03)	0.82*** (0.07)	0.94*** (0.03)	0.42*** (0.02)	0.31*** (0.05)	0.42*** (0.02)
Control mean	1.09	1.09	1.09	0.19	0.19	0.19
Observations	1,884,938					
Make FE	Yes	—	Yes	Yes	—	Yes
Model-year FE	Yes	—	Yes	Yes	—	Yes
Make $\times$ Mod. $\times$ MY FE	—	Yes	—	—	Yes	—
Capture-date FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	No	No	Yes	No	No	Yes
Driving controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: Each column reports coefficients from a regression of on-road emissions (g/kg-fuel) on a fraud indicator (estimated posterior $\hat{\tau}_i > 0.5$ from the detection model) and an unregistered vehicle indicator. Columns (1) and (4) reproduce the preferred specification from the main table (separate make, model-year, and capture-date fixed effects with driving controls). Columns (2) and (5) replace separate make and model-year FE with a single combined make $\times$ model $\times$ model-year fixed effect (e.g., Honda Civic 1999). Columns (3) and (6) add registered-county fixed effects (17 TCEQ emissions counties) to the preferred specification. Standard errors two-way clustered by vehicle and capture date in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.9 Health Damages: County-Level Detail

Table F4 disaggregates the metropolitan-area damages reported in the body (Table 3) to the individual county level. Counties are grouped by metropolitan area and ordered by annual fraud damage within each group. Unregistered damage estimates are available only for the 14 counties where TORVET remote sensing equipment is physically deployed (all DFW and Houston counties). Austin (Travis and Williamson) and El Paso lack local sensors; vehicles registered in these counties appear in the TORVET data only when traveling through DFW or Houston sensor sites, producing a selected sample with upward-biased unregistered shares. We therefore omit unregistered damage estimates for these three counties.

F.10 Age-Specific Damages Robustness

Table F5 presents a robustness check that allows both the treatment effect and VMT to vary by vehicle age. We estimate fraud $\times$ model-year bin interactions (as in Figure F2) and combine the bin-specific coefficients with bin-specific median VMT and fuel economy. Although older vehicles have dramatically larger treatment effects, they also drive substantially fewer miles per year; however, the net effect of accounting for this heterogeneity is

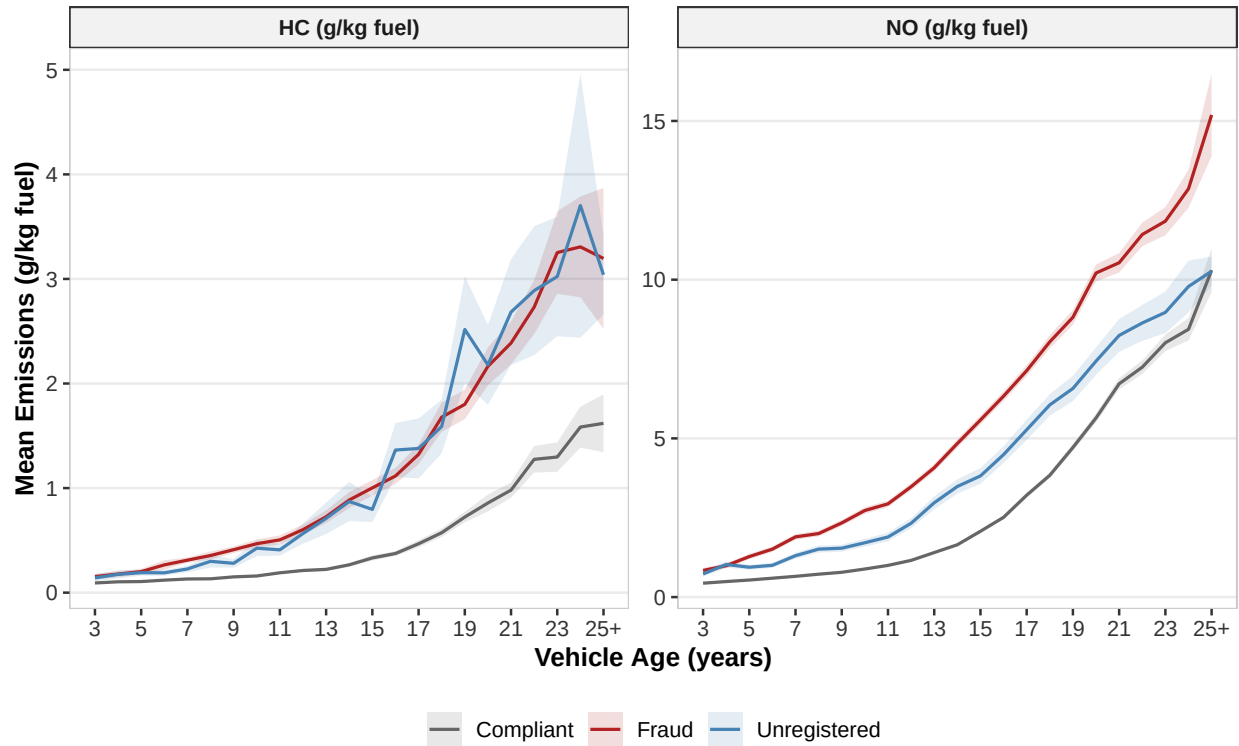


Figure F1. Mean On-Road Emissions by Vehicle Age and Treatment Group

Note: Each line plots the weighted mean emission rate (g/kg fuel) by vehicle age for compliant vehicles, fraud-flagged vehicles (repeat electronic profile), and unregistered vehicles. Shaded bands show 95% confidence intervals. Observations are weighted by the inverse of the number of remote sensing captures per vehicle per year. Vehicle ages 25 and above are pooled.

to *increase* the damage estimate relative to the uniform specification, because fraud-flagged vehicles disproportionately skew toward older age bins.

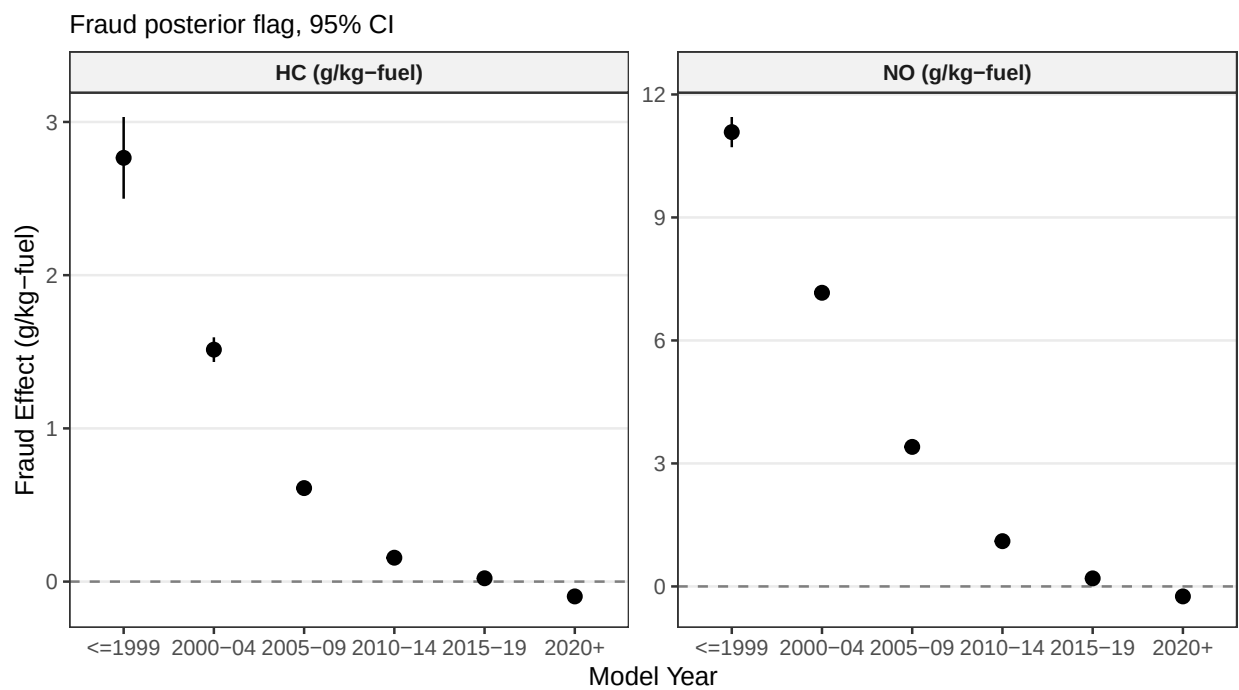


Figure F2. Fraud Effect on Emissions by Model-Year Bin

Note: Point estimates and 95% confidence intervals from a regression of on-road emissions (g/kg-fuel) on fraud flag $\times$ model-year bin interactions, controlling for unregistered status, driving conditions, and make and capture-date fixed effects. Clustering by VIN and capture date.

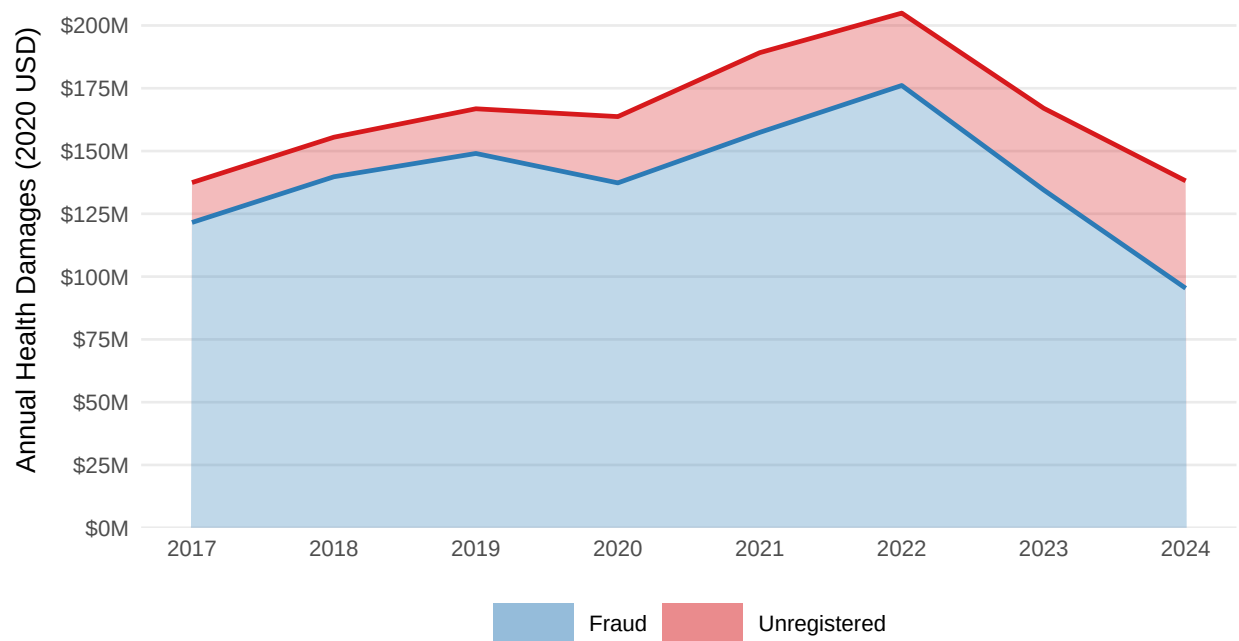


Figure F3. Statewide Health Damages by Year

Note: Stacked bars show annual health damages from fraud-flagged vehicles and unregistered vehicles (at 100% of registered-vehicle VMT). Damages computed using AP4 county-level marginal damages for NOx and VOC.

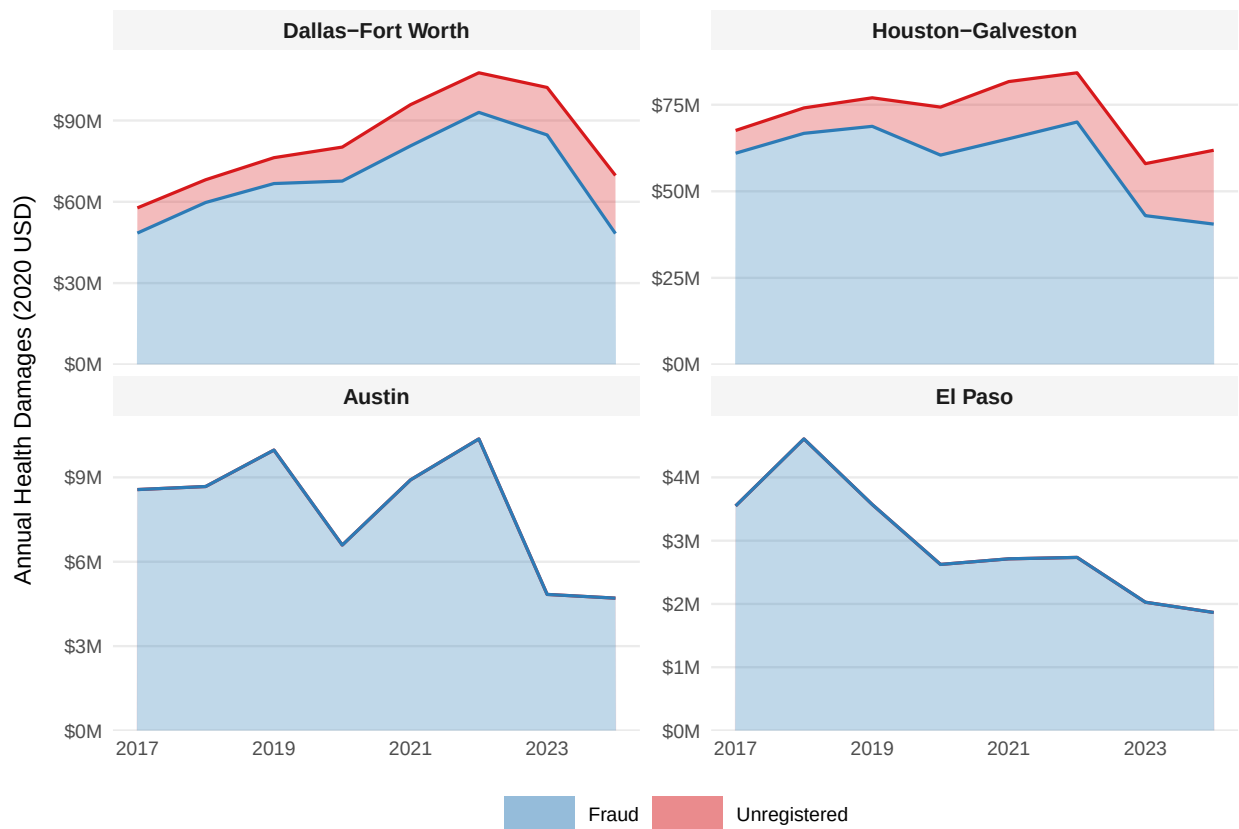


Figure F4. Health Damages by Metropolitan Area and Year

Note: Each panel shows annual health damages for one metropolitan area, decomposed into fraud-flagged and unregistered vehicle contributions. See Table 3 notes for metropolitan area definitions.

Table F4. Annual Health Damages from Excess Emissions by County

County	Fraud Vehicles/yr	Per-Vehicle Damage	Annual Fraud Damage	Unreg. (100% VMT)	Unreg. (50% VMT)
<i>Dallas-Fort Worth</i>					
Dallas	477,394	\$135	\$64.5 M	\$7.5 M	\$3.8 M
Collin	17,028	\$91	\$1.5 M	\$1.3 M	\$635,595
Tarrant	15,081	\$69	\$1.0 M	\$2.3 M	\$1.2 M
Denton	6,271	\$117	\$731,933	\$1.3 M	\$628,508
Ellis	2,443	\$188	\$458,634	\$362,677	\$181,338
Johnson	1,024	\$151	\$154,806	\$301,481	\$150,740
Kaufman	642	\$130	\$83,227	\$195,087	\$97,543
Parker	984	\$77	\$75,902	\$109,242	\$54,621
Rockwall	634	\$114	\$72,376	\$181,855	\$90,927
<i>Subtotal</i>	521,500		\$68.6 M	\$13.6 M	\$6.8 M
<i>Houston-Galveston</i>					
Harris	452,133	\$123	\$55.7 M	\$10.5 M	\$5.2 M
Fort Bend	17,065	\$95	\$1.6 M	\$545,132	\$272,566
Brazoria	8,700	\$159	\$1.4 M	\$704,535	\$352,268
Montgomery	4,010	\$134	\$537,929	\$811,348	\$405,674
Galveston	3,285	\$62	\$202,228	\$342,409	\$171,205
<i>Subtotal</i>	485,194		\$59.5 M	\$12.9 M	\$6.4 M
<i>Austin</i>					
Travis	68,926	\$94	\$6.5 M	— [†]	— [†]
Williamson	16,915	\$78	\$1.3 M	— [†]	— [†]
<i>Subtotal</i>	85,842		\$7.8 M	— [†]	— [†]
<i>El Paso</i>					
El Paso	55,394	\$53	\$3.0 M	— [†]	— [†]
<i>Subtotal</i>	55,394		\$3.0 M	— [†]	— [†]
Total	1,147,929	\$121	\$138.9 M	\$26.5 M	\$13.2 M

Note: See Table 3 notes. Counties grouped by metropolitan area and ordered by annual fraud damage within each group. [†]No local TORVET sensors; unregistered share cannot be reliably estimated.

Table F5. Age-Specific Damages: Treatment Effect and VMT by Model Year

Model Year	NO Effect (g/kg-fuel)	HC Effect (g/kg-fuel)	Median VMT (mi/yr)	Median MPG	Per-Vehicle Damage	Fleet Share
2000-04	7.16	1.51	7,341	19.0	\$272	19.7%
2005-09	3.40	0.61	9,042	19.0	\$154	29.1%
2010-14	1.10	0.16	11,277	21.0	\$54	26.5%
2015-19	0.20	0.02	11,965	22.5	\$9	17.6%
2020+	-0.24	-0.10	12,311	22.0	\$-16	2.3%
≤1999	11.08	2.77	5,599	19.0	\$335	4.9%
Weighted avg.					\$130	

Note: Treatment effects from interaction regression (fraud flag $\times$ model-year bin) with make and capture-date fixed effects plus driving controls. VMT and MPG are medians for observed-VMT vehicles in each bin. Per-vehicle damage uses county-weighted AP4 marginal damages. Fleet share is the fraction of fraud-flagged vehicles in each age bin. Weighted average per-vehicle damage: \$130 vs. \$121 from uniform specification.

G Demand Model Details

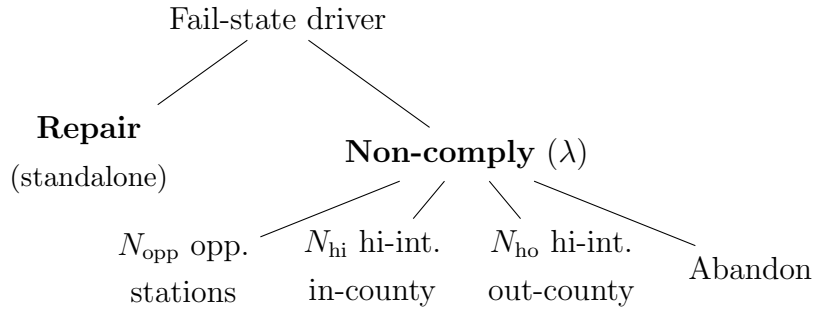
This appendix provides the full technical details underlying the demand model presented in Section 5. We present the notation, state distribution construction, likelihood derivation, model comparison, and the TORVET-based abandon decomposition.

G.1 Notation

Symbol	Definition
<i>Agent characteristics (fail-state vehicles only)</i>	
$\tau_i \in \{P, F\}$	Maintenance state: P = would pass; F = would fail
$r_i > 0$	Predicted repair cost (2024 dollars)
α_i	Repair cost sensitivity: $\alpha_i = -\exp(\mu + \sigma\nu_i)$, $\nu_i \sim N(0, 1)$
<i>Market characteristics</i>	
m	Market (county $\times$ year)
$N_{\text{opp},m}$	Count of opportunistic in-county fraud stations in market m
$N_{\text{hi},m}$	Count of high-intensity fraud stations in-county
$N_{\text{ho},m}$	Count of high-intensity fraud stations out-of-county
<i>Structural parameters</i>	
μ, σ	Log-normal location and scale for $ \alpha_i $
$\lambda \in (0, 1)$	Nesting parameter (within non-comply nest)
$\delta_{\text{opp}}, \delta_{\text{hi}}, \delta_{\text{ho}}$	Station type utilities (within nest)
$\beta_r, \beta_{\text{nc}}, \beta_a$	Coefficient vectors: repair, non-comply, and abandon equations

G.2 Nesting Structure

The three choices are organized as follows:



- **Repair** (standalone): The driver repairs the vehicle, pays r_i , and obtains a legitimate passing inspection. There is a single repair option per market—we abstract from which

valid station the driver visits.

- **Non-comply nest** (λ): Fraud stations (aggregated by type) plus abandon. The nesting parameter λ governs within-nest substitution.
- **Abandon**: The driver exits the inspection system. This is a composite of unregistered driving and vehicle exit (scrap/sell), decomposed post-estimation (Section 5.5).

The comply/non-comply structure is motivated by the descriptive evidence in Section 4.1: as repair costs increase, fraud and abandonment rise in parallel while their ratio remains approximately constant (Figure 3a). The constant ratio is consistent with within-nest IIA: conditional on choosing non-comply, the fraud/abandon split depends on station availability and outside options rather than repair cost.

G.3 Estimation Sample and Scope

The model is estimated on the *initial-fail sample*: the 4,381,031 vehicle-cycles whose inspection cycle begins with a failed OBD-II test. These vehicles are directly identified as fail-state, and we observe both their diagnostic-based predicted repair cost and their terminal outcome (repair, fraud, or abandon).

This sample excludes two populations. First, *direct-to-fraud* vehicles—drivers who proceed to a fraud station without first attempting a legitimate inspection. These vehicles are fail-state (pass-state vehicles have no incentive to purchase fraud) but lack an observed initial failure from which to compute predicted repair costs. Extending the model predictions to this group requires an additional assumption: that direct-to-fraud drivers have the same conditional distribution of preferences (α_i, ε_i) as tested fail-state drivers with the same observables. We do not impose this assumption in estimation, so the structural parameters are identified from the tested population.

Second, the model does not include pass-state vehicles, who face no repair cost and have no incentive to choose fraud or abandon. Their compliance decisions are driven by registration hassle costs rather than the inspection program, and they are invariant to the policy counterfactuals we consider.

G.4 Likelihood Function

For a single observation with choice $y_i \in \{\text{repair, fraud, abandon}\}$ and random coefficient $\alpha_i = -\exp(\mu + \sigma\nu_i)$, the conditional choice probability given ν_i is:

Repair:

$$P(\text{repair} \mid \nu_i) = \frac{\exp(V_{\text{repair},i})}{\exp(V_{\text{repair},i}) + \exp(V_{\text{nc},i})} \quad (\text{G1})$$

where $V_{\text{repair},i} = \mathbf{x}'_i \boldsymbol{\beta}_r + \alpha_i \cdot r_i$ and $V_{\text{nc},i} = \mathbf{x}'_i \boldsymbol{\beta}_{\text{nc}} + \text{IV}_i$ (equations 12–15).

Fraud:

$$P(\text{fraud} \mid \nu_i) = P(\text{nc} \mid \nu_i) \cdot s_f \quad (\text{G2})$$

where $P(\text{nc} \mid \nu_i) = 1 - P(\text{repair} \mid \nu_i)$ and the within-nest fraud share is:

$$s_f = \frac{N_{\text{opp}} e^{\delta_{\text{opp}}/\lambda} + N_{\text{hi}} e^{\delta_{\text{hi}}/\lambda} + N_{\text{ho}} e^{\delta_{\text{ho}}/\lambda}}{N_{\text{opp}} e^{\delta_{\text{opp}}/\lambda} + N_{\text{hi}} e^{\delta_{\text{hi}}/\lambda} + N_{\text{ho}} e^{\delta_{\text{ho}}/\lambda} + e^{\delta_{\text{abn},i}/\lambda}} \quad (\text{G3})$$

Abandon:

$$P(\text{abandon} \mid \nu_i) = P(\text{nc} \mid \nu_i) \cdot (1 - s_f) \quad (\text{G4})$$

The simulated log-likelihood integrates out ν_i using $S = 50$ Halton draws:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^N \log \left[\frac{1}{S} \sum_{s=1}^S P(y_i \mid \nu^{(s)}; \boldsymbol{\theta}) \right] \quad (\text{G5})$$

The nesting parameter is parameterized as $\lambda = (1 + e^{-\rho})^{-1}$ for unconstrained optimization. We maximize using L-BFGS with analytically derived gradients, verified against automatic differentiation (ForwardDiff) on a 500-observation subsample.

Inference. Standard errors are computed from the inverse of the numerical Hessian of the log-likelihood at the maximum. For the $K = 74$ parameter preferred specification, this requires $2K$ gradient evaluations using central finite differences. Delta method standard errors are reported for derived quantities: $\mathbb{E}[\alpha] = -\exp(\mu + \sigma^2/2)$, $\text{median}(\alpha) = -\exp(\mu)$, and $\lambda = (1 + e^{-\rho})^{-1}$.

G.5 Model Comparison

Table G1 reports estimation results for all 16 specifications. We estimate two model families—E (no nesting, standard 3-choice logit with log-normal repair cost sensitivity) and H (nested logit with station types)—crossed with four control sets and two sample definitions.

Key patterns. First, Model H consistently improves on Model E across all specifications: station types and nesting always improve fit, with log-likelihood improvements (ΔLL) ranging from -1,065 (H4 DFW, where county fixed effects absorb much of the station count

Table G1. Demand Model Comparison

		Nesting	K	Log-lik.	AIC	$\mathbb{E}[\alpha]$	λ	ΔLL
(1)	Age + metro	No	18	-2,632,010	5,264,056	-1.17	—	—
(2)	Age + metro	Yes	29	-2,602,046	5,204,151	-1.52	0.966	+29,964
(3)	Age + metro + time	No	22	-2,613,284	5,226,611	-0.81	—	—
(4)	Age + metro + time	Yes	35	-2,598,430	5,196,929	-2.06	0.821	+14,854
(5)	Age + county	No	44	-2,601,534	5,203,156	-1.19	—	—
(6)	Age + county	Yes	68	-2,585,320	5,170,776	-3.03	0.693	+16,214
(7)	Age + county + time	No	48	-2,581,970	5,164,036	-1.17	—	—
(8)	Age + county + time	Yes	74	-2,583,036	5,166,219	-2.04	0.857	+1,065

Note: Models E (no nesting) and H (nested logit with station types) estimated by simulated maximum likelihood on individual fail-cycle observations for the full 17-county sample. K = number of parameters. $\mathbb{E}[\alpha]$ = expected repair cost sensitivity (per \$100), from log-normal distribution $\alpha_i = -\exp(\mu + \sigma\nu_i)$. λ = nesting parameter (1 = no nesting within the non-comply nest, 0 = perfect substitution). ΔLL = improvement in log-likelihood from E to H (adding station types and nesting). Preferred specification (H4) in bold. Specifications without time trends (H1, H3) find lower λ because they exploit time-series variation in fraud station access within counties—a source of variation that also reflects general trends in fraud preferences and prices over time. The preferred specification controls for these trends with a quadratic time trend.

variation) to 29,964 (H1 full, where metro dummies leave more variation for the station structure to explain).

Second, λ is stable across specifications that include time trends: 0.82–0.97 for H1, H2, and H4. Specifications without time trends (H1, H3) find lower λ , notably 0.69 for H3. These specifications exploit time-series variation in fraud station access within counties as a source of variation for pinning down λ , but this variation also reflects general trends in fraud preferences and prices over time. The preferred specification (H4) controls for these trends with a quadratic time trend, isolating the cross-sectional variation in station counts that is more plausibly exogenous to within-nest substitution patterns.

Third, the mean repair cost sensitivity $\mathbb{E}[\alpha]$ is somewhat sensitive to the control set: county fixed effects generally yield larger (more negative) estimates than metro dummies, reflecting that within-county variation in repair costs is more plausibly exogenous than cross-metro variation. The preferred specification (H4 full) balances the richest controls with the full sample.

G.6 TORVET Abandon Decomposition: Details

The abandon decomposition in Section 5.5 recovers $P(\text{unreg} \mid \text{abandon}, \mathbf{x}_i)$ from TORVET roadside sensor data. We provide additional details here.

Sample. We use 3.3 million fail-state vehicles from DFW and Houston-Galveston counties, 2017–2023. Each vehicle is classified as repair, fraud, or abandon based on its inspection

cycle outcome. The outcome variable is whether the vehicle’s VIN appears in TORVET roadside sensor data in the following calendar year ($t + 1$).

Specification. We estimate a logit regression:

$$P(\text{in TORVET}_{t+1} \mid \mathbf{x}_i) = \Lambda(\boldsymbol{\phi}'\mathbf{x}_i + \text{abandon}_i \cdot (\boldsymbol{\phi}'_a\mathbf{x}_i)) \quad (\text{G6})$$

where $\mathbf{x}_i$ includes five age bins interacted with county $\times$ year fixed effects, and the interaction $\text{abandon}_i \cdot \mathbf{x}_i$ allows the TORVET detection rate among abandoners to differ from registered vehicles along every dimension. The ratio of predicted probabilities— $P(\text{in TORVET}_{t+1} \mid \text{abandon}, \mathbf{x})/P(\text{in TORVET}_{t+1} \mid \text{registered}, \mathbf{x})$ —equals $P(\text{unreg} \mid \text{abandon}, \mathbf{x})$ under the assumption that unregistered vehicles appear in TORVET at the same base capture rate as registered vehicles of the same type.

Results by age and year. Table G2 reports $P(\text{unreg} \mid \text{abandon})$ by age bin and year.

Table G2. Probability of Unregistered Driving Conditional on Abandonment

Year	0-5	11-15	16-20	20+	6-10	Overall
2016	100.0	99.5	95.0	—	100.0	98.7
2017	57.0	49.0	46.0	50.0	55.2	50.8
2018	43.7	36.0	33.8	36.8	41.4	37.7
2019	50.3	43.9	41.6	46.1	49.1	45.5
2020	56.2	47.0	44.3	47.7	53.7	49.1
2021	54.4	47.7	45.4	49.5	52.8	49.5
2022	58.6	49.9	46.7	50.7	56.3	52.0
2023	65.7	56.3	53.3	58.3	63.6	58.6

Note: Entries show $P(\text{unreg} \mid \text{abandon}, \text{age}, \text{year})$ in percent, estimated from a logit regression of next-year TORVET roadside sensor appearance on abandonment status, interacted with age bins and county $\times$ year fixed effects. $P(\text{unreg} \mid \text{abandon}) = P(\text{in TORVET}_{t+1} \mid \text{abandon})/P(\text{in TORVET}_{t+1} \mid \text{registered})$, where the denominator captures the base TORVET detection rate for registered vehicles with the same characteristics. Sample: DFW and Houston-Galveston counties, 2017–2023 (year t). Overall column is the weighted average across age bins.

The overall share is approximately 55.9 %, indicating that roughly half of abandoners drive unregistered while the other half exit. This share has increased over the sample period—from approximately 38 % in 2018 to 59 % in 2023—consistent with the enforcement crackdown (Section 4.2) pushing marginal drivers from fraud toward unregistered driving rather than vehicle exit. Younger vehicles (ages 0–10) have higher unregistered rates than older vehicles, reflecting the greater option value of continuing to drive a newer vehicle.

G.7 Assumptions Summary

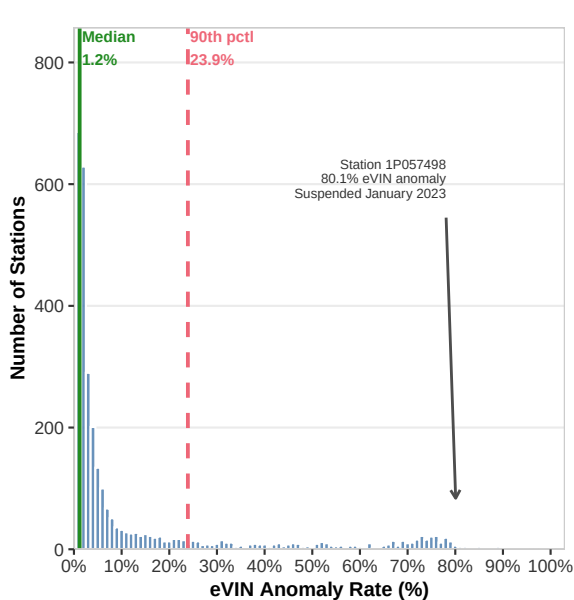
The key identifying assumption is diagnostic exogeneity (Assumption 4), which identifies μ and σ from within-type diagnostic variation in predicted repair cost. The model is estimated on vehicles with an observed initial failure, so maintenance state classification is immediate and requires no additional assumptions. Extending predictions to direct-to-fraud vehicles (not in the estimation sample) would require an assumption that their preferences are conditionally exchangeable with tested fail-state vehicles; see Section G.3.

H Additional Tables and Figures

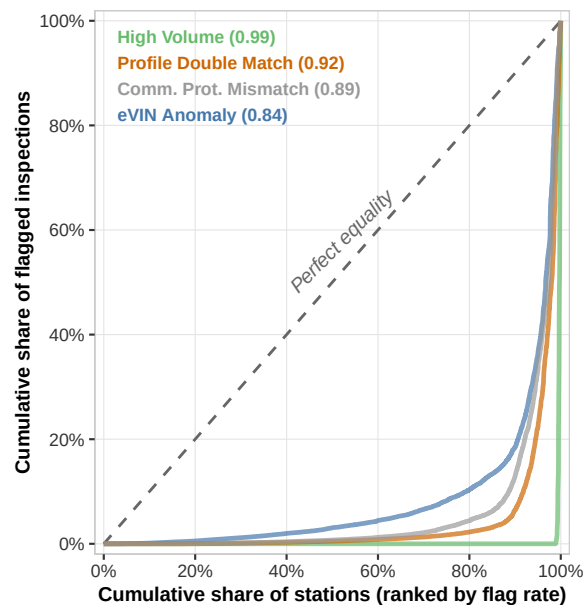
Table H1. Estimated Detection Rates by Indicator (Dallas–Fort Worth, 2022)

Indicator	Detection rate $\hat{\beta}_{j,v}$			Baseline $\hat{\Lambda}_{j,sv}$		V
	Median	[p25, p75]	Range	Median	Max	
eVIN anomaly	0.998	[0.001, 0.999]	[0.001, 0.999]	0.0042	0.0690	100
Not barcoded	0.977	[0.966, 0.985]	[0.662, 0.999]	0.2817	1.0000	100
OBD profile double match	0.935	[0.894, 0.950]	[0.001, 0.999]	0.0027	0.9628	100
Comm. protocol mismatch	0.908	[0.675, 0.965]	[0.001, 0.999]	0.0025	0.7072	100
High-volume analyzer	0.257	[0.248, 0.267]	[0.001, 0.995]	0.0000	0.4545	100

Notes: Each row summarizes the distribution of $\hat{\beta}_{j,v}$ across V vehicle types for indicator j . The detection rate is the probability that indicator j fires on a fraudulent inspection of vehicle type v . The baseline $\hat{\Lambda}_{j,sv}$ is the flag probability at honest inspections. Vehicle types are defined by make $\times$ three-year model-year bins.



(a) Distribution of eVIN Anomaly Rates



(b) Lorenz Curves

Figure H1. Concentration of Fraud Indicators Across Stations

Note: Panel (a) shows the distribution of eVIN anomaly rates across stations in 2022, restricted to stations with rates above 1%. The eVIN anomaly indicator flags tests where the electronic VIN is either missing or does not match the inspector-entered VIN. The solid green line marks the median; the dashed red line marks the 90th percentile. Panel (b) shows Lorenz curves for the cumulative distribution of flagged inspections across stations, with stations ranked by flag rate (lowest to highest), for four indicators spanning different measurement groups: High Volume (throughput), eVIN Anomaly (identity), Profile Double Match (OBD recycling), and Comm. Prot. Mismatch (communication protocol). Numbers in parentheses are Gini coefficients. Sample includes stations with at least 100 inspections in 2022.

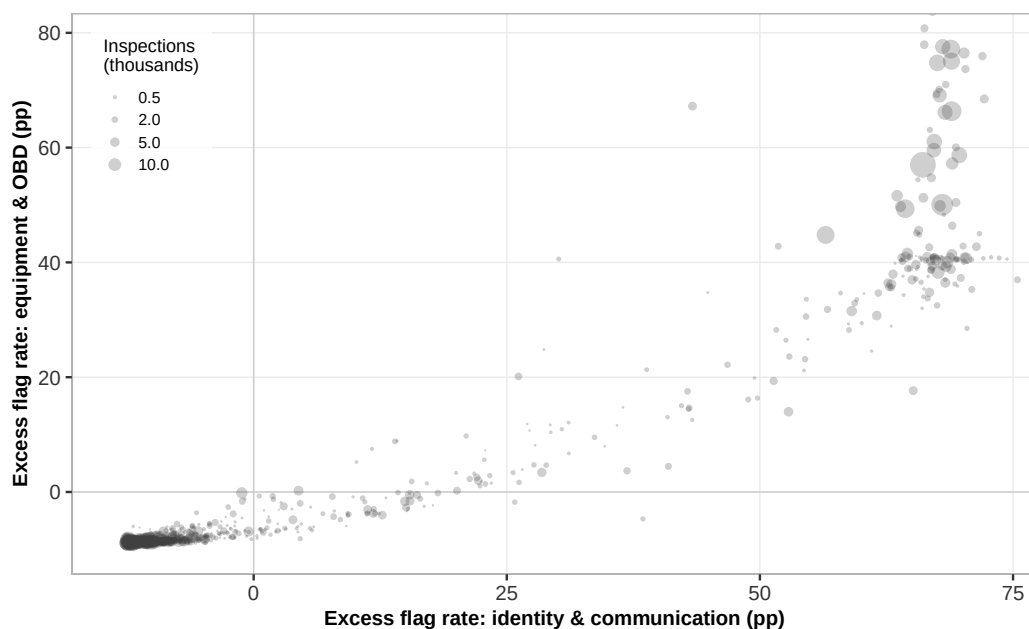


Figure H2. Correlation in Excess Flag Rates Across Distinct Indicator Groups

Note: Each point is a station in Dallas-Fort Worth, 2022. The x -axis plots the station's average excess flag rate for the identity and communication indicator group (eVIN anomaly, communication protocol mismatch); the y -axis plots the same for the equipment and OBD group (high-volume analyzer, profile double match). Excess flag rates are computed as the station's flag rate minus the inspection-weighted cross-station average for each indicator, averaged within each group. Point size is proportional to the number of inspections at the station. The cross-group correlation is 0.95. No estimation output is used; the figure shows the raw data structure the model exploits.

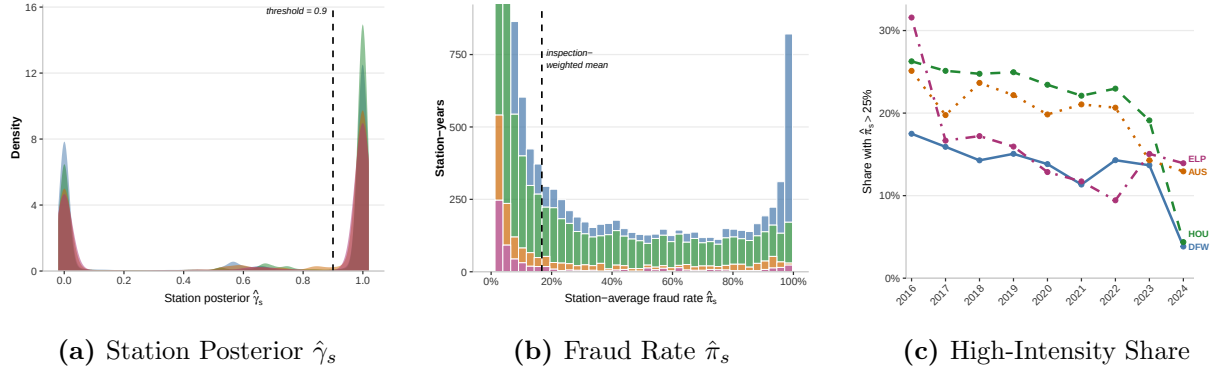
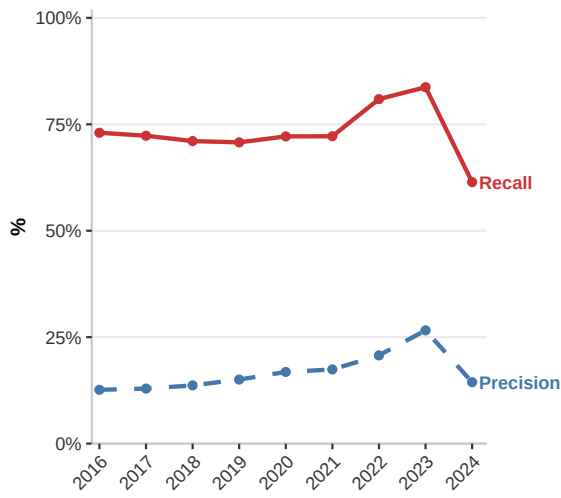
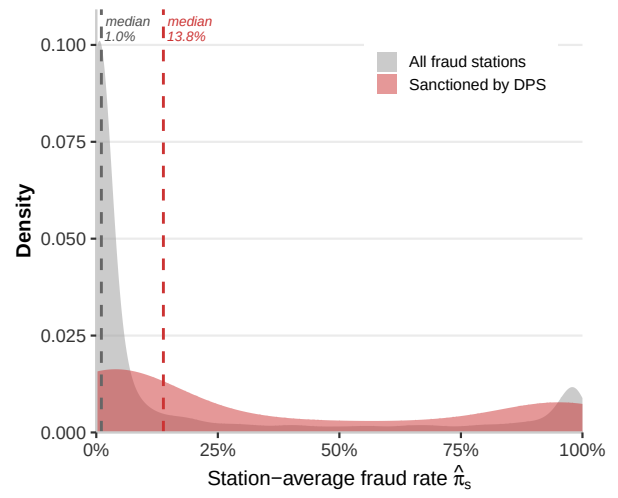


Figure H3. Station Classification and Fraud Intensity

Note: All panels pool across the four Texas metropolitan areas subject to the state’s vehicle emissions inspection program and nine years (2016–2024), yielding 48,985 station-year observations. Colors distinguish regions: Dallas–Fort Worth (DFW, blue), Houston (HOU, green), Austin (AUS, orange), and El Paso (ELP, purple). Panel (a) shows the distribution of the station-level posterior $\hat{\gamma}_s = P(b_s = 1 \mid \text{data})$. The dashed line marks the 0.9 classification threshold. Panel (b) shows the distribution of station-average fraud rates $\hat{\pi}_s$ (weighted by cell size n_{sv}) among fraud-classified stations ($\hat{\gamma}_s > 0.9$). The dashed line marks the inspection-weighted mean. Panel (c) plots the share of fraud-classified stations with station-average $\hat{\pi}_s > 25\%$.



(a) Recall and Precision Over Time



(b) Fraud Intensity by Sanction Status

Figure H4. Enforcement Validation

Note: Sanctioned stations are those with any suspension or revocation in the DPS enforcement records, 2016–2024. Panel (a) plots recall (share of sanctioned stations classified as fraud-capable) and precision (share of fraud-classified stations ever sanctioned) by year, pooling across all regions. Panel (b) compares the distribution of station-average fraud rates $\hat{\pi}_s$ among DPS-sanctioned fraud stations (red) versus all fraud-classified stations (grey), using each station’s most recent estimation year.

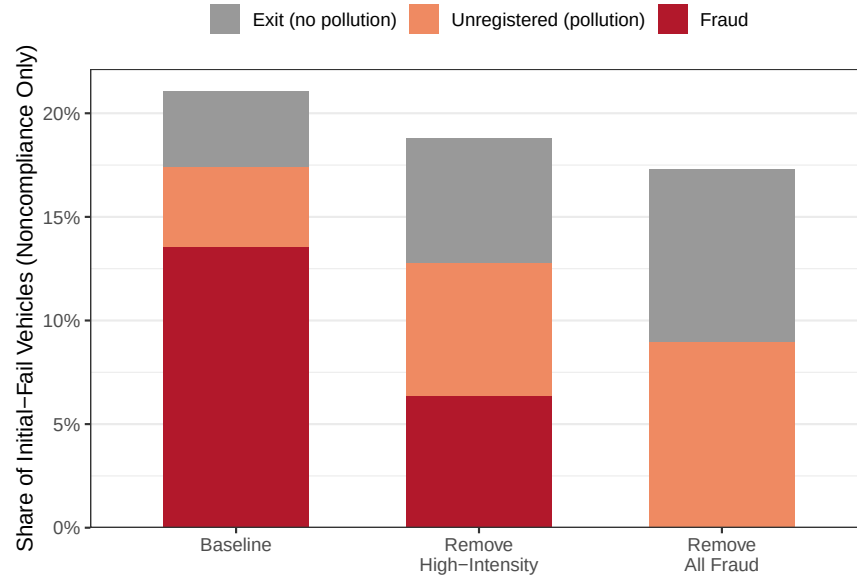


Figure H5. Counterfactual Choice Shares (2022)

Note: Predicted choice shares for initial-fail vehicles in the 17 TCEQ emissions counties, 2022. “Remove High-Intensity” eliminates dedicated fraud operations ($N_{hi} = N_{ho} = 0$), retaining opportunistic stations. “Remove All Fraud” eliminates all fraud stations. Predictions from the preferred specification (Model H4) with 50 Halton draws.

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